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技术规范

5G; NG-RAN; 架构描述 (3GPP TS 38.401 版本15.2.0 第15版)

ETSI TS 138 401 V15.2.0 (2018-07)

TECHNICAL SPECIFICATION

5G;
NG-RAN;
Architecture description
(3GPP TS 38.401 version 15.2.0 Release 15)

参考
DTS/TSGR-0338401vf20

关键词
5G

ETSI

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前言

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1 范围

本文件描述了下一代无线接入网（NG-RAN）的整体架构，包括NG接口、Xn接口和F1接口，以及它们与无线接口的交互。

2 参考文献

下列文件通过本文中的引用而构成本文件的规定。

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[1] 3GPP TR 21.905：“3GPP规范词汇表”。

[2] 3GPP TS 38.300: "新无线；总体描述；阶段2".

[3] 3GPP TS 23.501: "5G系统的系统架构".

[4] 3GPP TS 38.473: "下一代无线接入网；F1应用协议（F1应用协议）".

[5] 3GPP TS 38.414: "下一代无线接入网；NG数据传输".

[6] 3GPP TS 38.424: "下一代无线接入网；Xn数据传输".

[7] 3GPP TS 38.474: "下一代无线接入网；F1数据传输".

[8] ITU-T建议书G.823 (2000-03)："基于2048 kbit/s体系的数字网络中的抖动和漂移控制基于2048 kbit/s系列的网络。

[9] ITU-T建议书G.824 (2000-03)："基于1544 kbit/s体系的数字网络中的抖动和漂移控制基于1544 kbit/s体系的网络

[10] ITU-T建议书G.825 (2001-08)："基于同步数字体系(SDH)的数字网络中的抖动和漂移控制基于同步数字体系（SDH）的网络。

[11] ITU-T建议书G.8261/Y.1361 (2008-04): "定时与同步方面在分组网络".

[12] 3GPP TS 37.340: "NR；多连接；总体描述；阶段2".

[13] 3GPP TS 33.501: "5G系统的安全架构与流程".

[14] 3GPP TS 38.410: "NG-RAN；NG一般方面和原则".

[15] 3GPP TS 38.420: "NG-RAN；Xn一般方面和原则"

[16] 3GPP TS 38.470: "NG-RAN；F1一般方面和原则"。

[17] 3GPP TS 38.460: "NG-RAN；E1一般方面和原则"。

3 定义和缩略语

3.1 定义

根据本文件的目的，TR 21.905 [1] 中给出的术语和定义以及以下内容适用。本文件中定义的术语优先于TR 21.905 [1]中相同术语的定义（如有）。

1 Scope

The present document describes the overall architecture of the NG-RAN, including interfaces NG, Xn and F1 interfaces and their interaction with the radio interface.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 38.300: "NR; Overall description; Stage-2".

[3] 3GPP TS 23.501: "System Architecture for the 5G System".

[4] 3GPP TS 38.473: "NG-RAN; F1 application protocol (F1AP)".

[5] 3GPP TS 38.414: "NG-RAN; NG data transport".

[6] 3GPP TS 38.424: "NG-RAN; Xn data transport".

[7] 3GPP TS 38.474: "NG-RAN; F1 data transport".

[8] ITU-T Recommendation G.823 (2000-03): "The control of jitter and wander within digital networks which are based on the 2048 kbit/s hierarchy".

[9] ITU-T Recommendation G.824 (2000-03): "The control of jitter and wander within digital networks which are based on the 1544 kbit/s hierarchy".

[10] ITU-T Recommendation G.825 (2001-08): "The control of jitter and wander within digital networks which are based on the synchronous digital hierarchy (SDH)".

[11] ITU-T Recommendation G.8261/Y.1361 (2008-04): "Timing and Synchronization aspects in Packet networks".

[12] 3GPP TS 37.340: "NR; Multi-connectivity; Overall description; Stage-2".

[13] 3GPP TS 33.501: "Security Architecture and Procedures for 5G System".

[14] 3GPP TS 38.410: "NG-RAN; NG general aspect and principles".

[15] 3GPP TS 38.420: "NG-RAN; Xn general aspects and principles"

[16] 3GPP TS 38.470: "NG-RAN; F1 general aspects and principles".

[17] 3GPP TS 38.460: "NG-RAN; E1 general aspects and principles".

3 Definitions and abbreviations

3.1 Definitions

For the purposes of the present document, the terms and definitions given in TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in TR 21.905 [1].

en-gNB：如TS 37.340 [12]中所定义。

gNB: 如TS 38.300 [2]中所定义。

gNB集中单元（gNB-CU）：一个逻辑节点，承载gNB的RRC、SDAP和PDCP协议，或en-gNB的RRC和PDCP协议，控制一个或多个gNB-DU的操作。gNB-CU终止与gNB-DU连接的F1接口。

gNB分布单元（gNB-DU）：一个逻辑节点，承载gNB或en-gNB的RLC、MAC和PHY层，其操作部分由gNB-CU控制。一个gNB-DU支持一个或多个小区。一个小区仅由一个gNB-DU支持。gNB-DU终止与gNB-CU连接的F1接口。

gNB-CU控制平面 (gNB集中单元控制面): 一个逻辑节点，为en-gNB或gNB承载gNB-CU的RRC和PDCP协议的控制面部分。gNB集中单元控制面终止与gNB集中单元用户面连接的E1接口以及与gNB分布式单元连接的F1-C接口。

gNB-CU用户平面 (gNB集中单元用户面): 一个逻辑节点，为en-gNB承载gNB-CU的PDCP协议的用户平面部分，以及为gNB承载gNB-CU的PDCP协议的用户平面部分和SDAP协议。gNB集中单元用户面终止与gNB集中单元控制面连接的E1接口以及与gNB分布式单元连接的F1-U接口。

NG-RAN节点: 如TS 38.300 [2]中所定义。

PDU会话资源: 该术语用于NG、Xn和E1接口的规范。它表示支持PDU会话所提供的NG-RAN接口和无线资源。

3.2 缩写

对于本文件而言，TR 21.905 [1] 中给出的术语和定义以及以下内容适用。本文件中定义的术语优先于TR 21.905 [1]中相同术语的定义（如有）。

5GC 5G核心网 AMF 接入与移动性管理功能 AP 应用协议 AS 接入层
CM 连接管理 CMAS 商用移动警报服务 ETWS 地震与海啸预警系统
F1-U F1用户平面接口 F1-C F1控制平面接口 F1AP F1应用协议 F
DD 频分双工 GTP-U GPRS隧道协议 IP 互联网协议 NAS 非接入层
O&M 操作与维护 PWS 公共预警系统 QoS 服务质量 RNL 无线网络
层 RRC 无线资源控制 SAP 服务接入点 SCTP 流控制传输协议 SFN
系统帧号 SM 会话管理 SMF 会话管理功能 TDD 时分双工 TDM 时分
复用 TNL 传输网络层

en-gNB: as defined in TS 37.340 [12].

gNB: as defined in TS 38.300 [2].

gNB Central Unit (gNB-CU): a logical node hosting RRC, SDAP and PDCP protocols of the gNB or RRC and PDCP protocols of the en-gNB that controls the operation of one or more gNB-DUs. The gNB-CU terminates the F1 interface connected with the gNB-DU.

gNB Distributed Unit (gNB-DU): a logical node hosting RLC, MAC and PHY layers of the gNB or en-gNB, and its operation is partly controlled by gNB-CU. One gNB-DU supports one or multiple cells. One cell is supported by only one gNB-DU. The gNB-DU terminates the F1 interface connected with the gNB-CU.

gNB-CU-Control Plane (gNB-CU-CP): a logical node hosting the RRC and the control plane part of the PDCP protocol of the gNB-CU for an en-gNB or a gNB. The gNB-CU-CP terminates the E1 interface connected with the gNB-CU-UP and the F1-C interface connected with the gNB-DU.

gNB-CU-User Plane (gNB-CU-UP): a logical node hosting the user plane part of the PDCP protocol of the gNB-CU for an en-gNB, and the user plane part of the PDCP protocol and the SDAP protocol of the gNB-CU for a gNB. The gNB-CU-UP terminates the E1 interface connected with the gNB-CU-CP and the F1-U interface connected with the gNB-DU.

NG-RAN node: as defined in TS 38.300 [2].

PDU Session Resource: This term is used for specification of NG, Xn, and E1 interfaces. It denotes NG-RAN interface and radio resources provided to support a PDU Session.

3.2 Abbreviations

For the purposes of the present document, the terms and definitions given in TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in TR 21.905 [1].

5GC	5G Core Network
AMF	Access and Mobility Management Function
AP	Application Protocol
AS	Access Stratum
CM	Connection Management
CMAS	Commercial Mobile Alert Service
ETWS	Earthquake and Tsunami Warning System
F1-U	F1 User plane interface
F1-C	F1 Control plane interface
F1AP	F1 Application Protocol
FDD	Frequency Division Duplex
GTP-U	GPRS Tunnelling Protocol
IP	Internet Protocol
NAS	Non-Access Stratum
O&M	Operation and Maintenance
PWS	Public Warning System
QoS	Quality of Service
RNL	Radio Network Layer
RRC	Radio Resource Control
SAP	Service Access Point
SCTP	Stream Control Transmission Protocol
SFN	System Frame Number
SM	Session Management
SMF	Session Management Function
TDD	Time Division Duplex
TDM	Time Division Multiplexing
TNL	Transport Network Layer

4 一般原则

指导NG-RAN架构及NG-RAN接口定义的一般原则如下:

- 信令与数据传输网络的逻辑分离。 - NG-RAN和5GC功能与传输功能完全分离。NG-RAN和5GC中使用的寻址方案不应依赖于传输功能的寻址方案。某些NG-RAN或5GC功能与某些传输功能位于同一设备中，这一事实并不使传输功能成为NG-RAN或5GC的一部分。 - RRC连接的移动性完全由NG-RAN控制。 - NG-RAN接口按照以下原则定义： - 接口之间的功能划分应尽可能少选项。 - 接口基于通过该接口控制的实体的逻辑模型。 - 一个物理网元能实现多个逻辑节点。

5 总体架构

5.1 概述

Uu接口和NG接口上的协议分为两种结构:

- 用户面协议

  这些是实现实际PDU会话服务的协议，即通过接入层传输用户数据。

- 控制面协议

  这些是从不同方面控制PDU会话以及UE与网络之间连接的协议（包括请求服务、控制不同的传输资源、切换等）。此外，还包括一种用于透明传输NAS消息的机制。

5.2 用户平面

PDU会话资源服务由接入层从服务接入点提供给服务接入点。图5.2-1显示了Uu接口和NG接口上的协议，这些协议连接在一起提供此PDU会话资源服务。

4 General principles

The general principles guiding the definition of NG-RAN architecture as well as the NG-RAN interfaces are the following:

- Logical separation of signalling and data transport networks.
- NG-RAN and 5GC functions are fully separated from transport functions. Addressing scheme used in NG-RAN and 5GC shall not be tied to the addressing schemes of transport functions. The fact that some NG-RAN or 5GC functions reside in the same equipment as some transport functions does not make the transport functions part of the NG-RAN or the 5GC.
- Mobility for an RRC connection is fully controlled by the NG-RAN.
- The NG-RAN interfaces are defined along the following principles:
  - The functional division across the interfaces have as few options as possible.
  - Interfaces are based on a logical model of the entity controlled through this interface.
  - One physical network element can implement multiple logical nodes.

5 General architecture

5.1 General

The protocols over Uu and NG interfaces are divided into two structures:

- **User plane protocols**

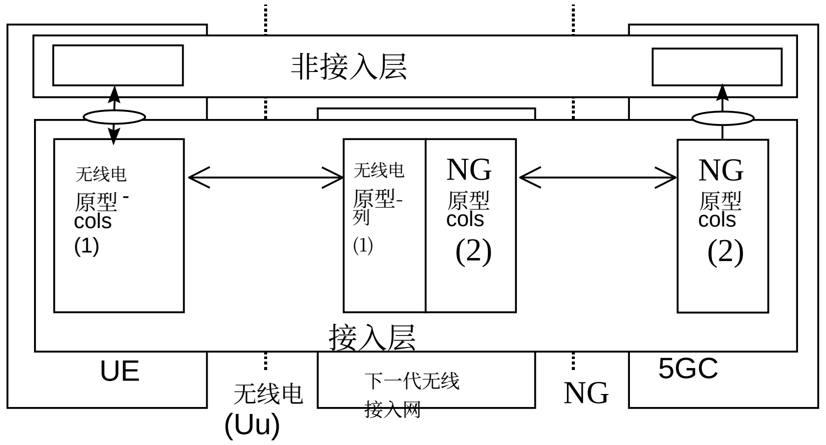
  These are the protocols implementing the actual PDU Session service, i.e. carrying user data through the access stratum.

- **Control plane protocols**

  These are the protocols for controlling the PDU Sessions and the connection between the UE and the network from different aspects (including requesting the service, controlling different transmission resources, handover etc.). Also a mechanism for transparent transfer of NAS messages is included.

5.2 User plane

The PDU Session Resource service is offered from SAP to SAP by the Access Stratum. Figure 5.2-1 shows the protocols on the Uu and the NG interfaces that linked together provide this PDU Session Resource service.

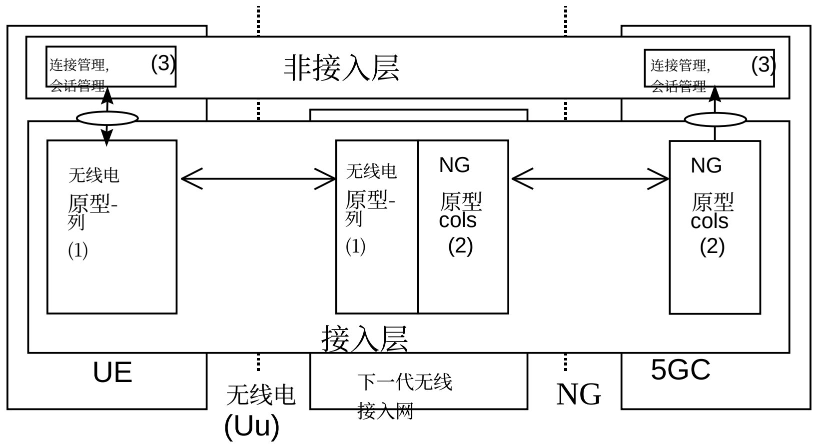


注意1：无线接口协议定义于3GPP TS 38.2xx和TS 38.3xx。注意2：NG接口协议定义于3GPP TS 38.41x。

图5.2-1：NG接口和Uu接口用户平面

5.3 控制面

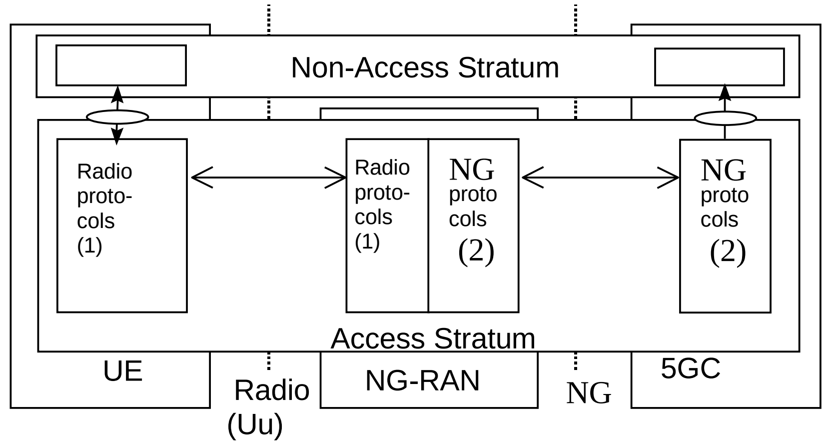
图5.3-1展示了NG和Uu接口上的控制面（信令）协议栈。



注1：无线接口协议定义在3GPP TS 38.2xx和TS 38.3xx中。注2：该协议定义在3GPP TS 38.41x中。（NG接口描述）。注3：CM、SM：这举例说明了 UE 与 5GC 之间的一组非接入层控制协议。这些协议的协议架构演进不在本文件范围内。

图5.3-1: NG和Uu控制面

注： 无线协议和NG协议都包含一种透明传输NA的机制。 S
消息。

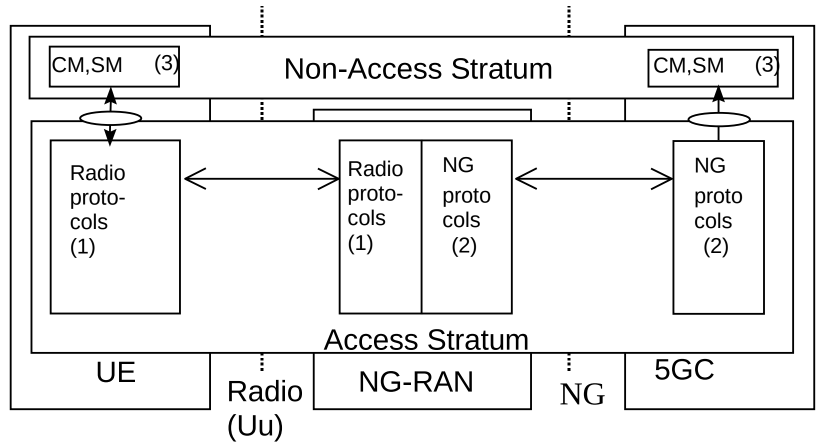


NOTE 1: The radio interface protocols are defined in 3GPP TS 38.2xx and TS 38.3xx.
NOTE 2: The NG interface protocols are defined in 3GPP TS 38.41x.

Figure 5.2-1: NG and Uu user plane

5.3 Control plane

Figure 5.3-1 shows the control plane (signalling) protocol stacks on NG and Uu interfaces.



NOTE 1: The radio interface protocols are defined in 3GPP TS 38.2xx and TS 38.3xx.
NOTE 2: The protocol is defined in 3GPP TS 38.41x. (Description of NG interface).
NOTE 3: CM, SM: This exemplifies a set of NAS control protocols between UE and 5GC. The evolution of the protocol architecture for these protocols is outside the scope of the present document.

Figure 5.3-1: NG and Uu control plane

NOTE: Both the Radio protocols and the NG protocols contain a mechanism to transparently transfer NAS messages.

6 NG-RAN架构

6.1 概述

6.1.1 NG-RAN总体架构

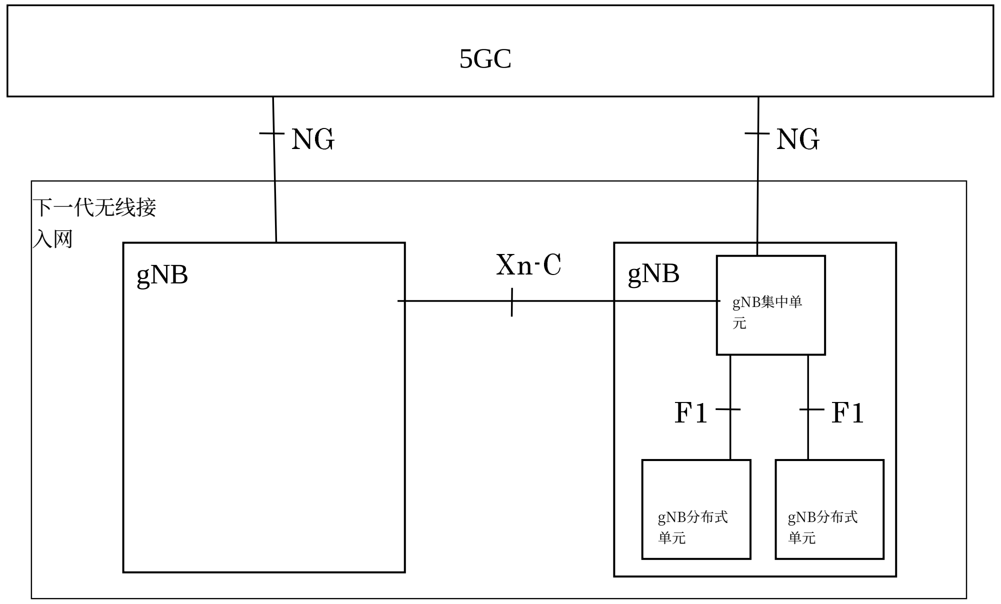


图6.1-1: 总体架构

下一代无线接入网由一组通过NG接口连接到5G核心网的gNB组成。

一个gNB能支持FDD模式、TDD模式或双模操作。

gNB能通过Xn接口互连。

一个gNB可由一个gNB-CU和一个或多个gNB-DU（多个）组成。gNB-CU和gNB-DU通过F1接口连接。

一个gNB-DU仅连接到一个gNB-CU。

注意：为实现弹性，通过适当实现，一个gNB-DU可连接到多个gNB-CU。

NG、Xn和F1是逻辑接口。

对于NG-RAN，由gNB-CU和gNB-DU组成的gNB的NG和Xn-C接口终止于gNB-CU。对于EN-DC，由gNB-CU和gNB-DU组成的gNB的S1-U和X2-C接口终止于gNB-CU。gNB-CU和连接的gNB-DU对于其他gNB和5GC仅视为一个gNB。可能的部署场景在附录A中描述。

承载NR PDCP用户面部分的节点（例如gNB-CU、gNB-CU-UP，对于EN-DC，取决于承载分离，可能是MeNB或SgNB）应执行用户不活动监视，并进一步将其不活动或（重新）激活通知给具有与核心网C平面连接的节点（例如通过E1、X2）。承载NR RLC的节点（例如gNB-DU）可执行用户不活动监视，并进一步将其不活动或（重新）激活通知给承载控制面的节点，例如gNB-CU或gNB-CU-CP。

UL PDCP配置（即UE如何在辅助节点使用UL）通过X2-C（对于EN-DC）、Xn-C（对于NG-RAN）和F1-C指示。用于DL和/或UL的无线链路中断/恢复通过X2-U（对于EN-DC）、Xn-U（对于NG-RAN）和F1-U指示。

下一代无线接入网（NG-RAN）分层为无线网络层（Radio Network Layer, RNL）和传输网络层（Transport Network Layer, TNL).

NG-RAN架构，即NG-RAN的逻辑节点及其之间的接口，被定义为RNL的一部分。

6 NG-RAN architecture

6.1 Overview

6.1.1 Overall Architecture of NG-RAN

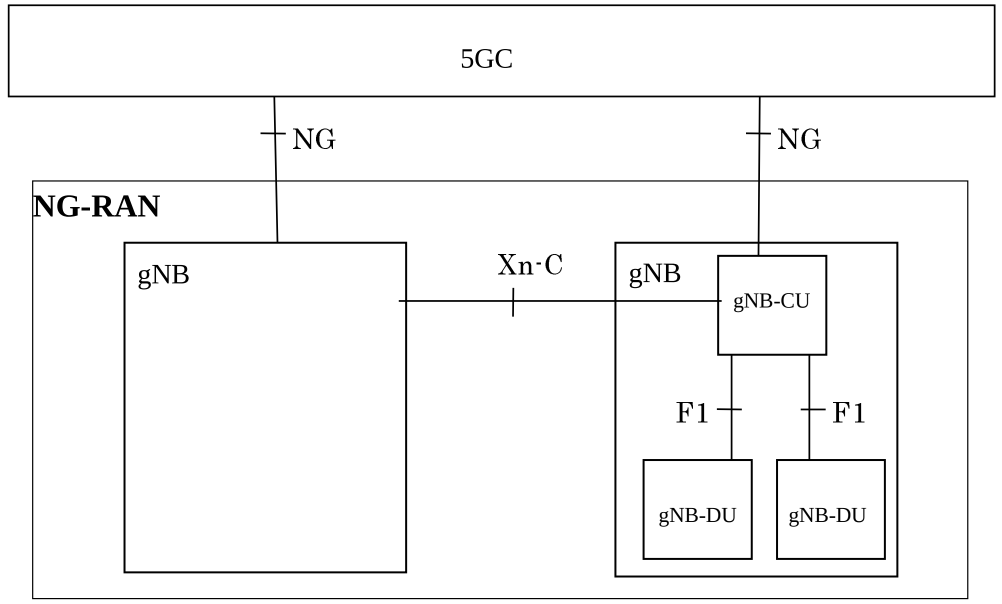


Figure 6.1-1: Overall architecture

The NG-RAN consists of a set of gNBs connected to the 5GC through the NG interface.

An gNB can support FDD mode, TDD mode or dual mode operation.

gNBs can be interconnected through the Xn interface.

A gNB may consist of a gNB-CU and one or more gNB-DU(s). A gNB-CU and a gNB-DU is connected via F1 interface.

One gNB-DU is connected to only one gNB-CU.

NOTE: For resiliency, a gNB-DU may be connected to multiple gNB-CUs by appropriate implementation.

NG, Xn and F1 are logical interfaces.

For NG-RAN, the NG and Xn-C interfaces for a gNB consisting of a gNB-CU and gNB-DUs, terminate in the gNB-CU. For EN-DC, the S1-U and X2-C interfaces for a gNB consisting of a gNB-CU and gNB-DUs, terminate in the gNB-CU. The gNB-CU and connected gNB-DUs are only visible to other gNBs and the 5GC as a gNB. A possible deployment scenario is described in Annex A.

The node hosting user plane part of NR PDCP (e.g. gNB-CU, gNB-CU-UP, and for EN-DC, MeNB or SgNB depending on the bearer split) shall perform user inactivity monitoring and further informs its inactivity or (re)activation to the node having C-plane connection towards the core network (e.g. over E1, X2). The node hosting NR RLC (e.g. gNB-DU) may perform user inactivity monitoring and further inform its inactivity or (re)activation to the node hosting control plane, e.g. gNB-CU or gNB-CU-CP.

UL PDCP configuration (i.e. how the UE uses the UL at the assisting node) is indicated via X2-C (for EN-DC), Xn-C (for NG-RAN) and F1-C. Radio Link Outage/Resume for DL and/or UL is indicated via X2-U (for EN-DC), Xn-U (for NG-RAN) and F1-U.

The NG-RAN is layered into a Radio Network Layer (RNL) and a Transport Network Layer (TNL).

The NG-RAN architecture, i.e. the NG-RAN logical nodes and interfaces between them, is defined as part of the RNL.

对于每个NG-RAN接口（NG接口、Xn接口、F1接口），规定了相关的TNL协议和功能。TNL提供用户平面传输、信令传输的服务。

在NG-Flex配置中，每个gNB连接到AMF区域内的所有AMF。AMF区域定义于3GPP技术规范23.501 [3]。

如果需要对NG-RAN接口的TNL上的控制面和用户平面数据提供安全保护，则应使用NDS/IP 3GPP技术规范33.501 [13] 。

6.1.2 gNB-CU-CP和gNB-CU-UP分离的总体架构

gNB-CU-CP和gNB-CU-UP分离的总体架构如图6.1.2-1所示。

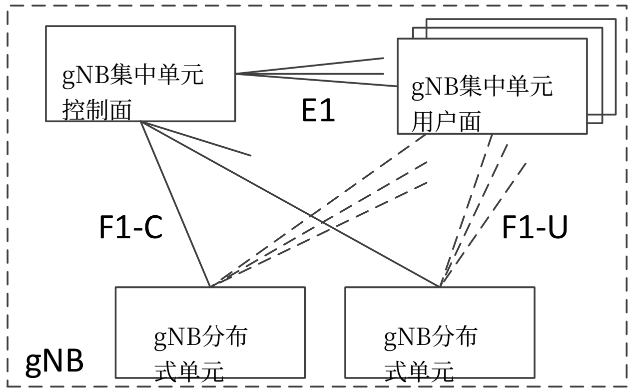


图6.1.2-1. gNB-CU-CP和gNB-CU-UP分离的总体架构

- 一个gNB可包含一个gNB-CU-CP、多个gNB-CU-UP和多个gNB-DU； - 一个gNB可包含一个gNB-CU-CP、多个gNB-CU-UP和多个gNB-DU； - gNB-CU-CP通过F1-C接口连接到gNB-DU； - gNB-CU-UP通过F1-U接口连接到gNB-DU； - gNB-CU-UP通过E1接口连接到gNB-CU-CP； - 一个gNB-DU仅连接到一个gNB-CU-CP； - 一个gNB-CU-UP仅连接到一个gNB-CU-CP； 注1：为增强弹性，一个gNB-DU和/或一个gNB-CU-UP可通过适当的实现连接到多个gNB-CU-CP。 - 一个gNB-DU能在同一个gNB-CU-CP的控制下连接到多个gNB-CU-UP； - 一个gNB-CU-UP能在同一个gNB-CU-CP的控制下连接到多个DU； 注2：gNB-CU-UP和gNB-DU之间的连接由gNB-CU-CP使用承载上下文管理功能建立。 注3：gNB-CU-CP为UE请求的服务选择合适的gNB-CU-UP。 注4：在gNB内的gNB-CU-CP内部切换期间，gNB-CU-UP之间的数据转发可通过Xn-U支持。

6.2 NG-RAN标识符

6.2.1 应用协议标识处理原则

当一个NG-RAN节点或AMF中创建新的UE关联的逻辑连接时，会分配一个应用协议标识 (AP ID)。应用协议标识应唯一标识通过NG

For each NG-RAN interface (NG, Xn, F1) the related TNL protocol and the functionality are specified. The TNL provides services for user plane transport, signalling transport.

In NG-Flex configuration, each gNB is connected to all AMFs within an AMF Region. The AMF Region is defined in 3GPP TS 23.501 [3].

If security protection for control plane and user plane data on TNL of NG-RAN interfaces has to be supported, NDS/IP 3GPP TS 33.501 [13] shall be applied.

6.1.2 Overall architecture for separation of gNB-CU-CP and gNB-CU-UP

The overall architecture for separation of gNB-CU-CP and gNB-CU-UP is depicted in Figure 6.1.2-1.

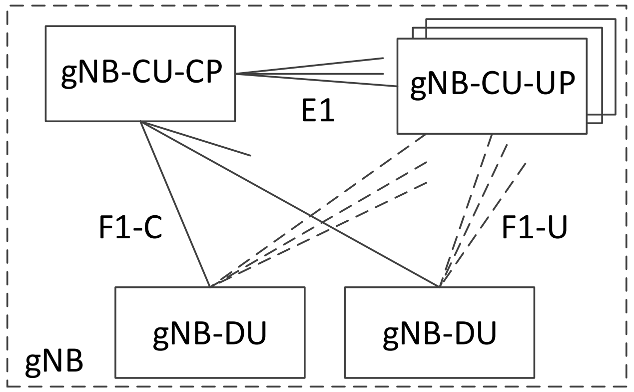


Figure 6.1.2-1. Overall architecture for separation of gNB-CU-CP and gNB-CU-UP

- A gNB may consist of a gNB-CU-CP, multiple gNB-CU-UPs and multiple gNB-DUs;
- A gNB may consist of a gNB-CU-CP, multiple gNB-CU-UPs and multiple gNB-DUs;
- The gNB-CU-CP is connected to the gNB-DU through the F1-C interface;
- The gNB-CU-UP is connected to the gNB-DU through the F1-U interface;
- The gNB-CU-UP is connected to the gNB-CU-CP through the E1 interface;
- One gNB-DU is connected to only one gNB-CU-CP;
- One gNB-CU-UP is connected to only one gNB-CU-CP;

NOTE 1: For resiliency, a gNB-DU and/or a gNB-CU-UP may be connected to multiple gNB-CU-CPs by appropriate implementation.

- One gNB-DU can be connected to multiple gNB-CU-UPs under the control of the same gNB-CU-CP;
- One gNB-CU-UP can be connected to multiple DUs under the control of the same gNB-CU-CP;

NOTE 2: The connectivity between a gNB-CU-UP and a gNB-DU is established by the gNB-CU-CP using Bearer Context Management functions.

NOTE 3: The gNB-CU-CP selects the appropriate gNB-CU-UP(s) for the requested services for the UE.

NOTE 4: Data forwarding between gNB-CU-UPs during intra-gNB-CU-CP handover within a gNB may be supported by Xn-U.

6.2 NG-RAN identifiers

6.2.1 Principle of handling Application Protocol Identities

An Application Protocol Identity (AP ID) is allocated when a new UE-associated logical connection is created in either an NG-RAN node or an AMF. An AP ID shall uniquely identify a logical connection associated to a UE over the NG

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接口或Xn接口在节点（NG-RAN节点或AMF）内，或通过F1接口或通过E1接口与用户设备关联的逻辑连接。在接收到来自发送节点的具有新的应用协议标识的消息后，接收节点应在逻辑连接持续期间存储发送节点的应用协议标识。接收节点应分配要用于标识与用户设备关联的逻辑连接的应用协议标识，并将其以及先前接收的来自发送节点的新的应用协议标识包含在返回给发送节点的第一条消息中。在之后所有与发送节点之间的消息中，发送节点和接收节点的两个应用协议标识均应包含在内。

定义 在NG接口、Xn接口、F1接口或E1接口上使用的AP标识如下所示 :

RAN UE NGAP标识:

应分配RAN UE NGAP标识，以便在gNB内的NG接口上唯一标识UE。当AMF接收到RAN UE NGAP标识时，应将其存储在该UE的与UE关联的逻辑NG连接持续时间内。一旦AMF知晓该标识，则将其包含在所有与UE相关的NGAP信令中。

RAN UE NGAP标识应在逻辑NG-RAN节点内唯一。

AMF UE NGAP标识:

应分配一个AMF UE NGAP标识，以便在AMF内通过NG接口唯一标识UE。当NG-RAN节点收到AMF UE NGAP标识时，应在此UE的与UE关联的逻辑NG连接持续期间存储该标识。一旦NG-RAN节点获知该标识，该标识将包含在所有与UE关联的NGAP信令中。

AMF UE NGAP标识应在AMF逻辑节点内唯一。

旧NG-RAN节点UE XnAP ID:

应分配一个旧NG-RAN节点UE XnAP ID，以便在源NG-RAN节点内通过Xn接口唯一标识UE。当目标NG-RAN节点收到旧NG-RAN节点UE XnAP ID时，应在此UE的与UE关联的逻辑Xn连接持续期间存储该标识。一旦目标NG-RAN节点获知该标识，该标识将包含在所有与UE关联的XnAP信令中。旧NG-RAN节点UE XnAP ID应在逻辑NG-RAN节点内唯一。

新NG-RAN节点UE XnAP ID:

新NG-RAN节点UE XnAP ID应被分配，以便在目标NG-RAN节点内通过Xn接口唯一标识该UE。当源NG-RAN节点接收到新NG-RAN节点UE XnAP ID时，应在该UE的与UE关联的逻辑Xn连接持续时间内存储该ID。一旦源NG-RAN节点获知此ID，该ID将包含在所有与UE关联的XnAP信令中。新NG-RAN节点UE XnAP ID应在逻辑NG-RAN节点内唯一。

M-NG-RAN节点UE XnAP ID:

M-NG-RAN节点UE XnAP ID应被分配，以便在双连接场景下，在M-NG-RAN节点内通过Xn接口唯一标识该UE。当S-NG-RAN节点接收到M-NG-RAN节点UE XnAP ID时，应在该UE的与UE关联的逻辑Xn连接持续时间内存储该ID。一旦S-NG-RAN节点获知此ID，该ID将包含在所有与UE关联的XnAP信令中。M-NG-RAN节点UE XnAP ID应在逻辑NG-RAN节点内唯一。

S-NG-RAN节点UE XnAP ID:

S-NG-RAN节点UE XnAP ID应被分配，以便在S-NG-RAN节点内的Xn接口上唯一标识用于双连接的UE。当M-NG-RAN节点接收到S-NG-RAN节点UE XnAP ID时，应在该UE的与UE关联的逻辑Xn连接存续期间存储该ID。一旦M-NG-RAN节点获知此ID，该ID将包含在所有与UE关联的XnAP信令中。S-NG-RAN节点UE XnAP ID应在逻辑NG-RAN节点内唯一。

gNB-CU UE F1AP标识:

应分配gNB-CU UE F1AP标识，以便在gNB-CU内的F1接口上唯一标识UE。当gNB-DU接收到gNB-CU UE F1AP标识时，它应在该UE的UE关联的逻辑F1连接持续期间内存储该标识。gNB-CU UE F1AP标识应在gNB-CU逻辑节点内唯一。

gNB-DU UE F1AP标识:

interface or Xn interface within a node (NG-RAN node or AMF) or over the F1 interface or over the E1 interface. Upon receipt of a message that has a new AP ID from the sending node, the receiving node shall store the AP ID of the sending node for the duration of the logical connection. The receiving node shall assign the AP ID to be used to identify the logical connection associated to the UE and include it as well as the previously received new AP ID from the sending node, in the first returned message to the sending node. In all subsequent messages to and from sending node, both AP IDs of sending node and receiving node shall be included.

The definitions of AP IDs as used on NG interface or Xn interface or F1 interface or E1 interface are shown below:

RAN UE NGAP ID:

A RAN UE NGAP ID shall be allocated so as to uniquely identify the UE over the NG interface within an gNB. When an AMF receives an RAN UE NGAP ID it shall store it for the duration of the UE-associated logical NG-connection for this UE. Once known to an AMF this is included in all UE associated NGAP signalling.

The RAN UE NGAP ID shall be unique within the logical NG-RAN node.

AMF UE NGAP ID:

An AMF UE NGAP ID shall be allocated so as to uniquely identify the UE over the NG interface within the AMF. When a NG-RAN node receives an AMF UE NGAP ID it shall store it for the duration of the UE-associated logical NG-connection for this UE. Once known to a NG-RAN node this ID is included in all UE associated NGAP signalling.

The AMF UE NGAP ID shall be unique within the AMF logical node.

Old NG-RAN node UE XnAP ID:

An Old NG-RAN node UE XnAP ID shall be allocated so as to uniquely identify the UE over the Xn interface within a source NG-RAN node. When a target NG-RAN node receives an Old NG-RAN node UE XnAP ID it shall store it for the duration of the UE-associated logical Xn-connection for this UE. Once known to a target NG-RAN node this ID is included in all UE associated XnAP signalling. The Old NG-RAN node UE XnAP ID shall be unique within the logical NG-RAN node.

New NG-RAN node UE XnAP ID:

A New NG-RAN node UE XnAP ID shall be allocated so as to uniquely identify the UE over the Xn interface within a target NG-RAN node. When a source NG-RAN node receives a New NG-RAN node UE XnAP ID it shall store it for the duration of the UE-associated logical Xn-connection for this UE. Once known to a source NG-RAN node this ID is included in all UE associated XnAP signalling. The New NG-RAN node UE XnAP ID shall be unique within the logical NG-RAN node.

M-NG-RAN node UE XnAP ID:

An M-NG-RAN node UE XnAP ID shall be allocated so as to uniquely identify the UE over the Xn interface within an M-NG-RAN node for dual connectivity. When an S-NG-RAN node receives an M-NG-RAN node UE XnAP ID it shall store it for the duration of the UE-associated logical Xn-connection for this UE. Once known to an S-NG-RAN node this ID is included in all UE associated XnAP signalling. The M-NG-RAN node UE XnAP ID shall be unique within the logical NG-RAN node.

S-NG-RAN node UE XnAP ID:

A S-NG-RAN node UE XnAP ID shall be allocated so as to uniquely identify the UE over the Xn interface within an S-NG-RAN node for dual connectivity. When an M-NG-RAN node receives a S-NG-RAN node UE XnAP ID it shall store it for the duration of the UE-associated logical Xn-connection for this UE. Once known to an M-NG-RAN node this ID is included in all UE associated XnAP signalling. The S-NG-RAN node UE XnAP ID shall be unique within the logical NG-RAN node.

gNB-CU UE F1AP ID:

A gNB-CU UE F1AP ID shall be allocated so as to uniquely identify the UE over the F1 interface within a gNB-CU. When a gNB-DU receives a gNB-CU UE F1AP ID it shall store it for the duration of the UE-associated logical F1-connection for this UE. The gNB-CU UE F1AP ID shall be unique within the gNB-CU logical node.

gNB-DU UE F1AP ID:

应分配gNB-DU UE F1AP标识，以便在gNB-DU内的F1接口上唯一标识UE。当gNB-CU接收到gNB-DU UE F1AP标识时，它应在该UE的UE关联的逻辑F1连接持续期间内存储该标识。gNB-DU UE F1AP标识应在gNB-DU逻辑节点内唯一。

gNB-CU-CP用户设备E1AP标识:

gNB-CU-CP用户设备E1AP标识应被分配，以便在gNB-CU-CP内的E1接口上唯一标识用户设备。当gNB-CU-UP接收到gNB-CU-CP用户设备E1AP标识时，它应在其对应的与用户设备关联的逻辑E1连接持续期间存储该标识。gNB-CU-CP用户设备E1AP标识应在gNB-CU-CP逻辑节点内唯一。

gNB-CU-UP用户设备E1AP标识:

gNB-CU-UP用户设备E1AP标识应被分配，以便在gNB-CU-UP内的E1接口上唯一标识用户设备。当gNB-CU-CP接收到gNB-CU-UP用户设备E1AP标识时，它应在其对应的与用户设备关联的逻辑E1连接持续期间存储该标识。gNB-CU-UP用户设备E1AP标识应在gNB-CU-UP逻辑节点内唯一。

6.2.2 gNB-DU标识

gNB-DU标识配置在gNB分布式单元上，用于至少在gNB集中单元内唯一标识gNB分布式单元。gNB分布式单元在F1建立过程中向gNB集中单元提供其gNB-DU标识。gNB-DU标识仅在F1AP过程中使用。

6.3 传输地址

传输层地址参数在导致传输承载连接建立的无线网络应用信令过程中传输。

传输层地址参数不应在无线网络应用协议中被解释，也不应揭示传输层中使用的寻址格式。

传输层地址的格式在3GPP TS 38.414 [5], 3GPP TS 38.424 [6] 和3GPP TS 38.474 [7]中有进一步描述。

6.4 NG-RAN节点中的UE关联

NG-RAN节点中需要几种类型的UE关联：用于存储UE所需所有信息的"NG-RAN节点UE上下文"，以及用于NG/XnAP UE关联消息的UE与逻辑NG和Xn连接之间的关联。对于处于CM_CONNECTED状态的UE，存在一个"NG-RAN节点UE上下文"。

定义:

NG-RAN节点UE上下文:

NG-RAN节点UE上下文是与一个UE相关联的NG-RAN节点中的信息块。该信息块包含维持面向激活UE的NG-RAN服务所需的必要信息。当UE完成到RRC连接态的转换时，或在切换准备期间完成切换资源分配后，在目标NG-RAN节点中，建立NG-RAN节点UE上下文，在这种情况下，至少UE状态信息、安全信息、UE能力信息以及UE关联的逻辑NG连接的标识应包含在NG-RAN节点UE上下文中。

对于双连接，在完成S-NG-RAN节点添加准备流程后，在S-NG-RAN节点中也建立了NG-RAN节点UE上下文。

承载上下文:

承载上下文是与一个用户设备关联的gNB-CU-UP节点中的信息块，用于通过E1接口进行通信。它可包括与该用户设备相关联的数据无线承载、PDU会话和QoS流的信息。该信息块包含维护面向用户设备的用户面服务所需的必要信息。

A gNB-DU UE F1AP ID shall be allocated so as to uniquely identify the UE over the F1 interface within a gNB-DU. When a gNB-CU receives a gNB-DU UE F1AP ID it shall store it for the duration of the UE-associated logical F1-connection for this UE. The gNB-DU UE F1AP ID shall be unique within the gNB-DU logical node.

gNB-CU-CP UE E1AP ID:

A gNB-CU-CP UE E1AP ID shall be allocated so as to uniquely identify the UE over the E1 interface within a gNB-CU-CP. When a gNB-CU-UP receives a gNB-CU-CP UE E1AP ID it shall store it for the duration of the UE-associated logical E1-connection for this UE. The gNB-CU-CP UE E1AP ID shall be unique within the gNB-CU-CP logical node.

gNB-CU-UP UE E1AP ID:

A gNB-CU-UP UE E1AP ID shall be allocated so as to uniquely identify the UE over the E1 interface within a gNB-CU-UP. When a gNB-CU-CP receives a gNB-CU-UP UE E1AP ID it shall store it for the duration of the UE-associated logical E1-connection for this UE. The gNB-CU-UP UE E1AP ID shall be unique within the gNB-CU-UP logical node.

6.2.2 gNB-DU ID

The gNB-DU ID is configured at the gNB-DU and used to uniquely identify the gNB-DU at least within a gNB-CU. The gNB-DU provides its gNB-DU ID to the gNB-CU during the F1 Setup procedure. The gNB-DU ID is used only within F1AP procedures.

6.3 Transport addresses

The transport layer address parameter is transported in the radio network application signalling procedures that result in establishment of transport bearer connections.

The transport layer address parameter shall not be interpreted in the radio network application protocols and reveal the addressing format used in the transport layer.

The formats of the transport layer addresses are further described in 3GPP TS 38.414 [5], 3GPP TS 38.424 [6] and 3GPP TS 38.474 [7].

6.4 UE associations in NG-RAN Node

There are several types of UE associations needed in the NG-RAN node: the "NG-RAN node UE context" used to store all information needed for a UE and the associations between the UE and the logical NG and Xn connections used for NG/XnAP UE associated messages. An "NG-RAN node UE context" exists for a UE in CM_CONNECTED.

Definitions:

NG-RAN node UE context:

An NG-RAN node UE context is a block of information in an NG-RAN node associated to one UE. The block of information contains the necessary information required to maintain the NG-RAN services towards the active UE. An NG-RAN node UE context is established when the transition to RRC CONNECTED for a UE is completed or in the target NG-RAN node after completion of handover resource allocation during handover preparation, in which case at least UE state information, security information, UE capability information and the identities of the UE-associated logical NG-connection shall be included in the NG-RAN node UE context.

For Dual Connectivity an NG-RAN node UE context is also established in the S-NG-RAN node after completion of S-NG-RAN node Addition Preparation procedure.

Bearer context:

A bearer context is a block of information in a gNB-CU-UP node associated to one UE that is used for the sake of communication over the E1 interface. It may include the information about data radio bearers, PDU sessions and QoS-flows associated to the UE. The block of information contains the necessary information required to maintain user-plane services toward the UE.

UE关联的逻辑NG/Xn/F1/E1连接:

NGAP、XnAP、F1AP和E1AP提供了通过相应的NG-C、Xn-C、F1-C或E1接口交换与用户设备相关联的控制面消息的手段。

UE关联的逻辑连接是在NG/Xn/F1对等节点之间的首次NGAP/XnAP/F1AP消息交换期间建立的。

只要需要在NG/Xn/F1接口上交换UE关联的NG/XnAP/F1AP消息，该连接就保持。

UE关联的逻辑NG连接使用AMF UE NGAP标识和RAN UE NGAP标识。

与UE关联的逻辑Xn连接使用身份标识旧NG-RAN节点UE XnAP ID和新NG-RAN节点UE XnAP ID，或M-NG-RAN节点UE XnAP ID和S-NG-RAN节点UE XnAP ID。

UE关联的逻辑F1连接使用身份标识gNB-CU UE F1AP标识和gNB-DU UE F1AP标识。

当一个节点（AMF或gNB）接收到UE关联的NGAP/XnAP/F1AP消息时，该节点基于NGAP/XnAP/F1AP标识检索关联的UE。

UE关联信令:

UE关联信令是在与UE关联的逻辑NG/Xn/F1连接上交换与一个UE关联的NGAP/XnAP/F1AP消息的过程。

注意：UE关联的逻辑NG连接可在NG-RAN节点中建立NG-RAN节点UE上下文之前存在。

UE关联的逻辑F1连接可在gNB分布式单元中建立UE上下文之前存在。

7 NG-RAN功能描述

7.0 概述

功能列表请参见技术规范38.300 [2]。

7.1 空

保留供将来使用。

8 gNB-CU/gNB-DU架构中的总体流程

8.1 UE初始接入

UE初始接入的信令流程如图8.1-1所示。

UE-associated logical NG/Xn/F1/E1 -connection:

NGAP, XnAP, F1AP and E1AP provide means to exchange control plane messages associated with the UE over the respectively NG-C, Xn-C, F1-C or E1 interface.

A UE-associated logical connection is established during the first NGAP/XnAP/F1AP message exchange between the NG/Xn/F1 peer nodes.

The connection is maintained as long as UE associated NG/XnAP/F1AP messages need to be exchanged over the NG/Xn/F1 interface.

The UE-associated logical NG-connection uses the identities AMF UE NGAP ID and RAN UE NGAP ID.

The UE-associated logical Xn-connection uses the identities Old NG-RAN node UE XnAP ID and New NG-RAN node UE XnAP ID, or M-NG-RAN node UE XnAP ID and S-NG-RAN node UE XnAP ID.

The UE-associated logical F1-connection uses the identities gNB-CU UE F1AP ID and gNB-DU UE F1AP ID.

When a node (AMF or gNB) receives a UE associated NGAP/XnAP/F1AP message the node retrieves the associated UE based on the NGAP/XnAP/F1AP ID.

UE-associated signalling:

UE-associated signalling is an exchange of NGAP/XnAP/F1AP messages associated with one UE over the UE-associated logical NG/Xn/F1-connection.

NOTE: The UE-associated logical NG-connection may exist before the NG-RAN node UE context is setup in the NG-RAN node.

The UE-associated logical F1-connection may exist before the UE context is setup in the gNB-DU.

7 NG-RAN functions description

7.0 General

For the list of functions refer to TS 38.300 [2].

7.1 Void

Reserve for future use.

8 Overall procedures in gNB-CU/gNB-DU Architecture

8.1 UE Initial Access

The signalling flow for UE Initial access is shown in Figure 8.1-1.

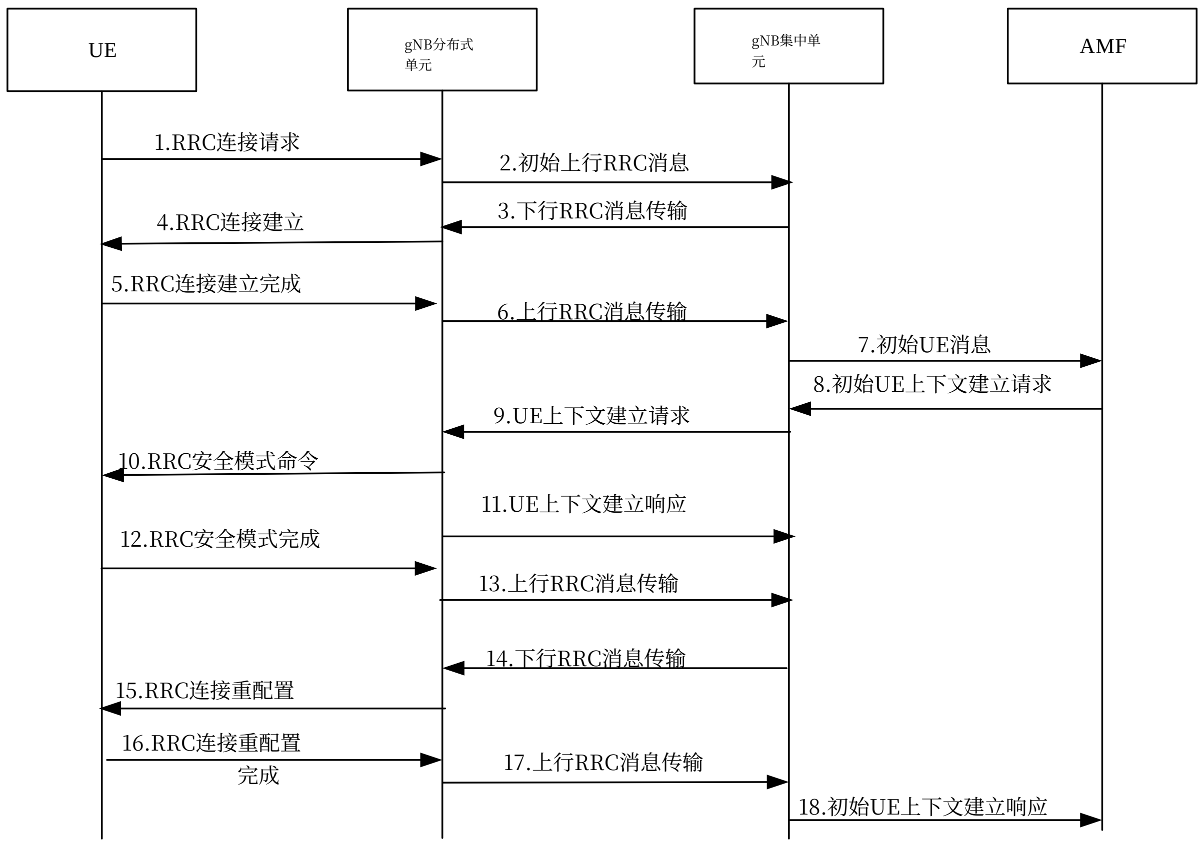

图8.1-1: UE初始接入过程

1. UE向gNB-DU发送RRC连接请求消息。2. gNB-DU在F1AP初始上行RRC消息传输消息中包含该RRC消息，并且如果UE被接纳，还包含UE对应的低层配置，然后传输给gNB-CU。初始上行RRC消息传输消息包含由gNB-DU分配的C-RNTI。3. gNB-CU为UE分配gNB-CU UE F1AP标识，并生成面向UE的RRC连接建立消息。该RRC消息封装在F1AP下行RRC消息传输消息中。4. gNB-DU向UE发送RRC连接建立消息。5. UE向gNB-DU发送RRC连接建立完成消息。6. gNB-DU将RRC消息封装在F1AP上行RRC消息传输消息中，并发送给gNB-CU。7. gNB-CU向AMF发送初始UE消息。8. AMF向gNB-CU发送初始UE上下文建立请求消息。9. gNB-CU发送UE上下文建立请求消息以在gNB-DU中建立UE上下文。在该消息中，gNB-CU还可封装RRC安全模式命令消息。10. gNB-DU向UE发送RRC安全模式命令消息。11. gNB-DU向gNB-CU发送UE上下文建立响应消息。12. UE以RRC安全模式完成消息进行响应。13. gNB-DU将RRC消息封装在F1AP上行RRC消息传输消息中，并发送给gNB-CU。

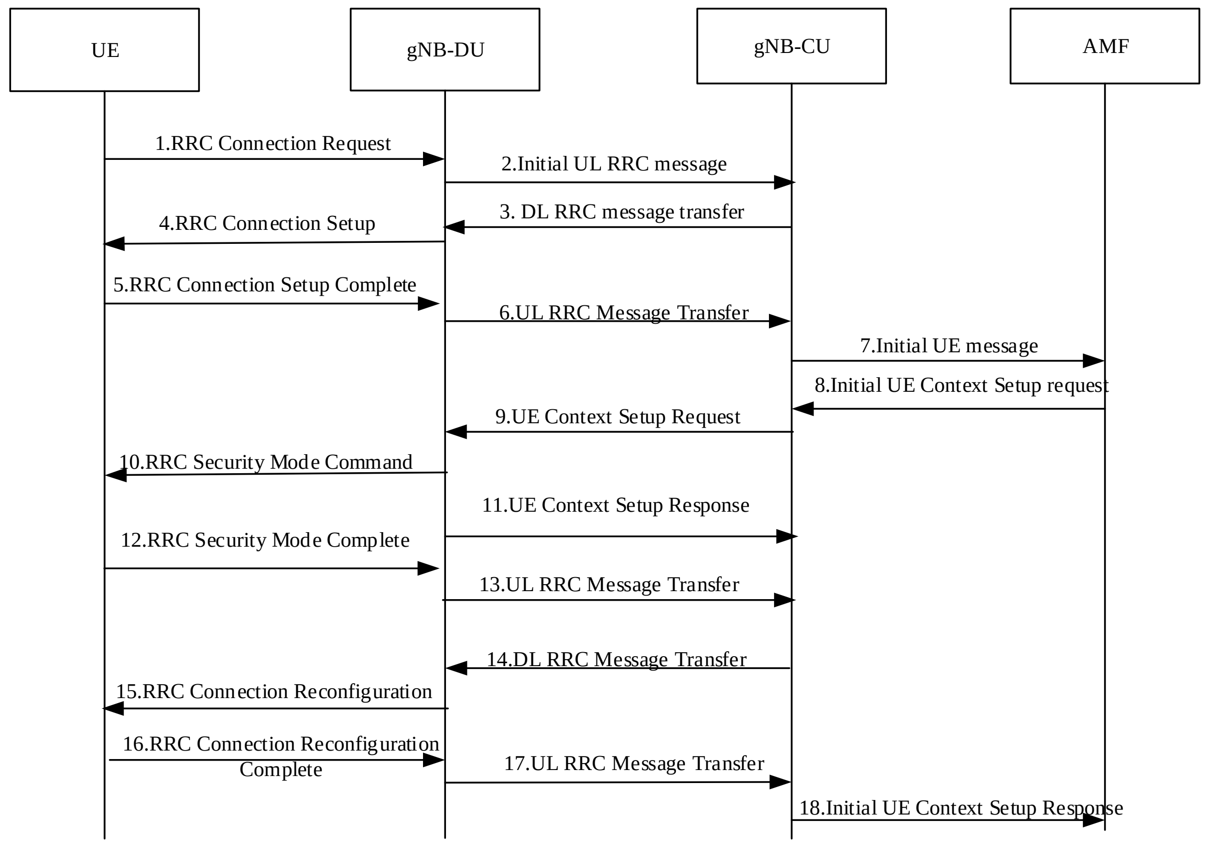

Figure 8.1-1: UE Initial Access procedure

1. The UE sends RRC Connection Request message to the gNB-DU.
2. The gNB-DU includes the RRC message and, if the UE is admitted, the corresponding low layer configuration for the UE in the F1AP INITIAL UL RRC MESSAGE TRANSFER message and transfers to the gNB-CU. The INITIAL UL RRC MESSAGE TRANSFER message includes the C-RNTI allocated by the gNB-DU.
3. The gNB-CU allocates a gNB-CU UE F1AP ID for the UE and generates RRC CONNECTION SETUP message towards UE. The RRC message is encapsulated in -the F1AP DL RRC MESSAGE TRANSFER message.
4. The gNB-DU sends the RRC CONNECTION SETUP message to the UE.
5. The UE sends the RRC CONNECTION SETUP COMPLETE message to the gNB-DU.
6. The gNB-DU encapsulates the RRC message in the F1AP UL RRC MESSAGE TRANSFER message and sends it to the gNB-CU.
7. The gNB-CU sends the INITIAL UE MESSAGE message to the AMF.
8. The AMF sends the INITIAL UE CONTEXT SETUP REQUEST message to the gNB-CU.
9. The gNB-CU sends the UE CONTEXT SETUP REQUEST message to establish the UE context in the gNB-DU. In this message, it may also encapsulate the RRC SECURITY MODE COMMAND message.
10. The gNB-DU sends the RRC SECURITY MODE COMMAND message to the UE.
11. The gNB-DU sends the UE CONTEXT SETUP RESPONSE message to the gNB-CU.
12. The UE responds with the RRC SECURITY MODE COMPLETE message
13. The gNB-DU encapsulates the RRC message in the F1AP UL RRC MESSAGE TRANSFER message and sends it to the gNB-CU.

14. gNB集中单元生成RRC连接重配置消息并将其封装在F1AP下行RRC消息传输消息中

15. gNB分布式单元向用户设备发送RRC连接重配置消息。

16. The UE向gNB分布式单元发送RRC连接重配置完成消息。

17. gNB-DU将RRC消息封装在F1AP上行RRC消息传输消息中，并将其发送给gNB集中单元。

18. gNB集中单元向接入和移动性管理功能发送初始UE上下文建立响应消息。

8.2 gNB-CU内移动性

8.2.1 NR内移动性

8.2.1.1 gNB-DU间移动性

此流程用于用户设备在NR操作期间从同一个gNB-CU内的一个gNB分布式单元移动到另一个gNB分布式单元的情况。图8.2.1.1-1展示了NR内gNB-DU间移动性过程。

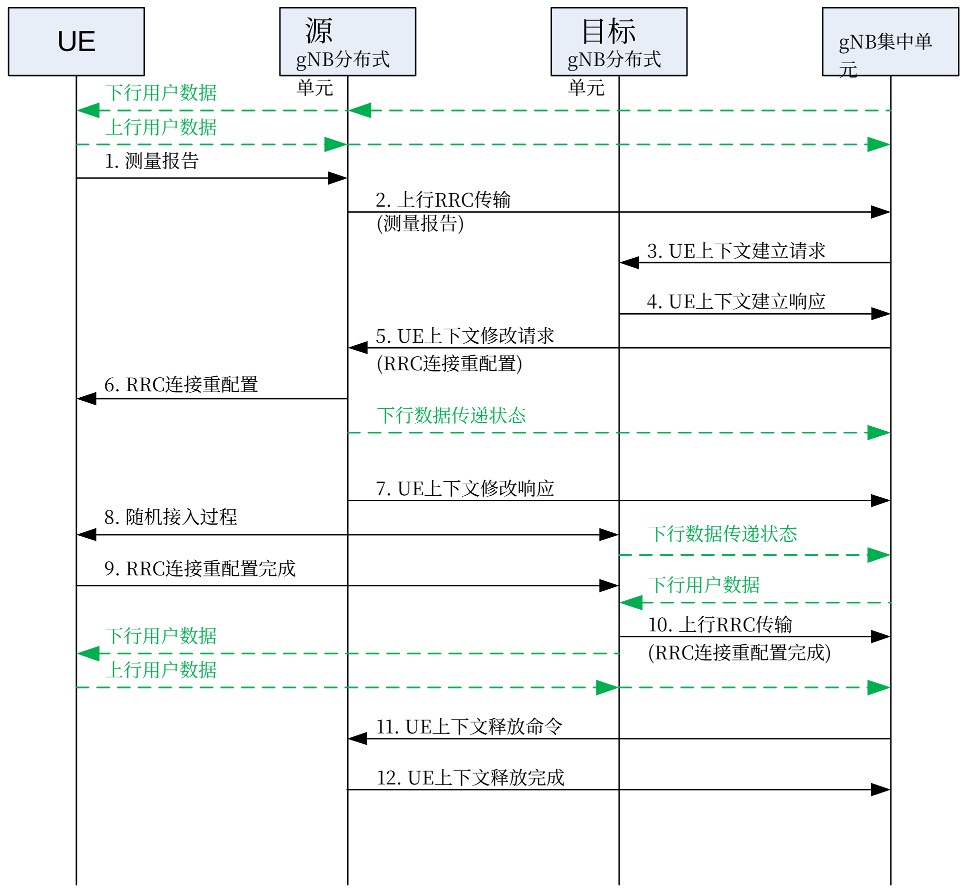

图8.2.1.1-1: intra-NR下gNB分布式单元间移动性

1. UE向源gNB-DU发送测量报告消息。

2. 源gNB-DU向gNB-CU发送上行RRC传输消息，以传递收到的测量报告。 3. gNB

-铜发送一个UE上下文建立请求消息给目标gNB-DU以创建一个UE上下文
并设置更多承载。

14. The gNB-CU generates the RRC CONNECTION RECONFIGURATION message and encapsulates it in the F1AP DL RRC MESSAGE TRANSFER message

15. The gNB-DU sends RRC CONNECTION RECONFIGURATION message to the UE.

16. The UE sends RRC CONNECTION RECONFIGURATION COMPLETE message to the gNB-DU.

17. The gNB-DU encapsulates the RRC message in the F1AP UL RRC MESSAGE TRANSFER message and send it to the gNB-CU.

18. The gNB-CU sends the INITIAL UE CONTEXT SETUP RESPONSE message to the AMF.

8.2 Intra-gNB-CU Mobility

8.2.1 Intra-NR Mobility

8.2.1.1 Inter-gNB-DU Mobility

This procedure is used for the case when the UE moves from one gNB-DU to another gNB-DU within the same gNB-CU during NR operation. Figure 8.2.1.1-1 shows the inter-gNB-DU mobility procedure for intra-NR.

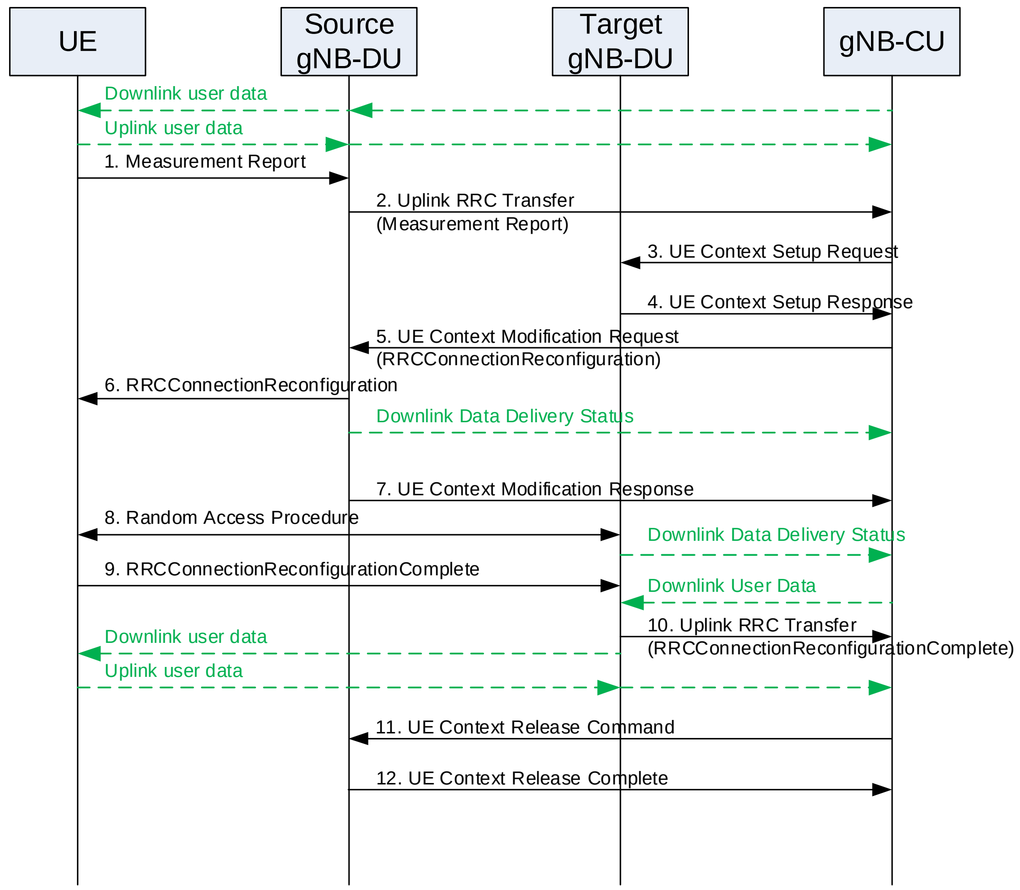

Figure 8.2.1.1-1: Inter-gNB-DU Mobility for intra-NR

1. The UE sends a *Measurement Report* message to the source gNB-DU.

2. The source gNB-DU sends an Uplink RRC Transfer message to the gNB-CU to convey the received *Measurement Report*.

3. The gNB-CU sends an UE Context Setup Request message to the target gNB-DU to create an UE context and setup one or more bearers.

4. 目标gNB-DU向gNB集中单元发送UE上下文建立响应消息。

5. gNB集中单元向源gNB-DU发送UE上下文修改请求消息，其中包含生成的RRC连接重配置消息，并指示停止该用户设备的数据传输。源gNB-DU还发送下行数据传送状态帧，以告知gNB集中单元未成功传输给用户设备的下行数据。

6. 源gNB-DU向用户设备转发接收到的RRC连接重配置消息 .

7. 源gNB-DU向gNB集中单元回复UE上下文修改响应消息。

8. 在目标处执行随机接入过程。目标gNB-DU发送下行数据传送状态帧以告知gNB集中单元。下行数据包（可能包括在源gNB-DU中未成功传输的PDCP协议数据单元）从gNB集中单元发送到目标gNB-DU。

注：是否在下行数据传递状态接收之前或之后开始向gNB-DU发送下行用户数据，取决于gNB-CU的实现。

gNB分布式单元。

9. UE通过RRC连接重配置完成消息响应目标gNB-DU。 e.

10. 目标gNB-DU向gNB-CU发送一条上行RRC传输消息，以传达接收到的RRC连接重配置完成消息。下行数据包被发送到UE。同时，上行数据包从UE发出，通过目标gNB-DU转发到gNB-CU。

11. gNB-CU向源gNB-DU发送一条UE上下文释放命令消息。

12. 源gNB-DU释放UE上下文，并通过一条UE上下文释放完成消息响应gNB-CU。

8.2.1.2 gNB-DU内切换

此流程用于UE在同一gNB-DU内从一个小区移动到另一个小区，或NR操作期间执行小区内切换的情况，并受TS 38.473 [4]中规定的UE上下文修改（gNB-CU发起）流程支持。当执行小区内切换时，gNB-CU向gNB-DU提供新的上行GTP TEID，gNB-DU向gNB-CU提供新的下行GTP TEID。gNB-DU应继续使用先前的上行GTP TEID向gNB-CU发送上行PDCP PDU，直至其重建RLC，并在RLC重建后使用新的上行GTP TEID。gNB-CU应继续使用先前的下行GTP TEID向gNB-DU发送下行PDCP PDU，直至其执行PDCP重建或PDCP数据恢复，并从PDCP重建或数据恢复开始使用新的下行GTP TEID。

8.2.2 EN-DC移动性

8.2.2.1 使用MCG SRB的gNB-DU间移动性

本过程用于在EN-DC操作中只有MCG SRB可用时，UE从一个gNB-DU移动到同一gNB-CU内的另一个gNB-DU的情况。图8.2.2.1-1显示了在EN-DC中使用MCG SRB的gNB-DU间移动性过程。

4. The target gNB-DU responds to the gNB-CU with an UE Context Setup Response message.

5. The gNB-CU sends a UE Context Modification Request message to the source gNB-DU, which includes a generated *RRCConnectionReconfiguration* message and indicates to stop the data transmission for the UE. The source gNB-DU also sends a Downlink Data Delivery Status frame to inform the gNB-CU about the unsuccessfully transmitted downlink data to the UE.

6. The source gNB-DU forwards the received *RRCConnectionReconfiguration* message to the UE.

7. The source gNB-DU responds to the gNB-CU with the UE Context Modification Response message.

8. A Random Access procedure is performed at the target The target gNB-DU sends a Downlink Data Delivery Status frame to inform the gNB-CU. Downlink packets, which may include PDCP PDUs not successfully transmitted in the source gNB-DU, are sent from the gNB-CU to the target gNB-DU.

NOTE: It is up to gNB-CU implementation whether to start sending DL User Data to gNB-DU before or after reception of the Downlink Data Delivery Status.

gNB-DU.

9. The UE responds to the target gNB-DU with an *RRCConnectionReconfigurationComplete* message.

10. The target gNB-DU sends an Uplink RRC Transfer message to the gNB-CU to convey the received *RRCConnectionReconfigurationComplete* message. Downlink packets are sent to the UE. Also, uplink packets are sent from the UE, which are forwarded to the gNB-CU through the target gNB-DU.

11. The gNB-CU sends an UE Context Release Command message to the source gNB-DU.

12. The source gNB-DU releases the UE context and responds the gNB-CU with an UE Context Release Complete message.

8.2.1.2 Intra-gNB-DU handover

This procedure is used for the case that UE moves from one cell to another cell within the same gNB-DU or for the case that intra-cell handover is performed during NR operation, and supported by UE Context Modification (gNB-CU initiated) procedure as specified in TS 38.473 [4]. When the intra-cell handover is performed, the gNB-CU provides new UL GTP TEID to the gNB-DU and gNB-DU provides new DL GTP TEID to the gNB-CU.The gNB-DU shall continue sending UL PDCP PDUs to the gNB-CU with the previous UL GTP TEID until it re-establishes the RLC, and use the new UL GTP TEID after RLC re-establishment. The gNB-CU shall continue sending DL PDCP PDUs to the gNB-DU with the previous DL GTP TEID until it performs PDCP re-establishment or PDCP data recovery, and use the new DL GTP TEID starting with the PDCP re-establishment or data recovery.

8.2.2 EN-DC Mobility

8.2.2.1 Inter-gNB-DU Mobility using MCG SRB

This procedure is used for the case the UE moves from one gNB-DU to another gNB-DU within the same gNB-CU when only MCG SRB is available during EN-DC operation. Figure 8.2.2.1-1 shows the inter-gNB-DU mobility procedure using MCG SRB in EN-DC.

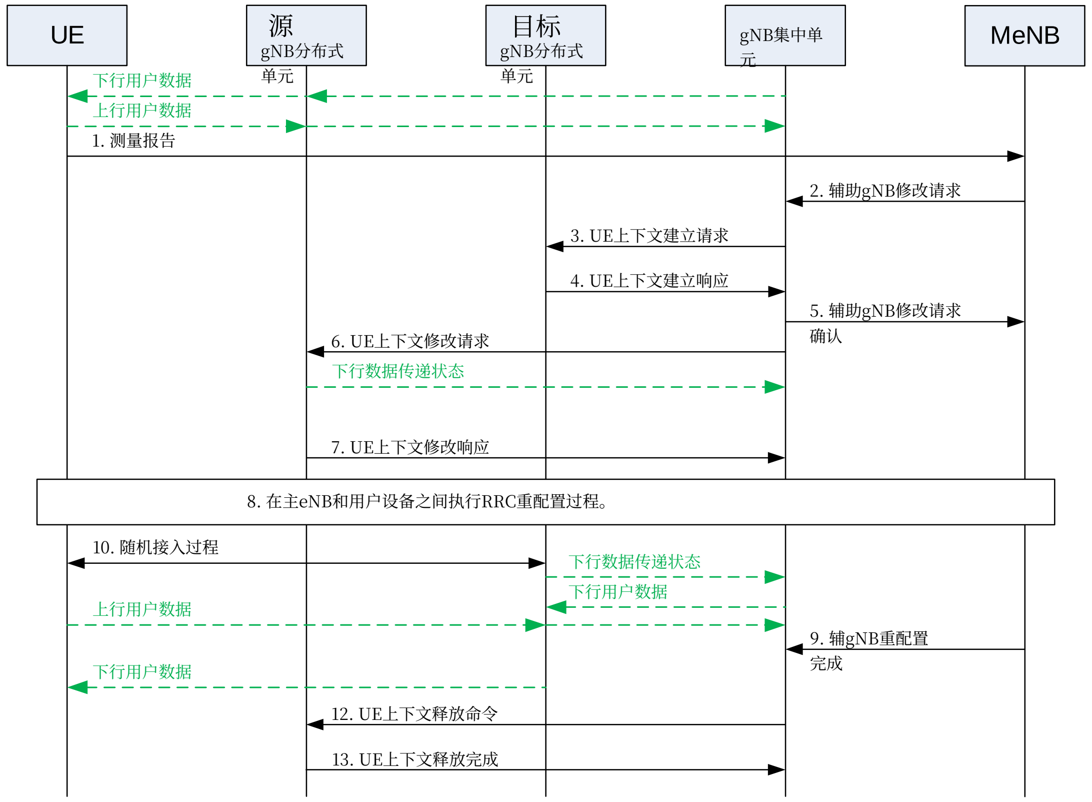

图8.2.2.1-1: EN-DC中使用MCG SRB的gNB-DU间移动性

1. UE向MeNB发送测量报告消息。2. MeNB发送SgNB修改请求消息。3. gNB-CU向目标gNB-DU发送UE上下文建立请求消息，以创建UE上下文并建立一个或多个承载。4. 目标gNB-DU向gNB-CU回应UE上下文建立响应消息。5. gNB-CU向MeNB回应SgNB修改请求确认消息。6. gNB-CU向源gNB-DU发送UE上下文修改请求消息，指示停止向UE的数据传输。源gNB-DU还发送下行数据传送状态帧，以通知gNB-CU未成功传输给UE的下行数据。7. 源gNB-DU向gNB-CU回应UE上下文修改响应消息。8. MeNB和UE执行RRC重配置过程。9. MeNB向gNB-CU发送SgNB重配置完成消息。10. 在目标gNB-DU处执行随机接入过程。目标gNB-DU发送下行数据传送状态帧以通知gNB-CU。下行数据包（可能包括在源gNB-DU未成功传输的PDCP PDU）从gNB-CU发送到目标gNB-DU。下行数据包被发送到UE。同时，上行数据包从UE发送，并通过目标gNB-DU转发到gNB-CU。

注意：是否在接收下行数据传递状态之前或之后开始向gNB-DU发送下行用户数据，取决于gNB中央单元实现。

11. gNB-CU向源gNB-DU发送UE上下文释放命令消息。

3GPP TS 38.401 version 15.2.0 Release 15 18 ETSI TS 138 401 V15.2.0 (2018-07)

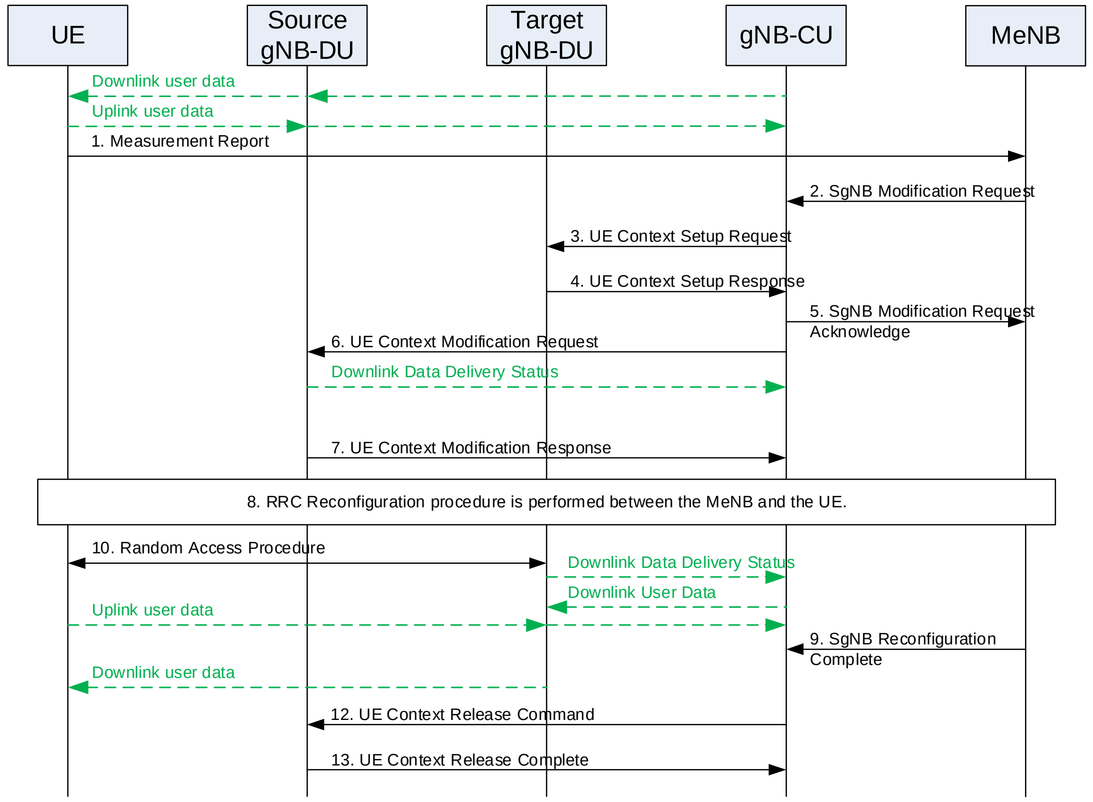

Figure 8.2.2.1-1: Inter-gNB-DU Mobility using MCG SRB in EN-DC

1. The UE sends a Measurement Report message to the MeNB.
2. The MeNB sends an SgNB Modification Request message.
3. The gNB-CU sends an UE Context Setup Request message to the target gNB-DU to create an UE context and setup one or more bearers.
4. The target gNB-DU responds the gNB-CU with an UE Context Setup Response message.
5. The gNB-CU responds the MeNB with an SgNB Modification Request Acknowledge message.
6. The gNB-CU sends a UE Context Modification Request message to the source gNB-DU indicating to stop the data transmission to the UE. The source gNB-DU also sends a Downlink Data Delivery Status frame to inform the gNB-CU about the unsuccessfully transmitted downlink data to the UE.
7. The source gNB-DU responds the gNB-CU with an UE Context Modification Response message.
8. The MeNB and the UE perform RRC Reconfiguration procedure.
9. The MeNB sends an SgNB Reconfiguration Complete message to the gNB-CU..
10. Random Access procedure is performed at the target gNB-DU. The target gNB-DU sends a Downlink Data Delivery Status frame to inform the gNB-CU. Downlink packets, which may include PDCP PDUs not successfully transmitted in the source gNB-DU, are sent from the gNB-CU to the target gNB-DU. Downlink packets are sent to the UE. Also, uplink packets are sent from the UE, which are forwarded to the gNB-CU through the target gNB-DU.

NOTE: It is up to gNB-CU implementation whether to start sending DL User Data to gNB-DU before or after reception of the Downlink Data Delivery Status.

11. The gNB-CU sends an UE Context Release Command message to the source gNB-DU.

12. 源gNB-DU释放UE上下文，并向gNB-CU响应UE上下文释放完成消息。

8.2.2.2 使用SCG SRB的gNB-DU间移动性

该流程用于在EN-DC操作期间，当SCG SRB可用时，UE从一个gNB-DU移动到另一个gNB-DU的情况。该流程与第8.2.1.1条定义的intra-NR下gNB分布式单元间移动性相同。

8.3 丢失PDU集中重传机制

8.3.1 gNB-CU内情况下的集中重传

该机制允许对由于朝向UE的对应无线链路中断而未被gNB-DU（gNB- DU1）传送的PDCP协议数据单元执行重传。当这种中断发生时，受中断影响的gNB- DU将此类事件通知gNB集中单元。gNB集中单元可以将数据流量传输，以及未交付的PDCP协议数据单元的重传，从受中断影响的gNB-DU切换到其他可用的、具有朝向UE的正常运行无线链路的gNB- DU（例如图8.3.1-1中的gNB-DU2）。该机制也适用于EN-DC和采用5GC的MR-DC，参见TS 37.340 [12]。

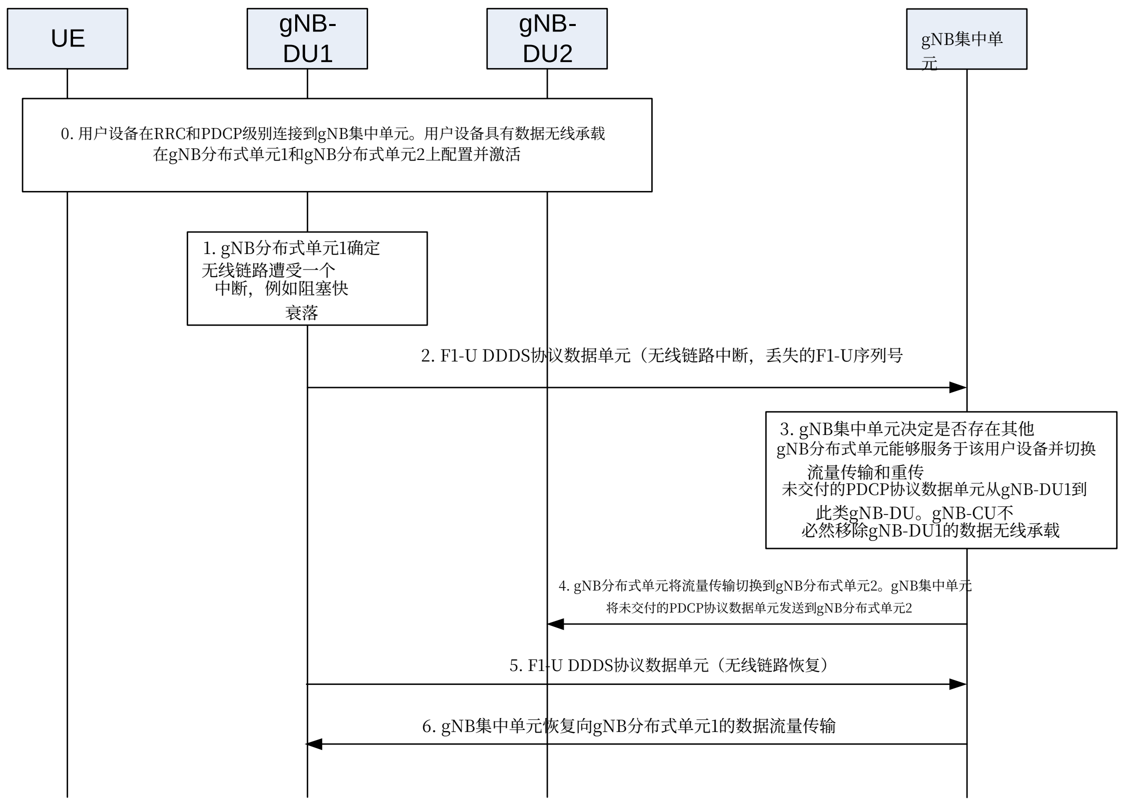

图8.3.1-1: gNB-CU内场景中的集中重传流程

该机制如图8.3.1-1所示，针对多连接场景，其中UE由至少建立在两个gNB-DU上的多个数据无线承载服务，并包括以下步骤:

0. UE已连接，并可以与gNB-DU1和gNB-DU2收发数据。

1. gNB分布式单元1 意识到与用户设备之间的无线链路正在经历中断。

2. gNB-DU1通过F1-U接口向gNB-CU发送“无线链路中断”通知消息，作为相关数据无线承载的DDDS PDU的一部分。该消息可包含对应于gNB-DU1中部分（和/或全部）未投递的PDCP PDU所报告的丢失的F1-U序列号范围数量。

12. The source gNB-DU releases the UE context and responds the gNB-CU with an UE Context Release Complete message.

8.2.2.2 Inter-gNB-DU Mobility using SCG SRB

This procedure is used for the case the UE moves from one gNB-DU to another gNB-DU when SCG SRB is available during EN-DC operation. The procedure is the same as inter-gNB-DU Mobility for intra-NR as defined in clause 8.2.1.1.

8.3 Mechanism of centralized retransmission of lost PDUs

8.3.1 Centralized Retransmission in Intra gNB-CU Cases

This mechanism allows to perform the retransmission of the PDCP PDUs that are not delivered by a gNB-DU (gNB-DU1) because the corresponding radio links toward the UE are subject to outage. When such outage occurs, the gNB-DU affected by outage informs the gNB-CU of such event. The gNB-CU can switch transmission of data traffic, as well as perform retransmission of undelivered PDCP PDUs, from the gNB-DU affected by outage to other available gNB-DUs (e.g. gNB-DU2 in Figure 8.3.1-1) with a well-functioning radio link toward the UE. The mechanism is also applicable in EN-DC and MR-DC with 5GC, refer to TS 37.340 [12].

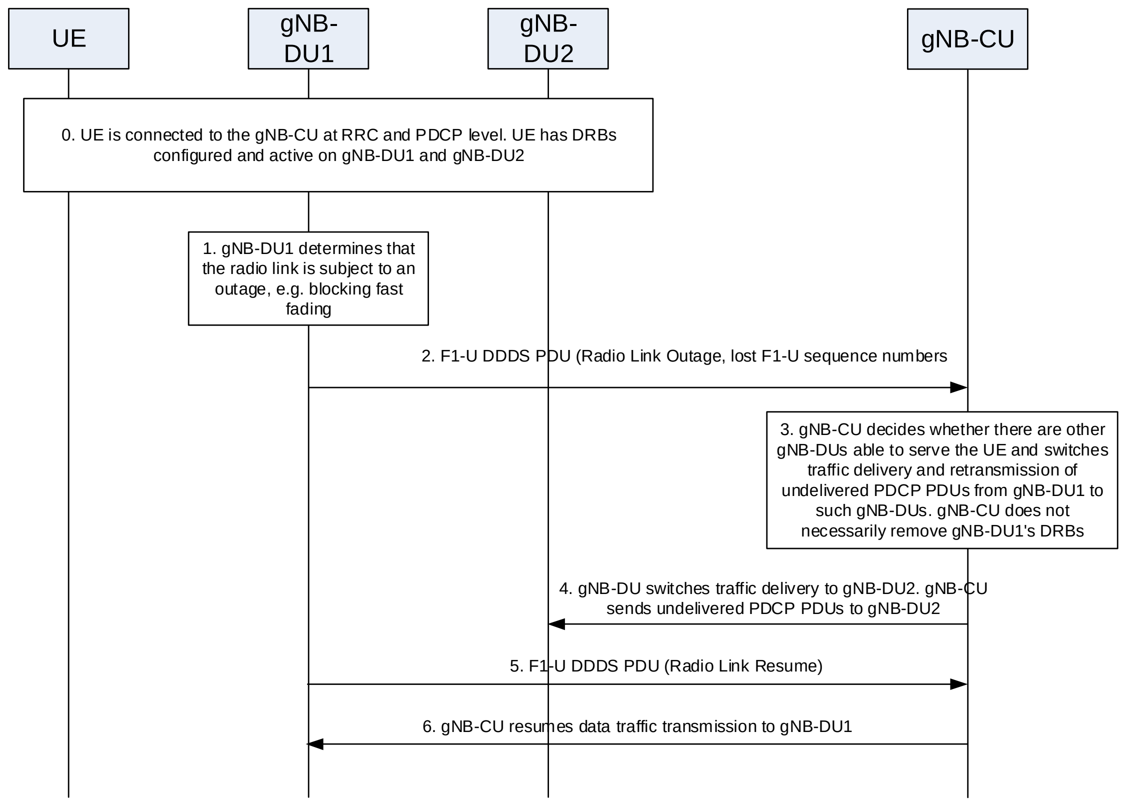

Figure 8.3.1-1: Procedure for centralized retransmission in intra gNB-CU scenarios

The mechanism is shown in Figure 8.3.1-1 and targets the multi-connectivity scenario, where a UE is served by multiple data radio bearers established at least on two gNB-DUs, and it includes the following steps:

0. The UE is connected and can transmit/receive data to/from gNB-DU1 and gNB-DU2.

1. gNB-DU1 realizes that the radio link towards the UE is experiencing outage.

2. gNB-DU1 sends the "Radio Link Outage" notification message to the gNB-CU over the F1-U interface, as part of the DDDS PDU of the concerned data radio bearer. The message may include the *Number of lost F1-U Sequence Number ranges reported* corresponding to the PDCP PDUs partially (and/or totally) undelivered in gNB-DU1.

3. gNB-CU决定将通过gNB-DU2切换数据流量传输以及重传在gNB-DU1中未投递的PDCP PDU。gNB-CU停止向gNB-DU1发送下行流量。gNB- DU1与UE之间的无线链路分支不一定被移除。

4. gNB集中单元开始向gNB分布式单元2发送流量（即新的PDU和可能重传的PDU）。

5. 如果gNB分布式单元1意识到相关数据无线承载的无线链路恢复正常运行，它可作为DDDS PDU的一部分通过F1-U接口发送“无线链路恢复”通知消息，告知gNB集中单元该无线链路可再次用于相关数据无线承载。

6. gNB集中单元可再次通过gNB分布式单元1开始发送相关数据无线承载的流量（即新的PDU和可能重传的PDU）。

8.4 多连接操作

8.4.1 辅节点添加

8.4.1.1 EN-DC

本节给出了在en-gNB由gNB-CU和gNB-DU（多个）组成的情况下，EN-DC中的辅gNB添加流程，如图8.4.1.1-1所示。

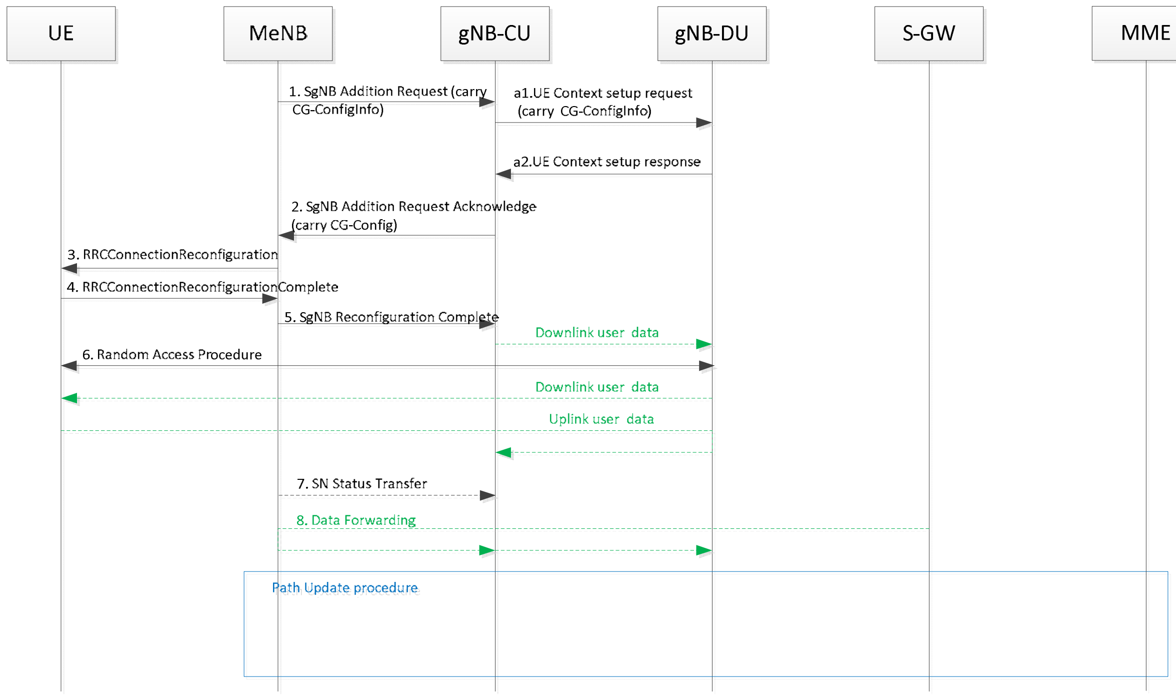

图8.4.1.1-1: EN-DC中的辅gNB添加流程

1~8: 参见TS 37.340 [12]

a1. 在接收到来自主eNB的辅gNB添加请求消息后，gNB-CU向gNB-DU发送UE上下文建立请求消息以创建UE上下文。如37.340 [12], 所述，在辅节点变更过程中，UE上下文建立请求消息可包含源小区组配置，以支持gNB-DU处的增量配置。

a2. gNB-DU以UE上下文建立响应消息向gNB-CU进行响应。如37.340 [12], 中所述，在辅节点变更过程中，如果gNB-DU在接收到源小区组配置后决定执行完整配置，则其应在UE上下文建立响应消息中指示其已应用了完整配置。

3. gNB-CU decides to switch data traffic transmission and retransmission of PDCP PDUs that were undelivered in gNB-DU1 via gNB-DU2. gNB-CU stops sending downlink traffic to gNB-DU1. The radio leg between gNB-DU1 and the UE is not necessarily removed.

4. gNB-CU starts sending traffic (namely new PDUs and potentially retransmitted PDUs) to gNB-DU2.

5. If gNB-DU1 realizes that the radio link of the concerned data radio bearer is back in normal operation, it may send a "Radio Link Resume" notification message as part of the DDDS PDU over the F1-U interface to inform the gNB-CU that the radio link can be used again for the concerned data radio bearer.

6. gNB-CU may start sending traffic (namely new PDUs and potentially retransmitted PDUs) of the concerned data radio bearer via gNB-DU1 again.

8.4 Multi-Connectivity operation

8.4.1 Secondary Node Addition

8.4.1.1 EN-DC

This clause gives the SgNB addition procedure in EN-DC given that en-gNB consists of gNB-CU and gNB-DU(s), as shown in Figure 8.4.1.1-1.

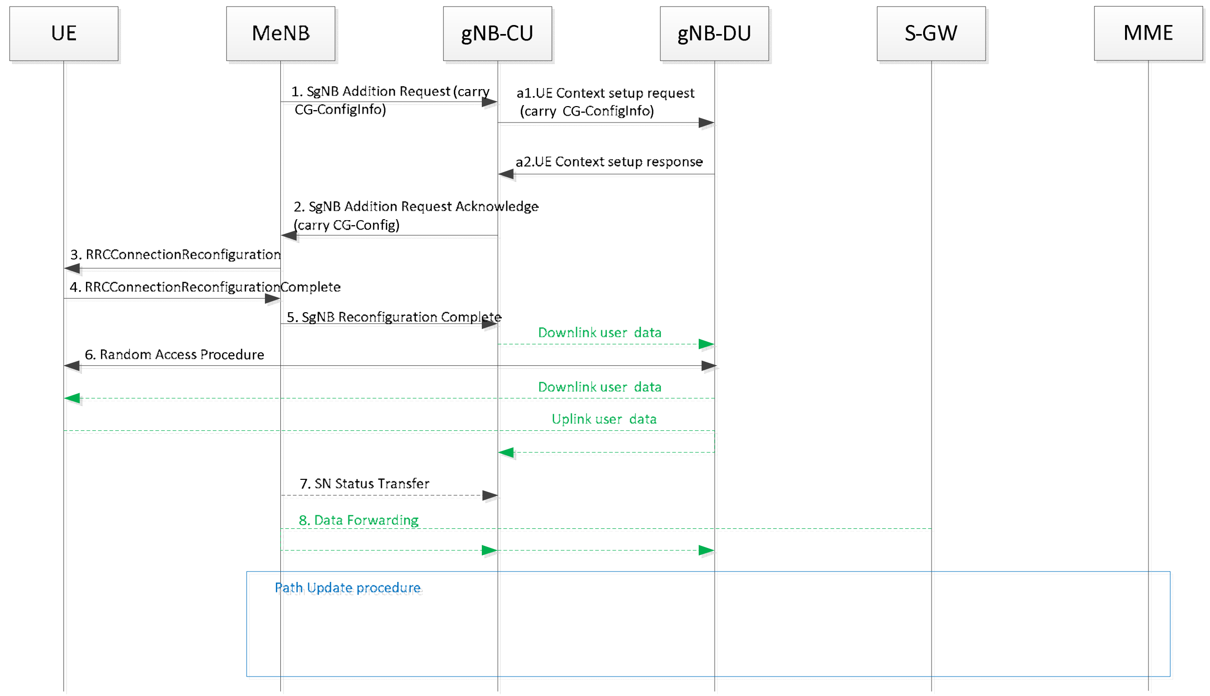

Figure 8.4.1.1-1: SgNB addition procedure in EN-DC

1~8: refer to TS 37.340 [12]

a1. After receiving the SgNB Addition Request message from MeNB, the gNB-CU sends the UE Context Setup Request message to the gNB-DU to create a UE context. As specified in 37.340 [12], in the course of a Secondary Node Change, the UE Context Setup Request message may contain source cell group configuration to support delta configuration at the gNB-DU.

a2. The gNB-DU responds to the gNB-CU with the UE Context Setup Response message. As specified in 37.340 [12], in the course of a Secondary Node Change, if the gNB-DU decides to perform full configuration after receiving the source cell group configuration, it shall indicate that it has applied full configuration in the UE Context Setup Response message.

8.4.2 辅节点释放（主节点/辅节点发起）

8.4.2.1 EN-DC

本条款给出 th　　e SgNB释放过程在EN-DC中，假设en-gNB由gNB集中单元和gNB分布式单元(s组成　　).

主节点发起的辅节点释放

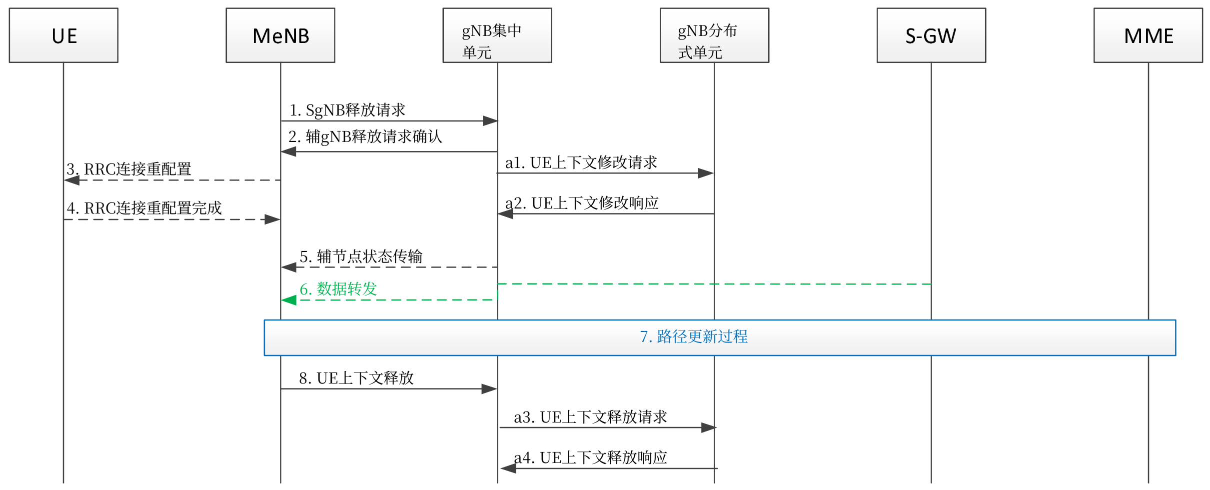

图8.4.2.1-1 EN-DC中的SgNB释放流程（主节点发起）

1~8：参见TS 37.340 [12]

注意：发送步骤2的SgNB释放请求确认消息的时机仅为示例，它可在步骤a1之后或a2之后发送，具体取决于实现。

a1. 在收到来自MeNB的SgNB释放请求消息后，gNB-CU向gNB-DU发送UE上下文修改请求消息以停止该UE的数据传输。gNB-DU何时停止UE调度取决于实现。

a2. gNB-DU 向 gNB-CU 响应 UE 上下文修改响应消息。

a3. 在从 MeNB 接收到 UE 上下文释放消息后，gNB-CU 向 gNB-DU 发送 UE 上下文释放请求消息以释放 UE 上下文。

a4. gNB分布式单元向gNB集中单元回复UE上下文释放响应消息。

辅节点发起的辅节点释放

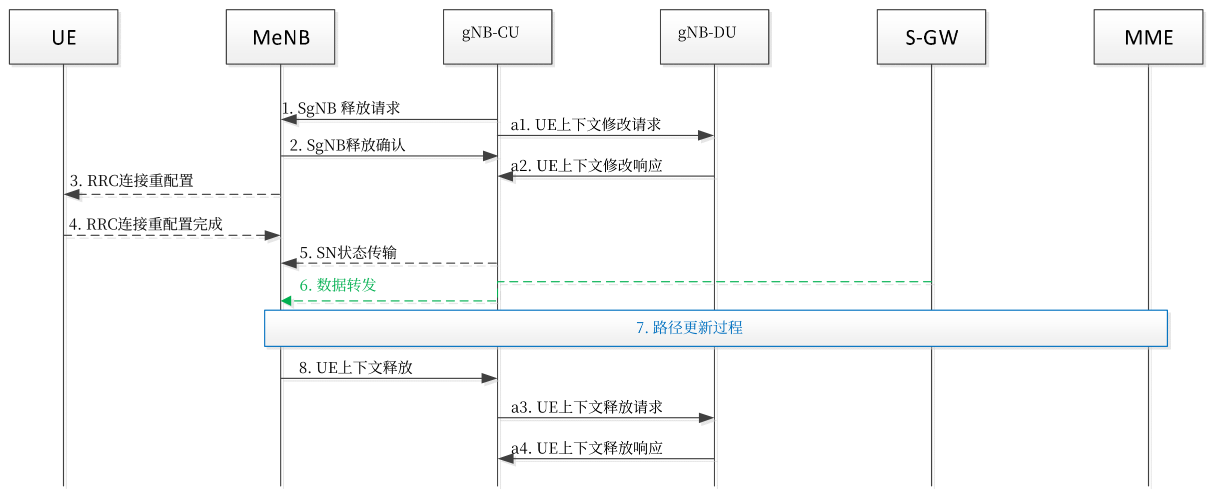

图8.4.2.1-2 EN-DC中的SgNB释放过程（SN发起）

8.4.2 Secondary Node Release (MN/SN initiated)

8.4.2.1 EN-DC

This clause gives the SgNB release procedure in EN-DC given that the en-gNB consists of a gNB-CU and gNB-DU(s).

MN initiated SN Release

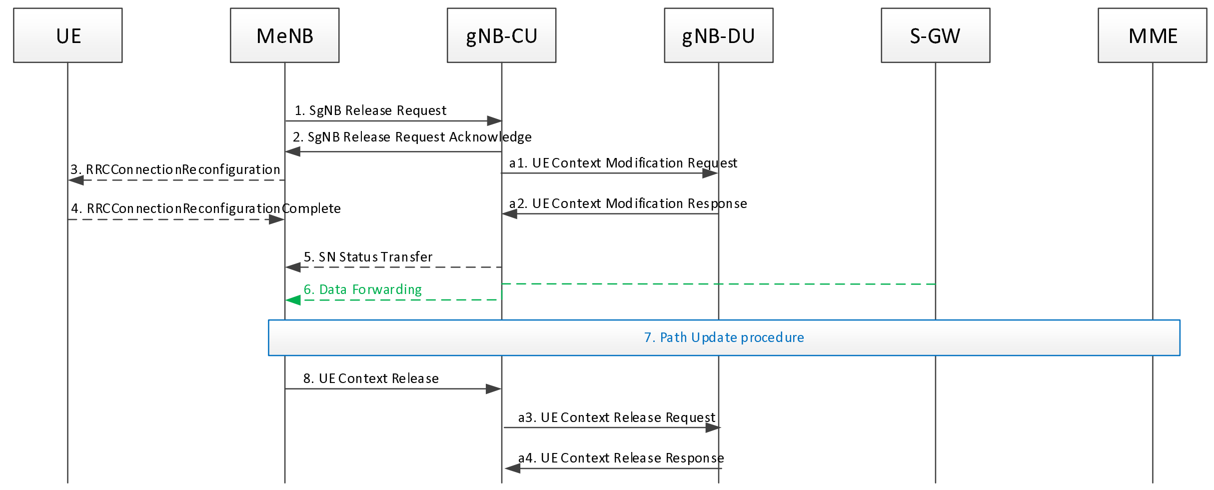

Figure 8.4.2.1-1 SgNB release procedure in EN-DC (MN initiated)

1~8: refer to TS 37.340 [12]

NOTE: The timing of sending the Step 2 SgNB Release Request Acknowledge message is an example, it may be sent e.g. after step a1 or after a2 and it is up to implementation.

a1. After receiving SgNB Release Request message from MeNB, gNB-CU sends the UE Context Modification Request message to gNB-DU to stop the data transmission for the UE. It is up to gNB-DU implementation when to stop the UE scheduling.

a2. gNB-DU responds to gNB-CU with UE Context Modification Response message.

a3. After receiving the UE Context Release message from MeNB, the gNB-CU sends the UE Context Release Request message to the gNB-DU to release the UE context.

a4. The gNB-DU responds to the gNB-CU with the UE Context Release Response message.

SN initiated SN Release

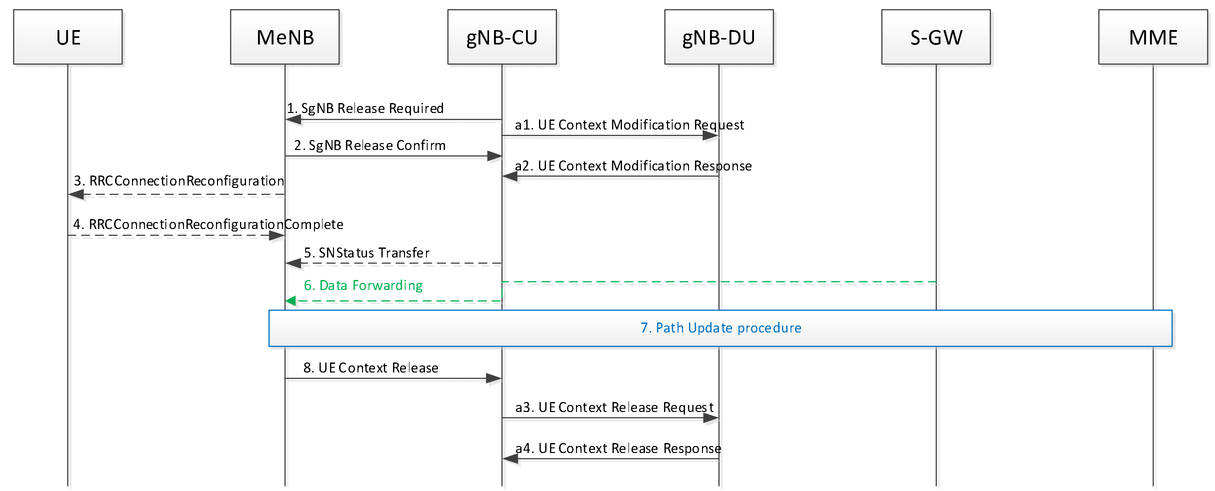

Figure 8.4.2.1-2 SgNB release procedure in EN-DC (SN initiated)

1~8: 参考技术规范37.340 [12]

a1. gNB集中单元向gNB分布式单元发送UE上下文修改请求消息，以停止该UE的数据传输。何时停止UE调度由gNB分布式单元实现决定。此步骤可在步骤1之前发生。

a2. gNB分布式单元以UE上下文修改响应消息响应gNB集中单元。

a3. 在从主eNB接收到UE上下文释放消息后，gNB集中单元向gNB分布式单元发送UE上下文释放请求消息以释放UE上下文。

a4. gNB分布式单元以UE上下文释放响应消息响应gNB集中单元。

8.5 F1启动和小区激活

该功能允许在gNB分布式单元和gNB集中单元之间建立F1接口，并允许激活gNB-DU小区。

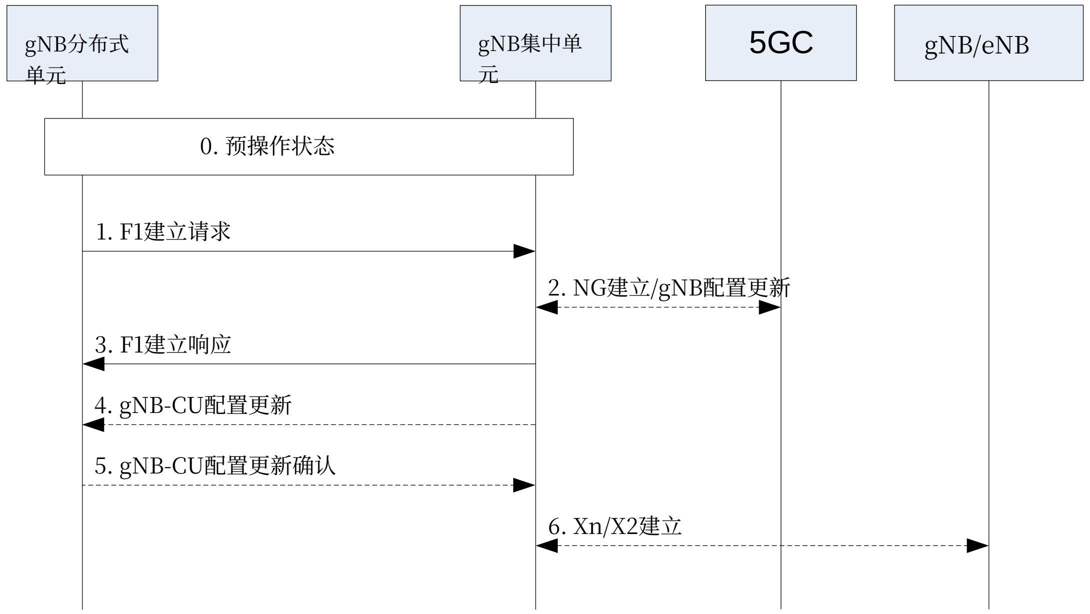

图8.5-1: F1启动与小区激活

0. gNB-DU及其小区由OAM在F1预操作状态下配置。gNB-DU与gNB-CU之间具有TNL连接。1. gNB-DU向gNB-CU发送F1建立请求消息，其中包含已配置并准备激活的小区列表。2. 在NG-RAN中，gNB-CU确保与核心网的连接。为此，gNB-CU可向5GC发起NG建立或gNB配置更新流程。3. gNB-CU向gNB-DU发送F1建立响应消息，该消息可选地包含待激活的小区列表。若gNB-DU成功激活这些小区，则小区变为可操作状态。若gNB-DU未能激活某些小区，则gNB-DU可向gNB-CU发起gNB-DU配置更新流程。gNB-DU在gNB-DU配置更新消息中包含活跃的小区（即gNB-DU应能够服务用户设备的小区）。gNB-DU也可指示应删除激活失败的小区，此时gNB-CU移除相应的小区信息。4. gNB-CU可向gNB-DU发送gNB-CU配置更新消息，该消息可选地包含待激活的小区列表，例如当这些小区未通过F1建立响应消息激活时。

5. gNB-DU回复一条gNB-DU配置更新确认消息，该消息可选地包含激活失败的小区列表。

1~8: refer to TS 37.340 [12]

a1. gNB-CU sends the UE Context Modification Request message to gNB-DU to stop the data transmission for the UE. It is up to gNB-DU implementation when to stop the UE scheduling. This step may occur before step 1.

a2. gNB-DU responds to gNB-CU with UE Context Modification Response message.

a3. After receiving the UE Context Release message from MeNB, the gNB-CU sends the UE Context Release Request message to the gNB-DU to release the UE context.

a4. The gNB-DU responds to the gNB-CU with the UE Context Release Response message.

8.5 F1 Startup and cells activation

This function allows to setup the F1 interface between a gNB-DU and a gNB-CU and it allows to activate the gNB-DU cells.

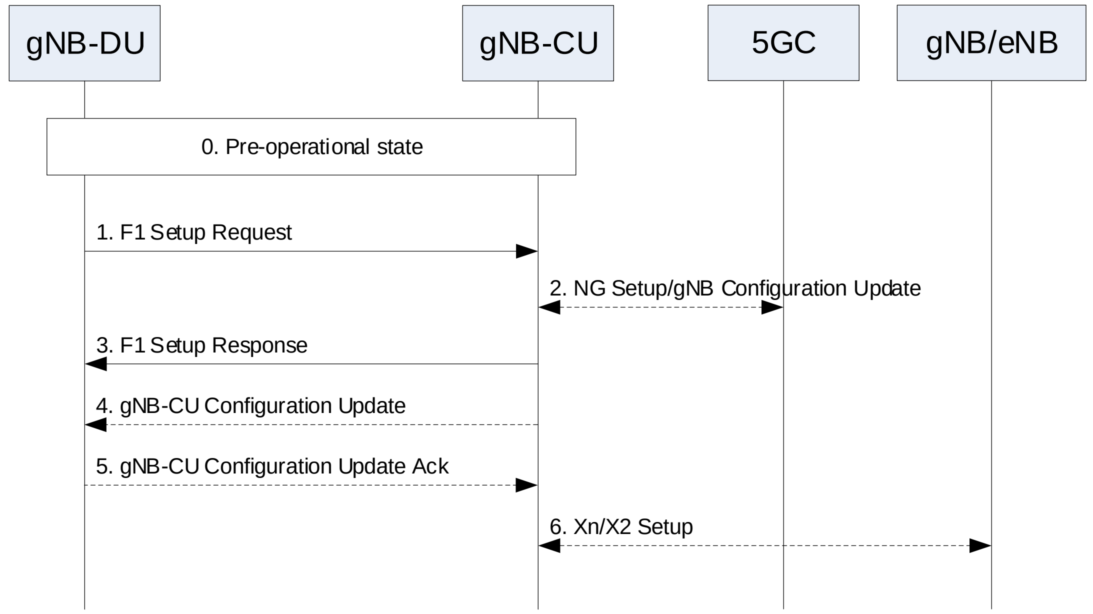

Figure 8.5-1: F1 startup and cell activation

0. The gNB-DU and its cells are configured by OAM in the F1 pre-operational state. The gNB-DU has TNL connectivity toward the gNB-CU.

1. The gNB-DU sends an F1 Setup Request message to the gNB-CU including a list of cells that are configured and ready to be activated.

2. In NG-RAN, the gNB-CU ensures the connectivity toward the core network. For this reason, the gNB-CU may initiate either the NG Setup or the gNB Configuration Update procedure towards 5GC.

3. The gNB-CU sends an F1 Setup Response message to the gNB-DU that optionally includes a list of cells to be activated. If the gNB-DU succeeds to activate the cell(s), then the cells become operational. If the gNB-DU fails to activate some cell(s), the gNB-DU may initiate the gNB-DU Configuration Update procedure towards the gNB-CU. The gNB-DU includes in the gNB-DU Configuration Update message the cell(s) that are active (i.e., the cell(s) for which the gNB-DU should be able to serve UEs). The gNB-DU may also indicate that the cell(s) that failed to activate should be deleted, in which case the gNB-CU removes the corresponding cell(s) information.

4. The gNB-CU may send a gNB-CU Configuration Update message to the gNB-DU that optionally includes a list of cells to activated, e.g., in case that these cells were not activated using the F1 Setup Response message.

5. The gNB-DU replies with a gNB-DU Configuration Update Acknowledge message that optionally includes a list of cells that failed to be activated.

6. gNB-CU可发起针对相邻NG-RAN节点的Xn建立流程，或针对相邻eNB的EN-DC X2建立流程。

注意: In c如果F1建立响应不用于激活任何小区，步骤2可在步骤3之后执行。

Over 在一个gNB-CU和gNB-DU对之间的F1接口上，以下两种小区状态是可能的:

- 非激活：该小区被gNB-DU和gNB-CU双方知晓。该小区不应服务UE； - 激活：该小区被gNB-DU和gNB-CU双方知晓。该小区宜能够服务UE。

gNB集中单元决定小区状态应为非激活还是激活。gNB集中单元能通过F1建立响应、gNB分布单元配置更新确认或gNB-CU配置更新消息请求gNB分布式单元更改小区状态。gNB分布式单元能通过gNB分布单元配置更新或gNB集中单元配置更新确认消息确认（或拒绝）更改小区状态的请求。

8.6 RRC状态转换

8.6.1 RRC连接态到RRC非激活态

本节描述了在gNB由gNB集中单元和gNB-DU(s)组成的情况下，RRC连接态到RRC非激活状态转换，如图8.6.1-1所示。

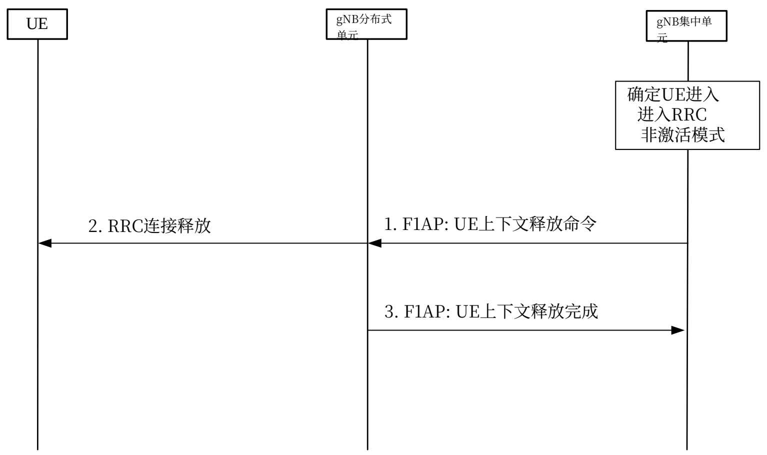

图8.6.1-1：RRC连接态到RRC非激活态状态转换流程

0. 首先，gNB集中单元决定用户设备从连接模式进入RRC非激活模式。

1. gNB集中单元生成向用户设备发送的RRC连接释放消息。该RRC消息被封装在发送给gNB分布式单元的F1AP UE上下文释放命令消息中。2. gNB分布式单元将RRC连接释放消息转发给用户设备。3. gNB分布式单元以F1AP UE上下文释放响应消息进行响应。

8.6.2 RRC非激活态到其他状态

本节给出在gNB由gNB集中单元和gNB分布式单元（多个）组成的情况下，RRC非活跃态到其他RRC状态的转换，如图8.6.2-1所示。

6. The gNB-CU may initiate either the Xn Setup towards a neighbour NG-RAN node or the EN-DC X2 Setup procedure towards a neighbour eNB.

NOTE: In case that the F1 Setup Response is not used to activate any cell, step 2 may be performed after step 3.

Over the F1 interface between a gNB-CU and a gNB-DU pair, the following two cell states are possible:

- *Inactive*: the cell is known by both the gNB-DU and the gNB-CU. The cell shall not serve UEs;
- *Active*: the cell is known by both the gNB-DU and the gNB-CU. The cell should be able to serve UEs.

The gNB-CU decides whether the cell state should be Inactive or Active. The gNB-CU can request the gNB-DU to change the cell state using the F1 Setup Response, the gNB-DU Configuration Update Acknowledge, or the gNB-CU Configuration Update messages. The gNB-DU can confirm (or reject) a request to change the cell state using the gNB-DU Configuration Update or the gNB-CU Configuration Update Acknowledge messages.

8.6 RRC state transition

8.6.1 RRC connected to RRC inactive

This section gives the RRC connected to RRC inactive state transition given that gNB consists of gNB-CU and gNB-DU(s), as shown in Figure 8.6.1-1.

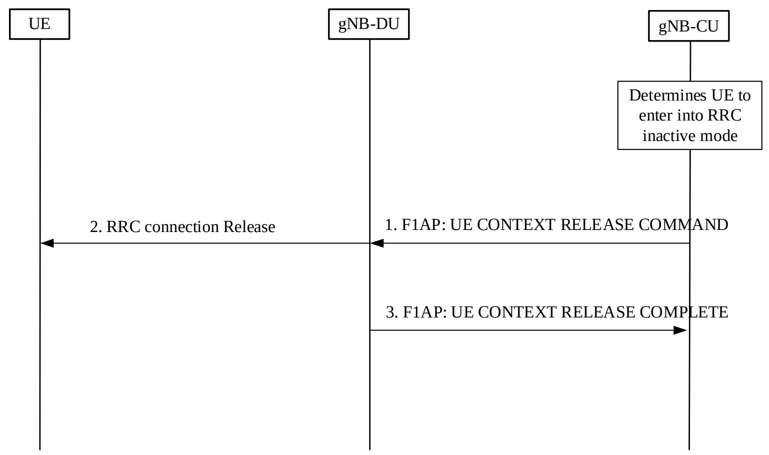

Figure 8.6.1-1: RRC connected to RRC inactive state transition procedure

0. At first, the gNB-CU determines the UE to enter into RRC inactive mode from connected mode.
1. The gNB-CU generates RRC Connection Release message towards UE. The RRC message is encapsulated in F1AP UE CONTEXT RELEASE COMMAND message to the gNB-DU.
2. The gNB-DU forwards RRC Connection Release message to UE.
3. gNB-DU responds with F1AP UE CONTEXT RELEASE RESPONSE message.

8.6.2 RRC inactive to other states

This section gives the RRC inactive to other RRC states transition given that gNB consists of gNB-CU and gNB-DU(s), as shown in Figure 8.6.2-1.

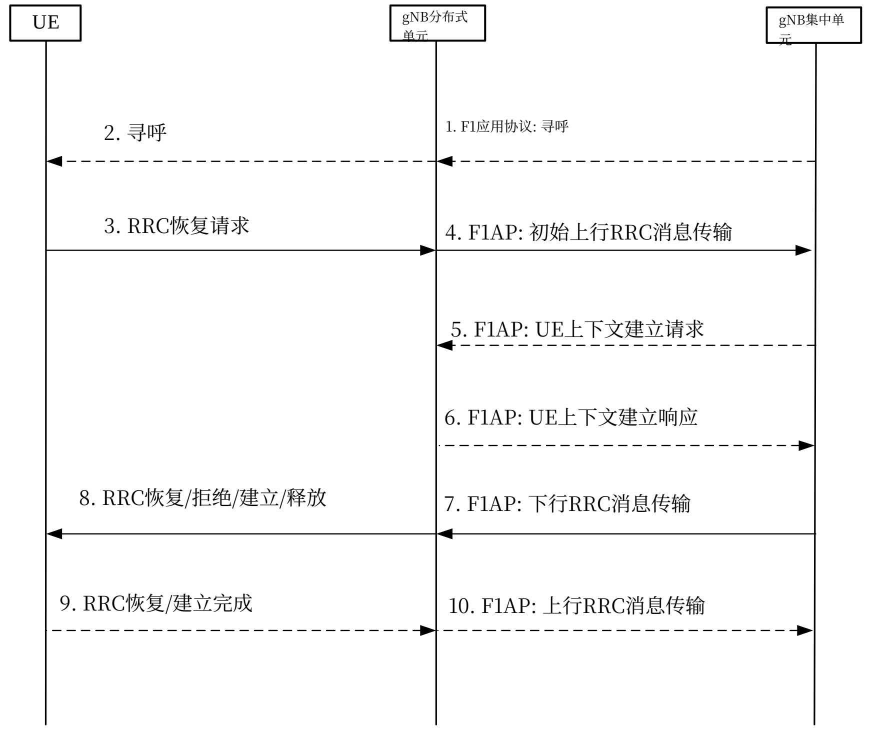

图8.6.2-1: RRC非激活到其他RRC状态转换过程

1. 如果从5G核心网收到数据，gNB集中单元向gNB分布式单元发送F1AP寻呼消息。

2. gNB分布式单元向用户设备发送RAN寻呼消息。

注意: 步骤1和2仅在下行数据到达时存在。

3. 用户设备在基于RAN的寻呼、上行数据到达或RNA更新时发送RRC恢复请求。 e.

4. gNB-DU将RRC恢复请求包含在非UE关联的F1AP初始上行RRC消息传输消息中，并传输给gNB-CU。

5. 对于UE非激活到UE激活转换（排除仅信令交换导致的转换），gNB-CU分配gNB-CU UE F1AP标识，并向gNB-DU发送F1AP UE上下文建立请求消息，该消息可包括待建立的SRB标识和DRB标识。

6. gNB-DU用F1AP UE上下文建立响应消息进行响应，该消息包含由gNB-DU提供的SRB和数据无线承载的RLC/MAC/PHY配置。

注意：步骤5和6存在于非激活到激活转换中，排除仅信令交换导致的转换。当gNB-CU成功检索并验证UE上下文时，它可决定让UE进入RRC激活模式。gNB-CU应触发gNB-CU与gNB-DU之间的UE上下文建立过程，在此期间，可建立信令无线承载1、SRB2和DRB。对于仅信令交换的转换，gNB-CU不触发UE上下文建立过程。对于非激活到空闲转换，gNB-CU不触发UE上下文建立过程。

7. gNB-CU生成面向UE的RRC恢复/建立/拒绝/释放消息。该RRC消息封装在F1AP下行RRC消息传输消息中，并带有SRB标识。

8. gNB-DU根据SRB指示的SRB0或SRB1，将RRC消息转发给UE。 ID.

注意：在第7步中，预期gNB-CU生成RRC恢复消息用于非活跃到活跃状态转换（仅信令交换和用户面数据交换两种情况），生成RRC建立消息用于回退以建立新的RRC连接，并且生成不带挂起配置的RRC释放消息用于非活跃到空闲状态转换，或者生成带有挂起配置的RRC释放消息以保持在非活跃状态。

如果 ste p 5 和 6 未执行，gNB-DU 推导出用于传递RRC消息的SRB
7在步骤来自SRB标识，即SRB标识“0”对应信令无线承载0，SRB标识“1”对应信令无线承载1。

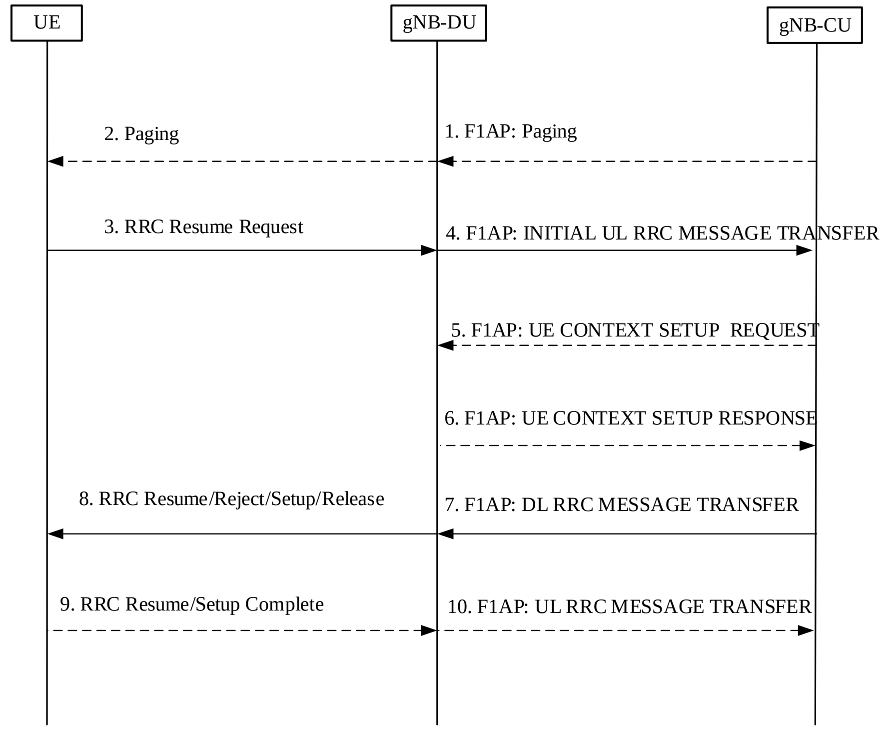

Figure 8.6.2-1: RRC inactive to other RRC states transition procedure

1. If data is received from 5GC, the gNB-CU sends F1AP Paging message to gNB-DU.
2. The gNB-DU sends RAN Paging message to UE.

NOTE: step 1 and 2 only exist in case of DL data arrival.

3. UE sends RRC resume request either upon RAN-based paging, UL data arrival or RNA update.
4. The gNB-DU includes RRC resume request in a non-UE associated F1AP INITIAL UL RRC MESSAGE TRANSFER message and transfer to the gNB-CU.
5. For UE Inactive to UE Active transitions, excluding transitions due to signalling exchange only, the gNB-CU allocates gNB-CU UE F1AP ID and sends F1AP UE CONTEXT SETUP REQUEST message to gNB-DU, which may include SRB ID(s) and DRB ID(s) to be setup.
6. The gNB-DU responds with F1AP UE CONTEXT SETUP RESPONSE message, which contains RLC/MAC/PHY configuration of SRB and DRBs provided by the gNB-DU.

NOTE: step 5 and 6 exist for inactive to active transitions, excluding transitions due to signalling exchange only. When gNB-CU successfully retrieves and verifies the UE context, it may decide to let the UE enter into RRC active mode. gNB-CU shall trigger UE context setup procedure between gNB-CU and gNB-DU, during which both SRB1, SRB2 and DRB(s) can be setup. For signalling exchange only transitions, gNB-CU does not trigger UE Context Setup procedure. For inactive to Idle transitions the gNB-CU does not trigger the UE Context Setup procedure.

7. The gNB-CU generates RRC resume/setup/reject/release message towards UE. The RRC message is encapsulated in F1AP DL RRC MESSAGE TRANSFER message together with SRB ID.
8. The gNB-DU forwards RRC message to UE either over SRB0 or SRB1 as indicated by the SRB ID.

NOTE: in step 7, it is expected that gNB-CU generates RRC resume message for inactive to active state transition(for both cases of signaling exchange only, and UP data exchange), generates RRC setup message for fallback to establish a new RRC connection, and generates either RRC release message without suspend configuration for inactive to idle state transition, or RRC release message with suspend configuration to remain in inactive state.
If step 5 and 6 are not performed, the gNB-DU deduces the SRB on which to deliver the RRC message in step 7 from the SRB ID, i.e. SRB ID “0” corresponds to SRB0, SRB ID “1” corresponds to SRB1.

9. 用户设备向gNB分布式单元发送RRC恢复/建立完成消息。

10. gNB-DU将RRC封装在F1AP上行RRC消息传输消息中，并发送给gNB-CU。

注意：步骤9和步骤10用于非活跃到活跃状态转换（仅信令交换和用户面数据交换两种情况）。UE生成RRC恢复/建立完成消息，分别用于恢复现有RRC连接或回退到新的RRC连接。

8.7 RRC连接重建

这个 pro 该过程用于UE尝试重建RRC连接的情况，如图8.7-1所示 .

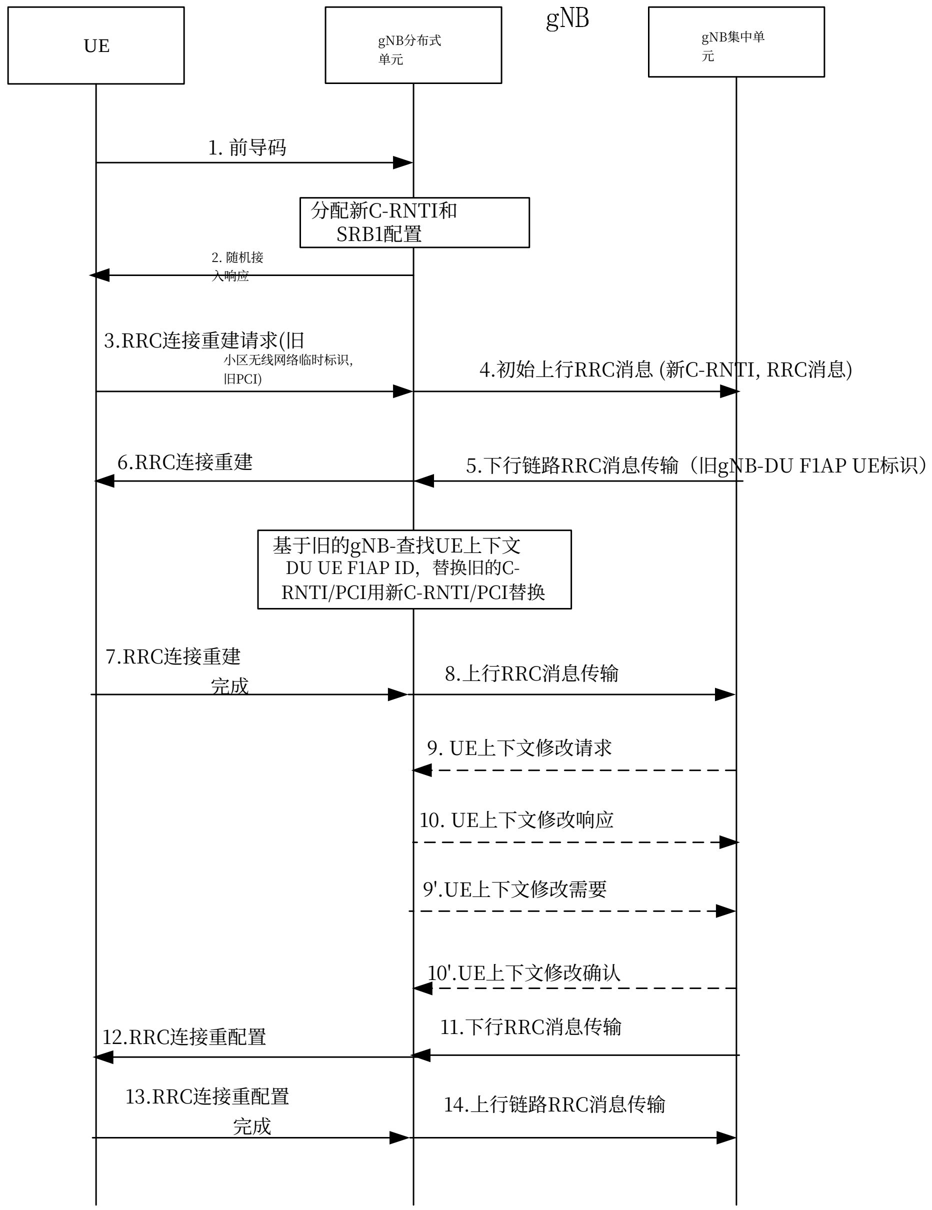

图8.7-1: RRC连接重建过程

1. UE向gNB-DU发送前导码。

2. gNB-DU分配新C-RNTI并通过RAR响应UE。

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9. UE sends RRC resume/setup complete message to the gNB-DU.

10. The gNB-DU encapsulates RRC in F1AP UL RRC MESSAGE TRANSFER message and send to the gNB-CU.

NOTE: step 9 and step 10 exist for inactive to active state transition (for both cases of signaling exchange only, and UP data exchange). UE generates RRC resume/setup complete message for resume the existing RRC connection or fallback to a new RRC connection respectively.

8.7 RRC connection reestablishment

This procedure is used for the case that UE tries to reestablish the RRC connection, as shown in Figure 8.7-1.

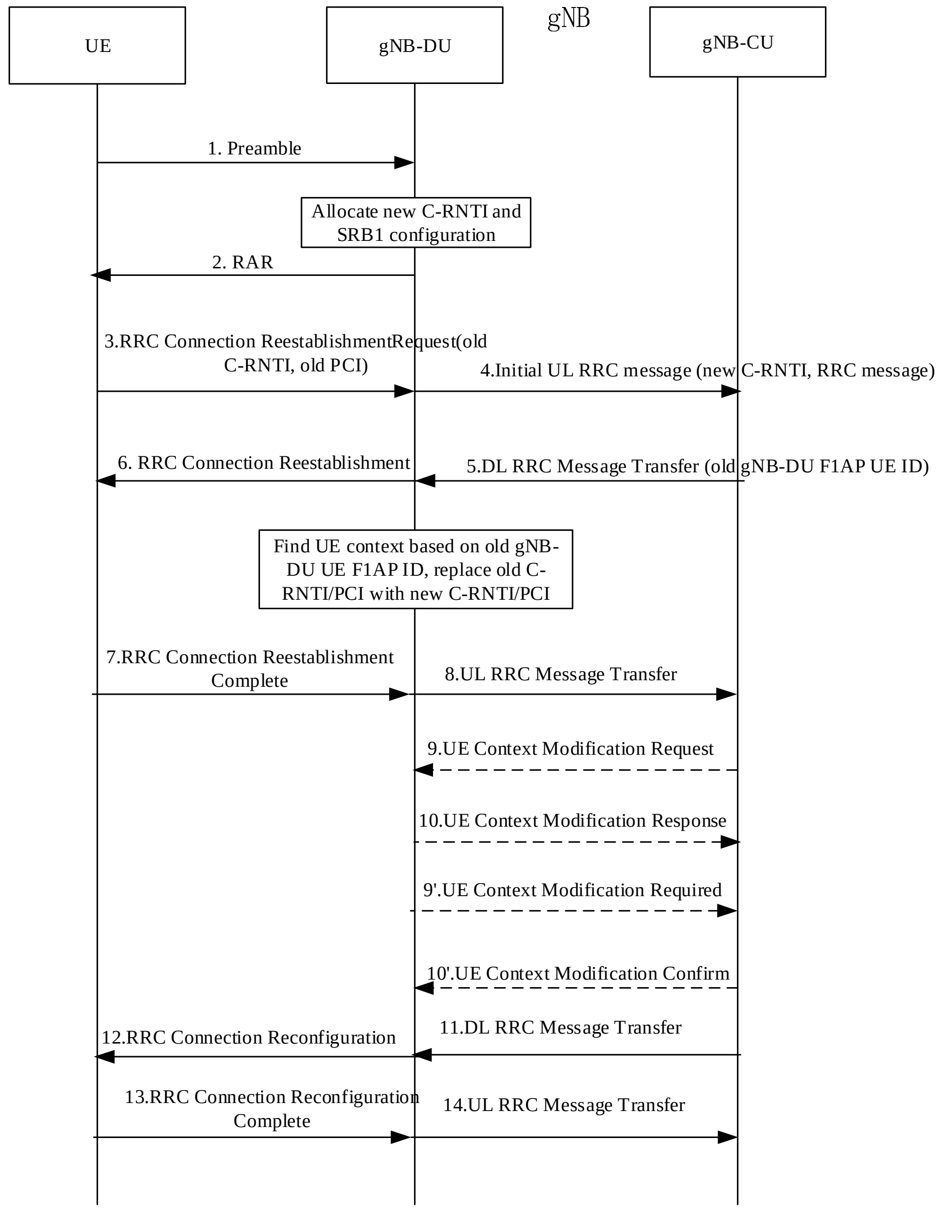

Figure 8.7-1: RRC connection reestablishment procedure

1. UE sends preamble to the gNB-DU.

2. The gNB-DU allocates new C-RNTI and responds UE with RAR.

3. UE向gNB-DU发送RRC连接重建请求消息，该消息包含旧C-RNTI和旧PCI。4. gNB-DU包含RRC消息，并且如果UE被接纳，则包含相应的低层配置

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对于F1-AP初始上行RRC消息传输消息中的UE，并传输到gNB-CU。T he INITAIL UL RRC MESSAGE TRANSFER message应包含C-RNTI。5. gNB-CU将RRC连接重建消息和旧gNB-DU F1AP UE标识包含到F1AP下行RRC消息传输消息中，并传输到gNB-DU。6. gNB-DU基于旧gNB-DU F1AP UE标识检索UE上下文，用新C替换旧C-RNTI/PCI

-

RNTI/PCI。它向UE发送RRC连接重建消息。7-8. UE向gNB-DU发送RRC连接重建完成消息。gNB-DU将RRC消息封装在F1AP上行RRC消息传输消息中并发送给gNB-CU。9-10. gNB-CU通过发送UE上下文修改请求触发UE上下文修改过程，该请求可包括要修改和释放的数据无线承载列表。gNB-DU以UE上下文修改响应消息响应。9'-10'. gNB-DU通过发送UE上下文修改要求触发UE上下文修改过程，该要求可包括要修改和释放的数据无线承载列表。gNB-CU以UE上下文修改确认消息响应。注意：这里假设UE从原始gNB-DU接入，其中UE上下文对该UE可用，并且步骤9-10或步骤9'和10'可能存在，或者两者都可跳过。注意：如果UE从非原始gNB-DU接入，gNB-CU应针对该新gNB-DU触发UE上下文建立过程。

11-12. gNB集中单元将RRC连接重配置消息包含到F1AP下行RRC消息传输消息中，并传输给gNB分布式单元。gNB分布式单元将其转发给用户设备。

13-14. 用户设备向gNB分布式单元发送RRC连接重配置完成消息，gNB分布式单元将其转发给gNB集中单元。

8.8 F1-C的多个TNLA

下面描述了管理F1-C的多个TNLA的流程。

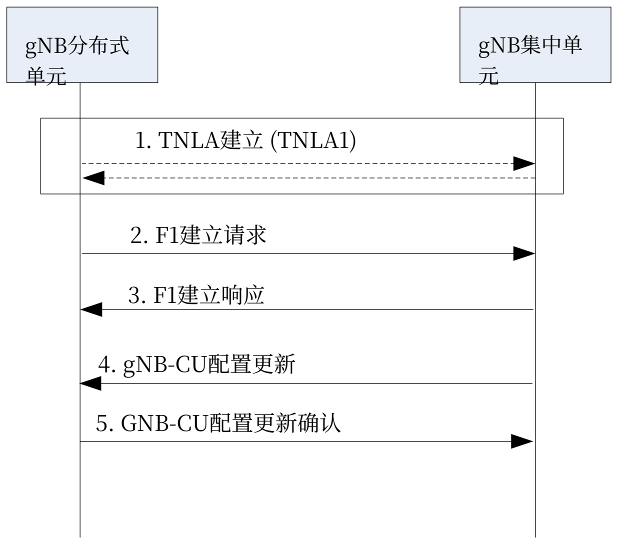


图8.8-1: 管理F1-C的多个TNLA。

1. gNB-DU使用配置的TNL地址与gNB-CU建立第一个TNLA。

3. The UE sends RRC CONNECTION REESTABLISHMENT REQUEST message to the gNB-DU, which contains old C-RNTI and old PCI.

4. The gNB-DU includes the RRC message and, if the UE is admitted, the corresponding low layer configuration for the UE in the F1-AP INITIAL UL RRC MESSAGE TRANSFER message and transfers to the gNB-CU. The INITAIL UL RRC MESSAGE TRANSFER message should include C-RNTI.

5. The gNB-CU includes RRC CONNECTION REESTABLISHMENT message and old gNB-DU F1AP UE ID into F1AP DL RRC MESSAGE TRANSFER message and transfers to the gNB-DU.

6. The gNB-DU retrieves UE context based on old gNB-DU F1AP UE ID, replaces old C-RNTI/PCI with new C-RNTI/PCI. It sends RRC CONNECTION REESTABLISHMENT message to UE.

7-8. The UE sends RRC CONNECTION REESTABLISHMENT COMPLETE message to the gNB-DU. The gNB-DU encapsulates the RRC message in F1AP UL RRC MESSAGE TRANSFER message and sends to gNB-CU.

9-10. The gNB-CU triggers UE context modification procedure by sending UE CONTEXT MODIFICIATION REQUEST, which may include DRBs to be modified and released list. The gNB-DU responses with UE CONTEXT MODIFICATION RESPONSE message.

9'-10'. The gNB-DU triggers UE context modification procedure by sending UE CONTEXT MODIFICIATION REQUIRED, which may include DRBs to be modified and released list. The gNB-CU responses with UE CONTEXT MODIFICATION CONFIRM message.

NOTE: Here it is assumed that UE accesses from the original gNB-DU where the UE contexts are available for that UE, and either step 9-10 or step 9’ and 10’ may exist or both could be skipped.

NOTE: If UE accesses from a gNB-DU other than the original one, gNB-CU should trigger UE CONTEXT SETUP procedure towards this new gNB-DU.

11-12. The gNB-CU includes RRC CONNECTION RECONFIGURATION message into F1AP DL RRC MESSAGE TRANSFER message and transfers to the gNB-DU. The gNB-DU forwards it to the UE.

13-14. The UE sends RRC CONNECTION RECONFIGURATION COMPLETE message to the gNB-DU, and gNB-DU forwards it to the gNB-CU.

8.8 Multiple TNLAs for F1-C

In the following, the procedure for managing multiple TNLAs for F1-C is described.

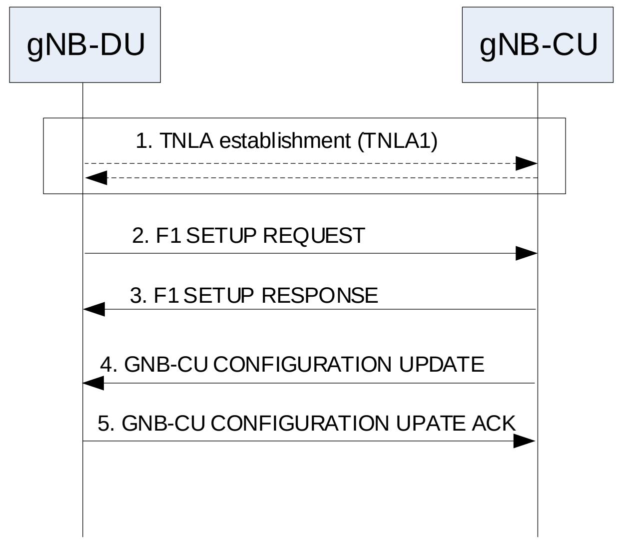


Figure 8.8-1: Managing multiple TNLAs for F1-C.

1. The gNB-DU establishes the first TNLA with the gNB-CU using a configured TNL address.

2-3. 一旦TNLA建立，gNB-DU启动F1建立过程以交换应用层配置数据。

4-5. 当需要时，gNB集中单元可通过gNB-CU配置更新过程添加额外的TNL端点，用于gNB集中单元与gNB分布式单元配对之间的F1-C信令。gNB-CU配置更新过程也允许gNB集中单元请求gNB分布式单元修改或释放传输网络层关联。

注意：关于传输网络层关联的建立需要进一步澄清。

F1AP UE TNLA绑定是针对特定UE的F1AP UE关联与特定TNL关联之间的绑定。在F1AP UE TNLA绑定创建后，gNB集中单元能通过不同的TNL关联向gNB分布式单元发送针对该UE的F1AP消息来更新UE TNLA绑定。gNB分布式单元应使用新的TNL关联更新F1AP UE TNLA绑定。

8.9 涉及E1和F1的总体流程

以下子条款描述了涉及E1和F1的总体流程。

8.9.1 UE初始接入

涉及E1和F1的UE初始接入的信令流程如图8.9.1-1所示。

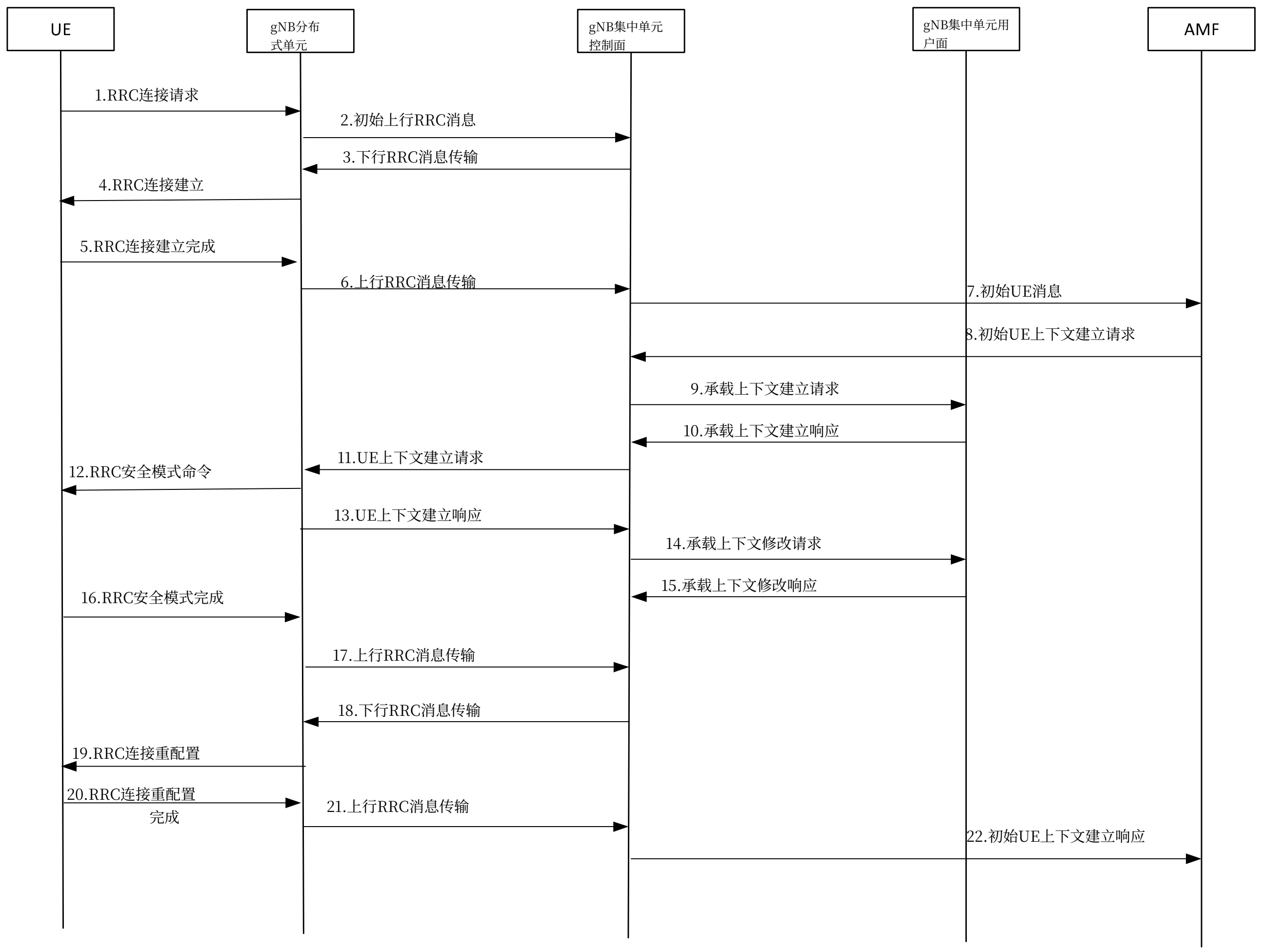


图8.9.1-1：涉及E1和F1的UE初始接入过程

步骤1-8在第8.6条中定义。

9. gNB-CU-CP发送E1AP承载上下文建立请求消息，在gNB-CU-UP中建立承载 上下文。

2-3. Once the TNLA has been established, the gNB-DU initiates the F1 Setup procedure to exchange application level configuration data

4-5. When needed, the gNB-CU may add additional TNL Endpoint(s) to be used for F1-C signalling between the gNB-CU and the gNB-DU pair using the gNB-CU Configuration Update procedure. The gNB-CU Configuration Update procedure also allows the gNB-CU to request the gNB-DU to modify or release TNLA(s).

Note: Further clarification regarding the establishment of the TNLA are needed.

The F1AP UE TNLA binding is a binding between a F1AP UE association and a specific TNL association for a given UE. After the F1AP UE TNLA binding is created, the gNB-CU can update the UE TNLA binding by sending the F1AP message for the UE to the gNB-DU via a different TNLA. The gNB-DU shall update the F1AP UE TNLA binding with the new TNLA.

8.9 Overall procedures involving E1 and F1

The following clauses describe the overall procedures involving E1 and F1.

8.9.1 UE Initial Access

The signalling flow for UE Initial access involving E1 and F1 is shown in Figure 8.9.1-1.

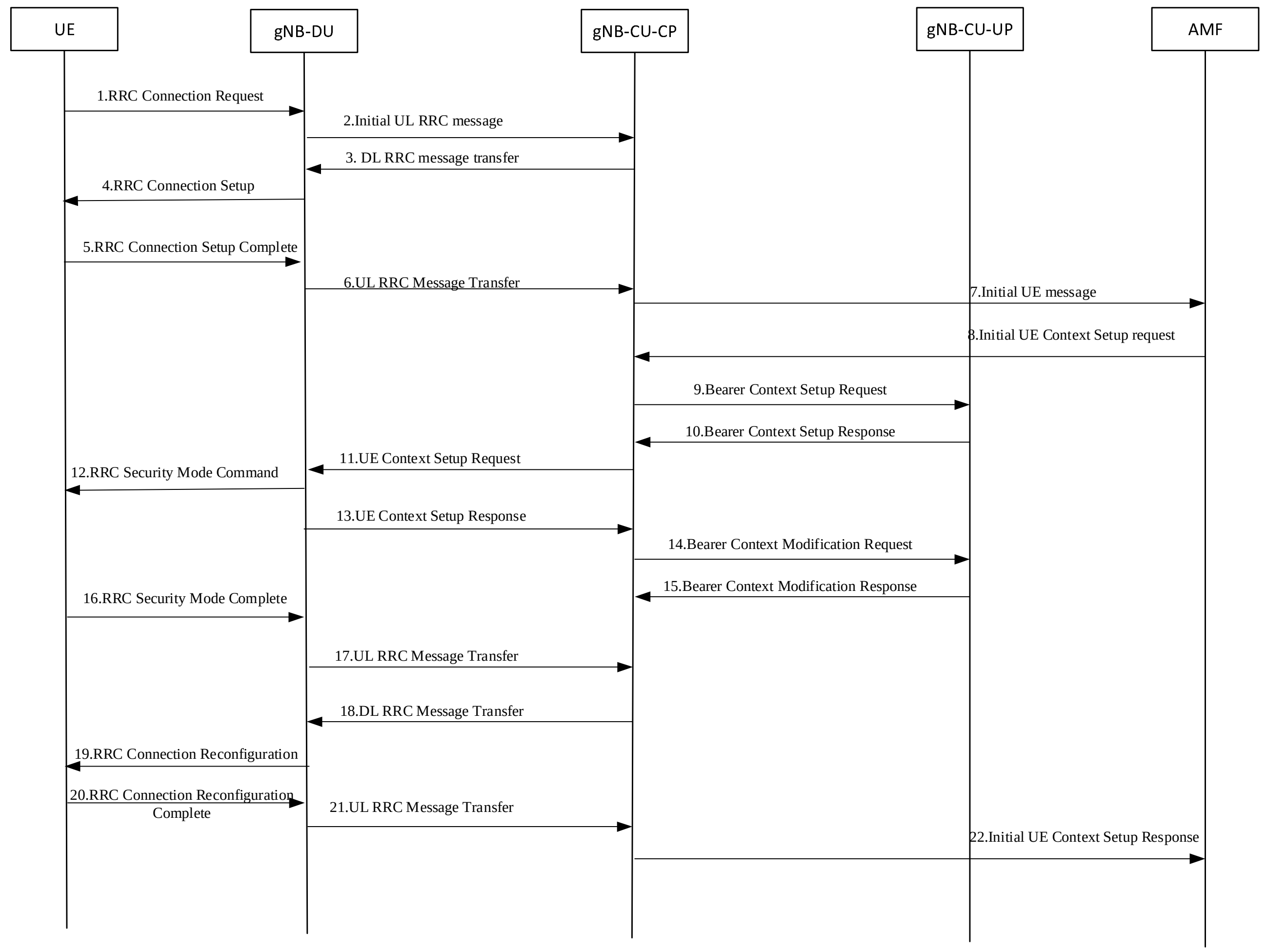


Figure 8.9.1-1: UE Initial Access procedure involving E1 and F1

Steps 1-8 are defined in clause 8.6.

9. The gNB-CU-CP sends the E1AP BEARER CONTEXT SETUP REQUEST message to establish the bearer context in the gNB-CU-UP.

10. gNB-CU-UP向gNB-CU-CP发送E1AP承载上下文建立响应消息，包括由gNB-CU-UP分配的F1-U上行隧道端点标识和传输层地址。

步骤11-13在第8.6条中定义。

14. gNB集中单元控制面向gNB集中单元用户面发送E1AP承载上下文修改请求消息，其中包括gNB分布式单元分配的F1-U下行隧道端点标识和传输层地址。

15. gNB集中单元用户面向gNB集中单元控制面发送E1AP承载上下文修改响应消息。

步骤16-22在第8.6条中定义。

注意：14-15和16-17能并行发生，但两者都在18之前。

8.9.2 通过F1-U的承载上下文建立

图8.9.2-1显示了用于在gNB集中单元用户面中建立承载上下文的流程。

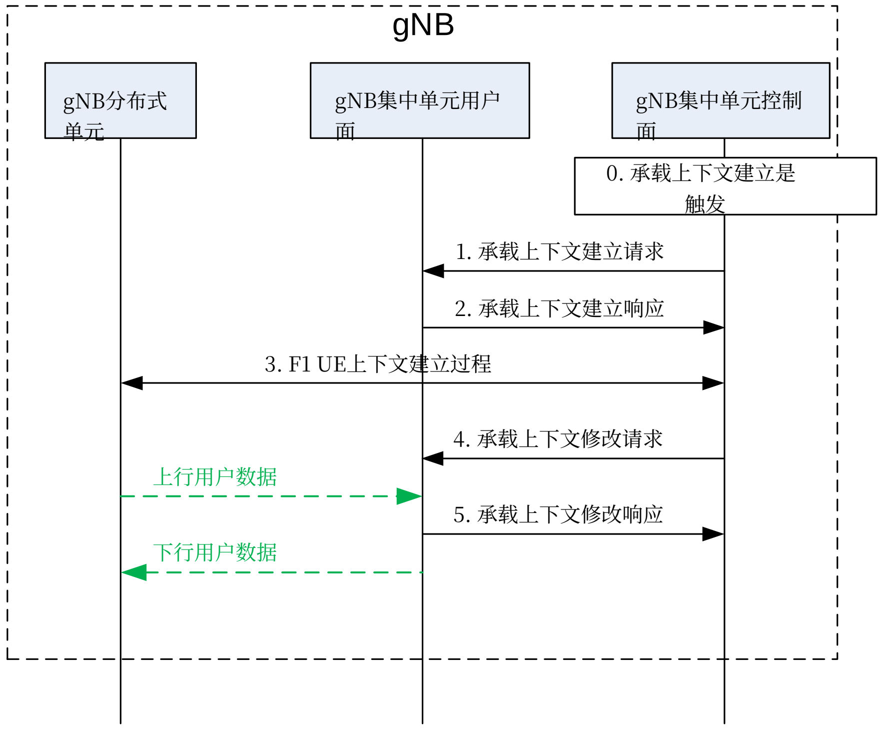


图8.9.2-1: 通过F1-U的承载上下文建立

0. 承载上下文建立（例如，在来自MeNB的SgNB添加请求之后）在gNB-CU-CP中触发。

1. gNB-CU-CP发送一条BEARER CONTEXT SETUP REQUEST消息，其中包含用于S1-U或NG-U的上行传输网络层地址信息，以及如果需要，用于X2-U或Xn-U的下行或上行传输网络层地址信息，以在gNB-CU-UP中建立承载上下文。对于NG-RAN，gNB-CU-CP决定流到数据无线承载映射，并将生成的SDAP和PDCP配置发送给gNB-CU-UP。

2. gNB集中单元用户面使用承载上下文建立响应消息进行响应，该消息包含用于F1-U接口的上行传输网络层地址信息、用于S1-U接口或NG-U的下行传输网络层地址信息，并且如果需要，还包含用于X2-U接口或Xn-U的下行或上行传输网络层地址信息。

3. 执行F1 UE上下文建立过程，以在gNB-DU中建立一个或多个承载。4. gNB-CU-CP发送一条包含用于F1-U的下行传输网络层地址信息和PDCP状态的承载上下文修改请求消息。5. gNB-CU-UP以承载上下文修改响应消息进行响应。

10 The gNB-CU-UP sends the E1AP BEARER CONTEXT SETUP RESPONSE message to gNB-CU-CP, including F1-U UL TEID and transport layer address allocated by gNB-CU-UP.

Steps 11-13 are defined in clause 8.6.

14. The gNB-CU-CP sends the E1AP BEARER CONTEXT MODIFICATION REQUEST message to the gNB-CU-UP, including F1-U DL TEID and transport layer address allocated by gNB-DU.

15. The gNB-CU-UP sends the E1AP BEARER CONTEXT MODIFICATION RESPONSE message to the gNB-CU-CP.

Steps 16-22 are defined in clause 8.6.

NOTE: 14-15 and 16-17 can happen in parallel, but both are before 18.

8.9.2 Bearer context setup over F1-U

Figure 8.9.2-1 shows the procedure used to setup the bearer context in the gNB-CU-UP.

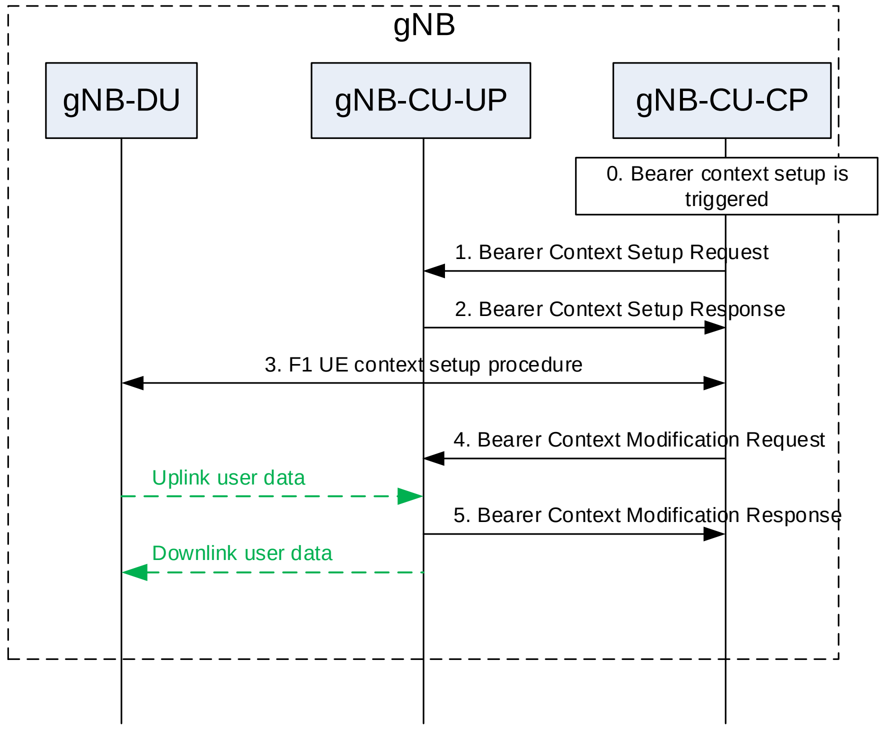


Figure 8.9.2-1: Bearer context setup over F1-U

0. Bearer context setup (e.g., following an SgNB Addition Request from the MeNB) is triggered in gNB-CU-CP.

1. The gNB-CU-CP sends a BEARER CONTEXT SETUP REQUEST message containing UL TNL address information for S1-U or NG-U, and if required, DL or UL TNL address information for X2-U or Xn-U to setup the bearer context in the gNB-CU-UP. For NG-RAN, the gNB-CU-CP decides flow-to-DRB mapping and sends the generated SDAP and PDCP configuration to the gNB-CU-UP.

2. The gNB-CU-UP responds with a BEARER CONTEXT SETUP RESPONSE message containing the UL TNL address information for F1-U, and DL TNL address information for S1-U or NG-U, and if required, DL or UL TNL address information for X2-U or Xn-U.

3. F1 UE context setup procedure is performed to setup one or more bearers in the gNB-DU.

4. The gNB-CU-CP sends a BEARER CONTEXT MODIFICATION REQUEST message containing the DL TNL address information for F1-U and PDCP status.

5. The gNB-CU-UP responds with a BEARER CONTEXT MODIFICATION RESPONSE message.

8.9.3 通过F1-U的承载上下文释放

8.9.3.1 gNB-CU-CP发起的承载上下文释放

图8.9.3.1- 1展示了由gNB-CU发起的在gNB-CU-UP中释放承载上下文所使用的流程 - CP.

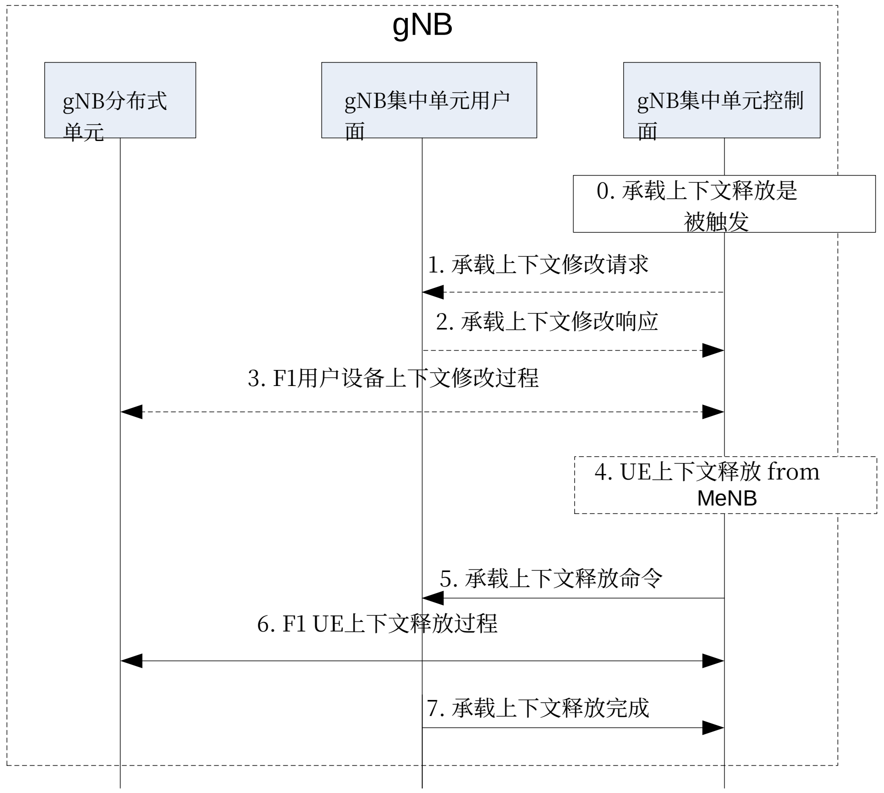


图8.9.3.1-1: 通过F1-U的承载上下文释放 – gNB-CU-CP发起

0. 承载上下文释放（例如，在从主eNB收到SgNB释放请求后）在gNB集中单元控制面中触发。

1. gNB集中单元控制面向gNB集中单元用户面发送承载上下文修改请求消息。2. gNB集中单元用户面以携带PDCP上行/下行状态的承载上下文修改响应进行应答。

3. 执行F1用户设备上下文修改过程以停止用户设备的数据传输。何时停止UE调度取决于gNB分布式单元的实现。

注意：仅当承载的PDCP状态需要保留时才执行步骤1-3，例如对于承载类型更改。

4. gNB集中单元控制面可如第8.4.2.1节所述在EN-DC操作中从主eNB接收用户设备上下文释放消息。5. 和7. 执行承载上下文释放过程。6. 执行F1 UE上下文释放过程以释放gNB-DU中的UE上下文。

8.9.3.2 gNB集中单元用户平面发起的承载上下文释放

图8.9.3.2-1展示了由gNB-CU-UP发起的、在gNB-CU-UP中释放承载上下文的流程。

8.9.3 Bearer context release over F1-U

8.9.3.1 gNB-CU-CP initiated bearer context release

Figure 8.9.3.1-1 shows the procedure used to release the bearer context in the gNB-CU-UP initiated by the gNB-CU-CP.

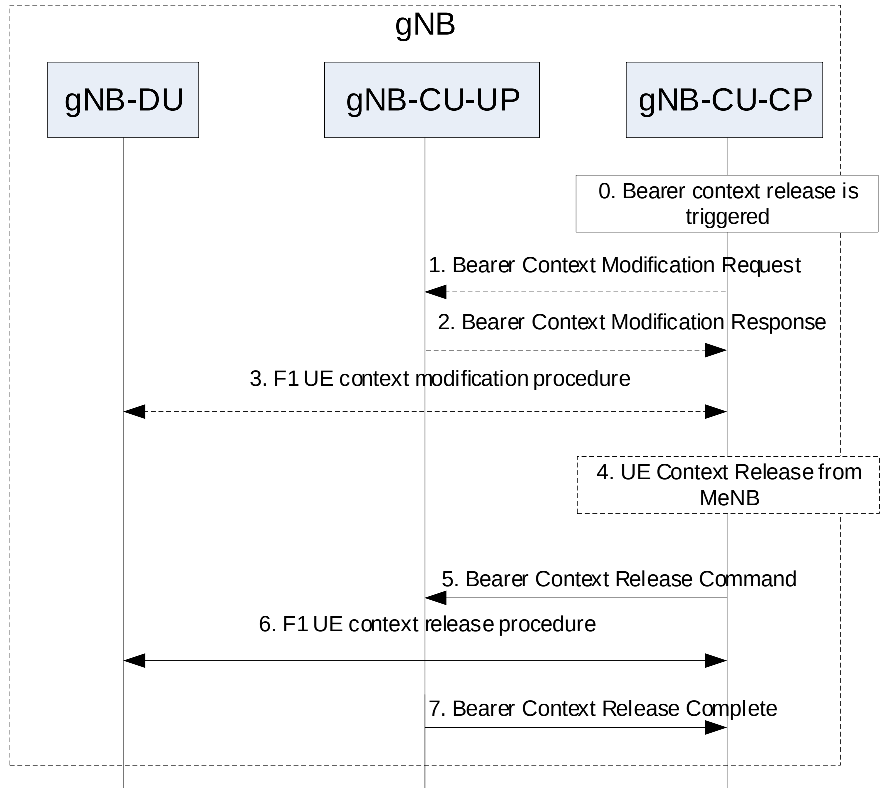


Figure 8.9.3.1-1: Bearer context release over F1-U – gNB-CU-CP initiated

0. Bearer context release (e.g., following an SgNB Release Request from the MeNB) is triggered in gNB-CU-CP.
1. The gNB-CU-CP sends a BEARER CONTEXT MODIFICATION REQUEST message to the gNB-CU-UP.
2. The gNB-CU-UP responds with a BEARER CONTEXT MODIFICATION RESPONSE carrying the PDCP UL/DL status.
3. F1 UE context modification procedure is performed to stop the data transmission for the UE. It is up to gNB-DU implementation when to stop the UE scheduling.

NOTE: step 1-3 are performed only if the PDCP status of the bearer(s) needs to be preserved e.g., for bearer type change.

4. The gNB-CU-CP may receive the UE CONTEXT RELEASE message from the MeNB in EN-DC operation as described in Section 8.4.2.1.
5. and 7. Bearer context release procedure is performed.
6. F1 UE context release procedure is performed to release the UE context in the gNB-DU.

8.9.3.2 gNB-CU-UP initiated bearer context release

Figure 8.9.3.2-1 shows the procedure used to release the bearer context in the gNB-CU-UP initiated by the gNB-CU-UP.

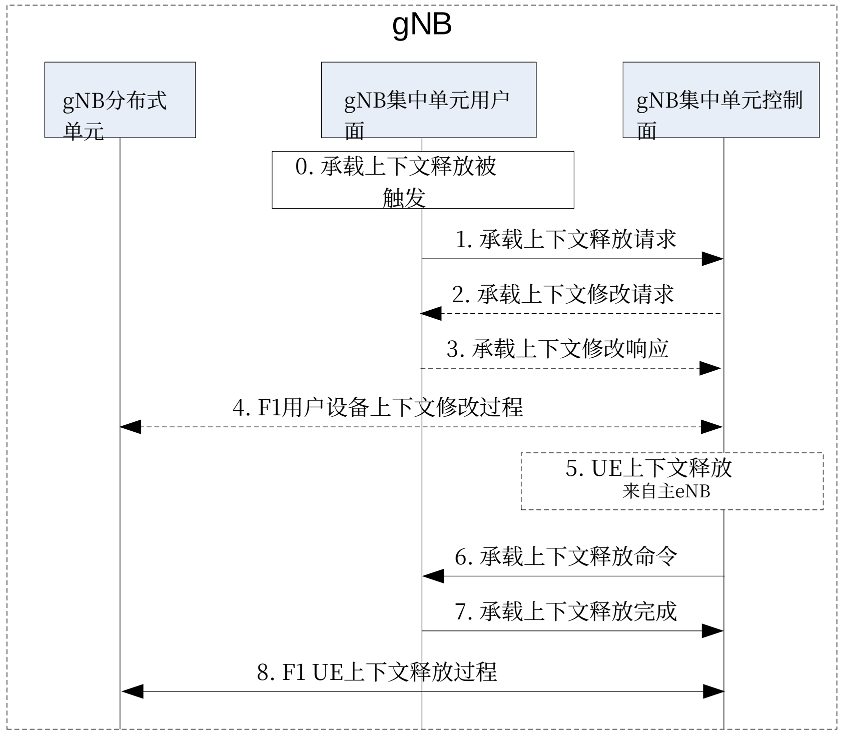


图8.9.3.2-1: 通过F1-U的承载上下文释放 – gNB-CU-UP发起

0. Bearer上下文重 lease is triggere d in gNB-CU-UP 例如 d由于 local fail ure.. 1. gNB集中单元用户面发送承载上下文释放请求消息，以请求释放gNB集中单元用户面中的承载上下文。此消息可包含PDCP状态。2.- 5. 如果需要保留PDCP状态，则执行E1承载上下文修改和F1 UE上下文修改流程。E1承载上下文修改流程用于向gNB集中单元用户面传输数据转发信息。gNB集中单元控制面可从主eNB接收UE上下文释放。6. gNB集中单元控制面发送承载上下文释放命令消息以释放承载上下文i

n

gNB集中单元用户面。7. gNB集中单元用户面用承载上下文释放完成消息进行响应，以确认承载上下文的释放，其中还包括数据转发信息。8. 可执行F1 UE上下文释放过程以释放gNB分布式单元中的UE上下文。

8.9.4 涉及gNB集中单元用户面变更的gNB间切换

图8.9.4-1 示出涉及gNB集中单元用户面变更的gNB间切换所使用的流程。总体gNB间切换流程 dure在TS 37.340 [12]中规定。

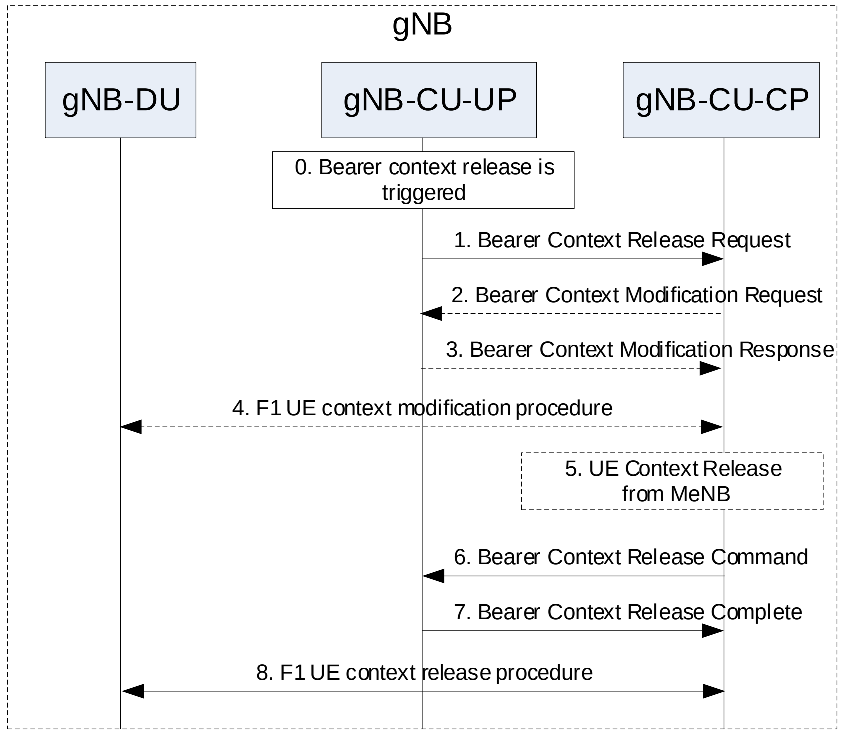


Figure 8.9.3.2-1: Bearer context release over F1-U – gNB-CU-UP initiated

0. Bearer context release is triggered in gNB-CU-UP e.g., due to local failure..

1. The gNB-CU-UP sends a BEARER CONTEXT RELEASE REQUEST message to request the release of the bearer context in the gNB-CU-UP. This message may contain the PDCP status.

2.- 5. If the PDCP status needs to be preserved, the E1 Bearer Context Modification and the F1 UE Context Modification procedures are performed. The E1 Bearer Context Modification procedure is used to convey data forwarding information to the gNB-CU-UP. The gNB-CU-CP may receive the UE Context Release from the MeNB.

6. The gNB-CU-CP sends a BEARER CONTEXT RELEASE COMMAND message to release the bearer context in the gNB-CU-UP.

7. The gNB-CU-UP responds with a BEARER CONTEXT RELEASE COMPLETE to confirm the release of the bearer context including also data forwarding information.

8. F1 UE context release procedure may be performed to release the UE context in the gNB-DU.

8.9.4 Inter-gNB handover involving gNB-CU-UP change

Figure 8.9.4-1 shows the procedure used for inter-gNB handover involving gNB-CU-UP change. Overall inter-gNB handover procedure is specified in TS 37.340 [12].

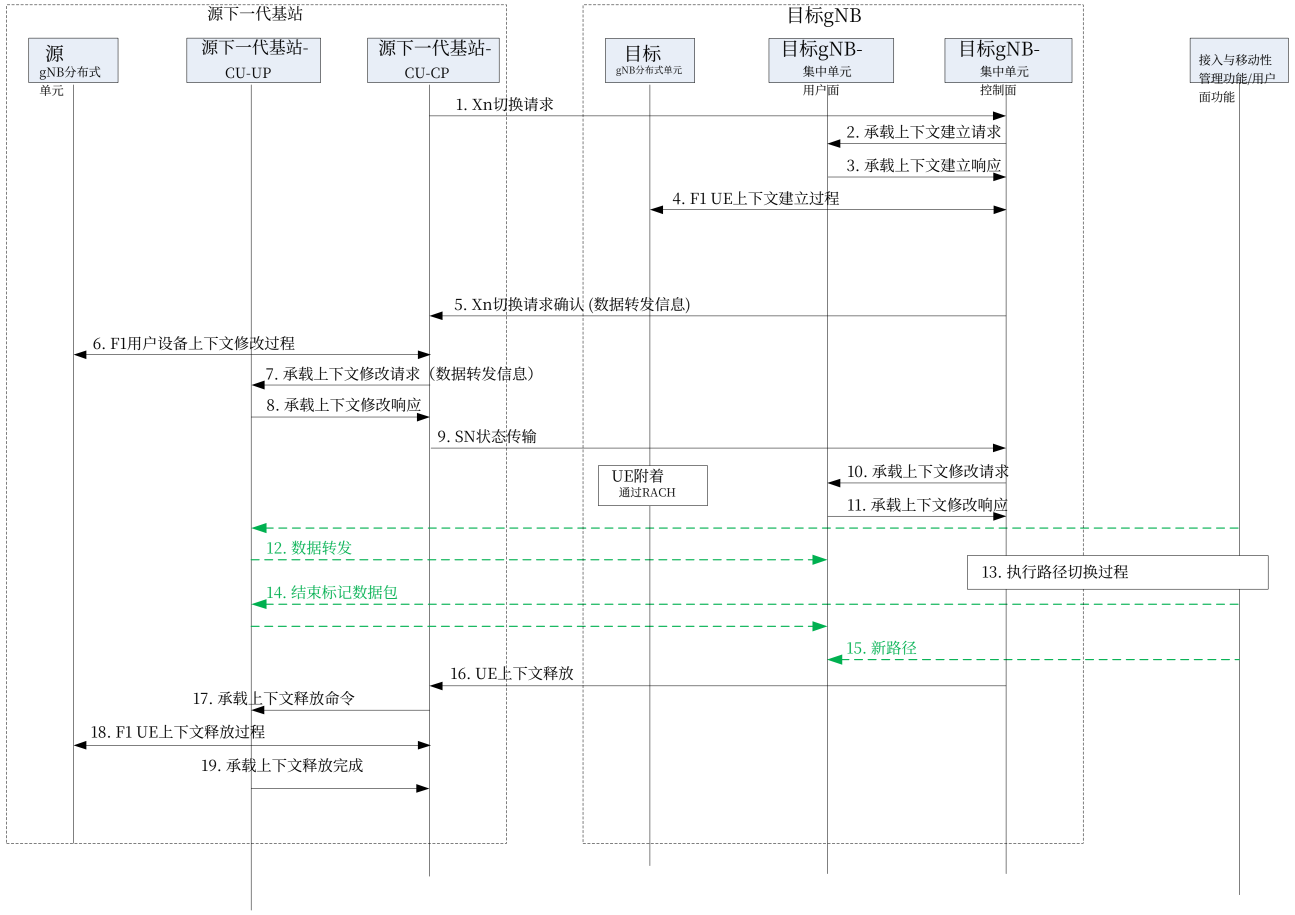


图8.9.4-1: 涉及gNB集中单元用户面变更的gNB间切换

1. 源下一代基站-集中单元-控制平面向目标gNB-CU-CP发送Xn切换请求消息。

2-4. 承载上下文建立过程按照第8.9.2节描述执行。

5. 目标gNB-CU-CP向源gNB-CU-CP响应一条Xn切换请求确认消息。6. 执行F1 UE上下文修改过程，以停止gNB-DU处的上行数据传输，并向UE发送切换命令。7-8. 执行承载上下文修改过程（由gNB-CU-CP发起），以使gNB-CU-CP能够检索PDCP上行/下行状态，并为该承载交换数据转发信息。9. 源gNB-CU-CP向目标gNB-CU-CP发送一条SN状态传输消息。10-11. 承载上下文修改过程按照第8.9.2节描述执行。12. 可执行从源gNB-CU-UP到目标gNB-CU-UP的数据转发。13-15. 执行路径切换过程，以更新朝向核心网的NG-U的下行TNL地址信息。16. 目标gNB-CU-CP向源gNB-CU-CP发送一条用户设备上下文释放消息。17. 和 19. 执行承载上下文释放过程。18. 执行F1 UE上下文释放过程，以释放源g NB-DU中的UE上下文。

8.9.5 gNB-CU-UP变更

图8.9.5-1显示了用于gNB内gNB-CU-UP变更的流程。

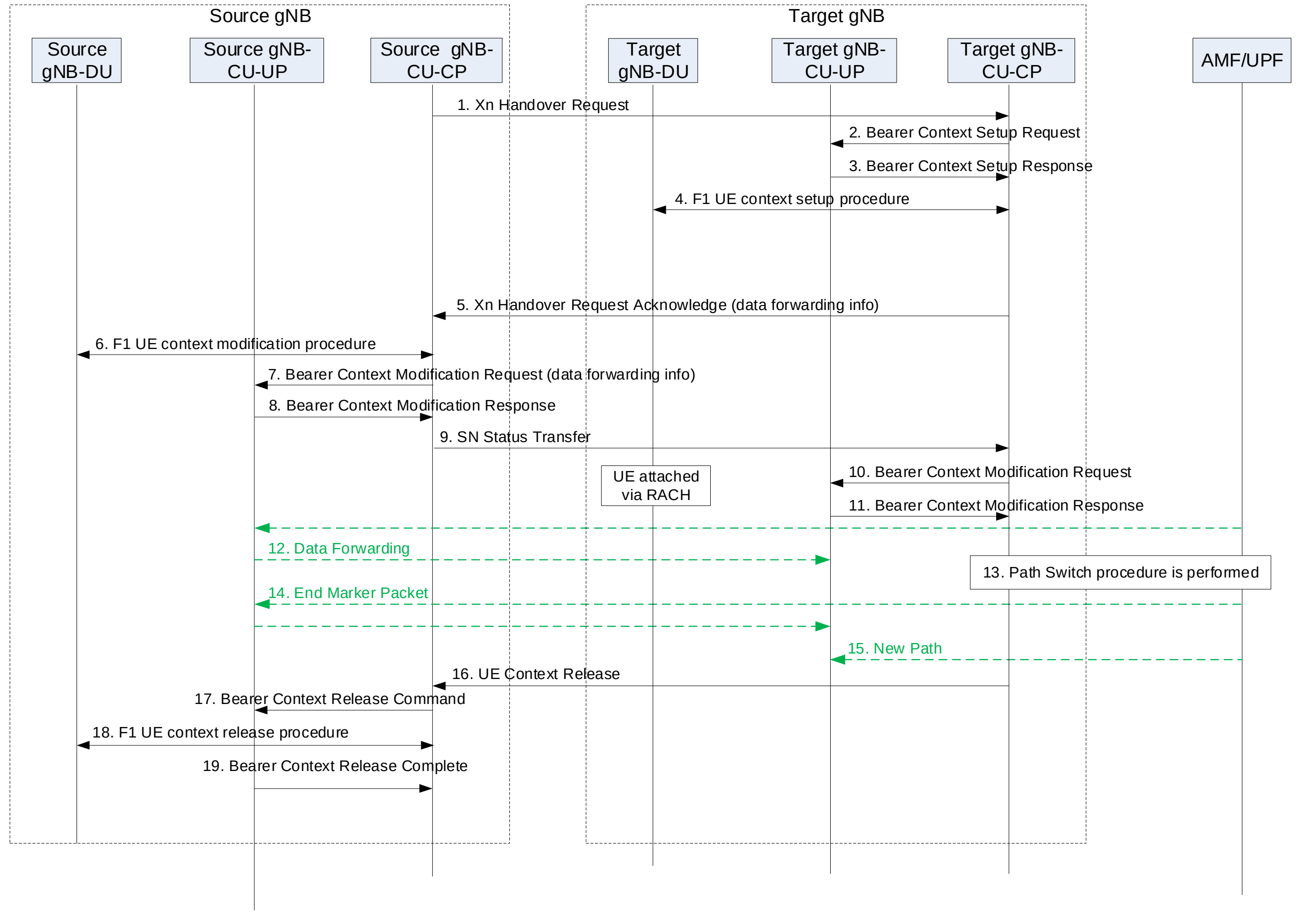


Figure 8.9.4-1: Inter-gNB handover involving gNB-CU-UP change

1. The source gNB-CU-CP sends XN HANDOVER REQUEST message to the target gNB-CU-CP.

2-4. Bearer context setup procedure is performed as described in Section 8.9.2.

5. The target gNB-CU-CP responds the source gNB-CU-CP with an XN HANDOVER REQUEST ACKNOWLEDGE message.

6. The F1 UE Context Modification procedure is performed to stop UL data transmission at the gNB-DU and to send the handover command to the UE.

7-8. Bearer context modification procedure (gNB-CU-CP initiated) is performed to enable the gNB-CU-CP to retrieve the PDCP UL/DL status and to exchange data forwarding information for the bearer.

9. The source gNB-CU-CP sends an SN STATUS TRANSFER message to the target gNB-CU-CP.

10-11. Bearer context modification procedure is performed as described in Section 8.9.2.

12. Data Forwarding may be performed from the source gNB-CU-UP to the target gNB-CU-UP.

13-15. Path Switch procedure is performed to update the DL TNL address information for the NG-U towards the core network.

16. The target gNB-CU-CP sends an UE CONTEXT RELEASE message to the source gNB-CU-CP.

17. and 19. Bearer context release procedure is performed.

18. F1 UE context release procedure is performed to release the UE context in the source gNB-DU.

8.9.5 Change of gNB-CU-UP

Figure 8.9.5-1 shows the procedure used for the change of gNB-CU-UP within a gNB.

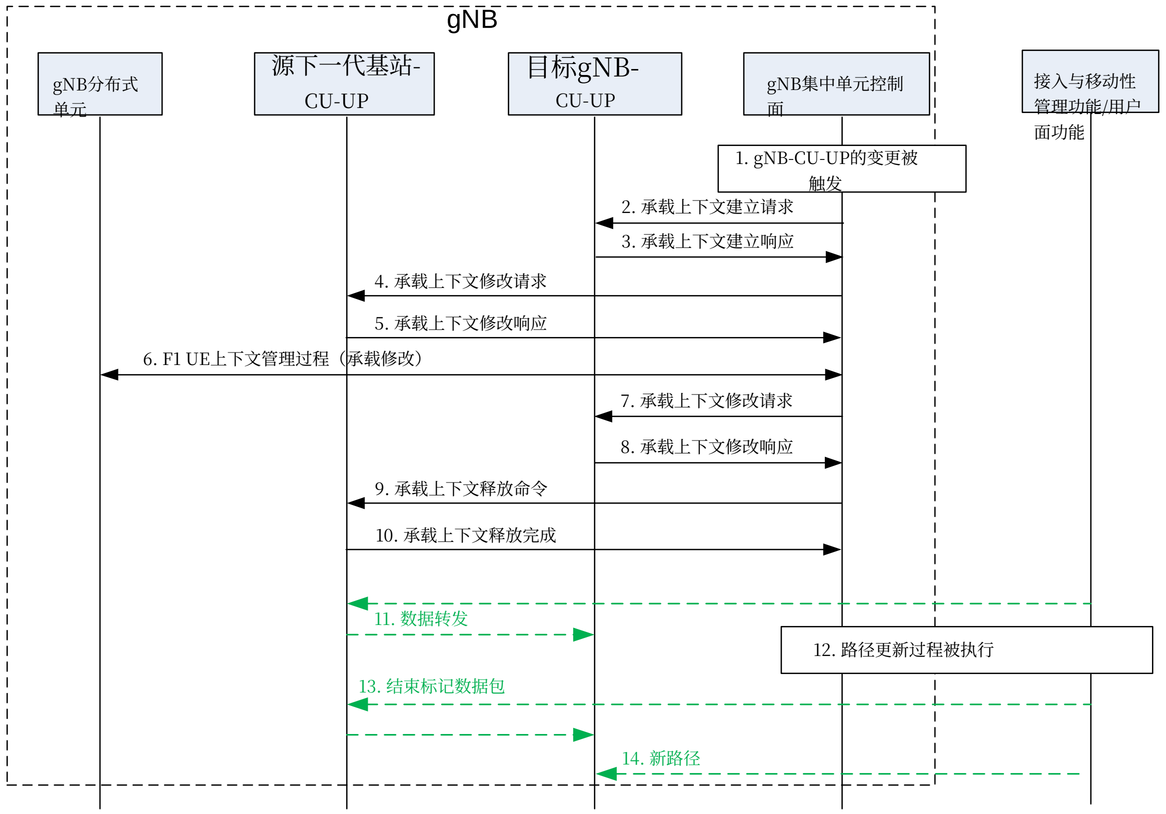


图8.9.5-1: gNB-CU-UP变更

1. 基于例如来自UE的测量报告，gNB-CU-UP的变更在gNB-CU-CP中被触发。

2-3. 承载上下文建立过程按第8.9.2节所述执行。4-5. 执行承载上下文修改过程（由gNB-CU-CP发起），以使gNB-CU-CP能够获取PDCP上行/下行状态，并交换承载的数据转发信息。6. 执行F1 UE上下文修改过程，以更改gNB-DU中一个或多个承载用于F1-U的上行传输网络层地址信息。7-8. 承载上下文修改过程按第8.9.2节所述执行。9-10. 承载上下文释放过程（由gNB-CU-CP发起）按第8.9.3节所述执行。11. 可从源gNB-CU-UP向目标gNB-CU-UP执行数据转发。12-14. 执行路径切换过程，以更新指向核心网的NG-U的下行传输网络层地址信息。

8.9.6 RRC状态转换

8.9.6.1 RRC连接态到RRC非激活态

该过程 用于将UE状态从RRC连接态更改为RRC非激活态的步骤如图8.9.6.1-1所示 .

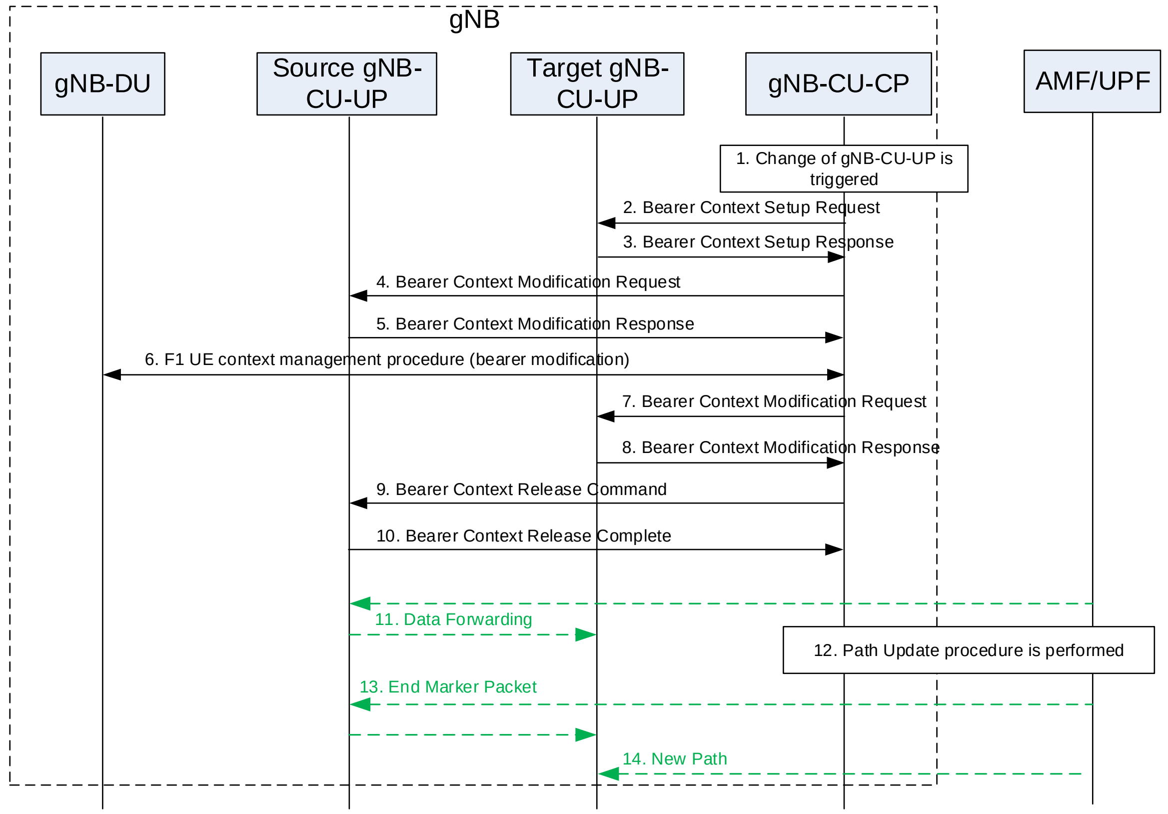


Figure 8.9.5-1: Change of gNB-CU-UP

1. Change of gNB-CU-UP is triggered in gNB-CU-CP based on e.g., measurement report from the UE.

2-3. Bearer context setup procedure is performed as described in Section 8.9.2.

4-5. Bearer context modification procedure (gNB-CU-CP initiated) is performed to enable the gNB-CU-CP to retrieve the PDCP UL/DL status and to exchange data forwarding information for the bearer.

6. F1 UE context modification procedure is performed to change the UL TNL address information for F1-U for one or more bearers in the gNB-DU.

7-8. Bearer context modification procedure is performed as described in Section 8.9.2.

9-10. Bearer context release procedure (gNB-CU-CP initiated) is performed as described in Section 8.9.3.

11. Data Forwarding may be performed from the source gNB-CU-UP to the target gNB-CU-UP.

12-14. Path Switch procedure is performed to update the DL TNL address information for the NG-U towards the core network.

8.9.6 RRC State transition

8.9.6.1 RRC Connected to RRC Inactive

The procedure for changing the UE state from RRC-connected to RRC-inactive is shown in Figure 8.9.6.1-1.

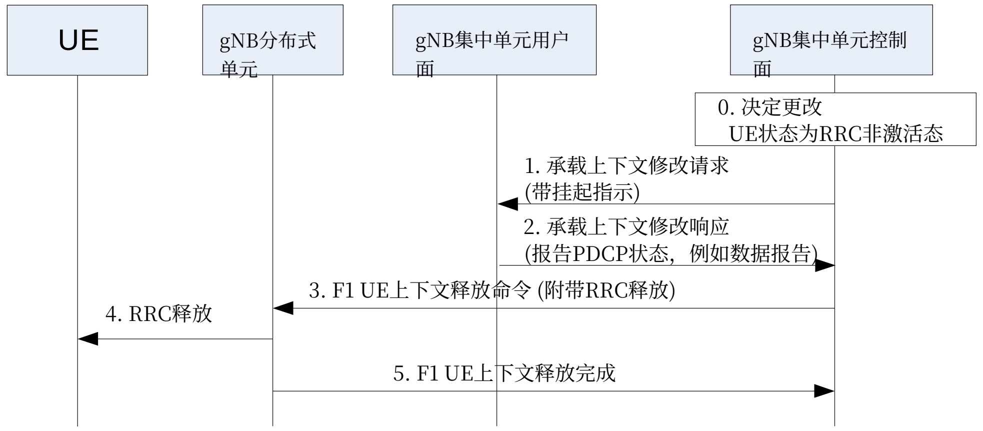


图8.9.6.1-1: RRC连接态到RRC非激活态状态转换。

0. gNB-CU-CP确定UE宜进入RRC非激活态。

1. gNB-CU-CP向gNB-CU-UP发送带有RRC挂起指示的E1承载上下文修改请求，指示UE正在进入RRC非激活态。gNB-CU-CP保留F1上行TEID。2. gNB-CU-UP发送包含PDCP上下行状态的E1承载上下文修改响应，该状态可能用于例如数据量上报。gNB-CU-UP保留承载上下文、UE关联逻辑E1连接、NG-U相关资源（例如NG-U下行TEID）和F1上行TEID。3. gNB-CU-CP向服务于UE的gNB-DU发送F1 UE上下文释放命令，连同th

e

待发送给UE的RRC释放消息。

注意：步骤1和步骤3可以同时执行。

4. gNB-DU向UE发送RRC释放消息。

5. gNB-DU向gNB-CU-CP发送F1 UE上下文释放完成消息。

8.9.6.2 RRC非激活态到其他状态

将UE状态从RRC非激活态更改为RRC连接态的过程如图8.9.6.2-1所示。

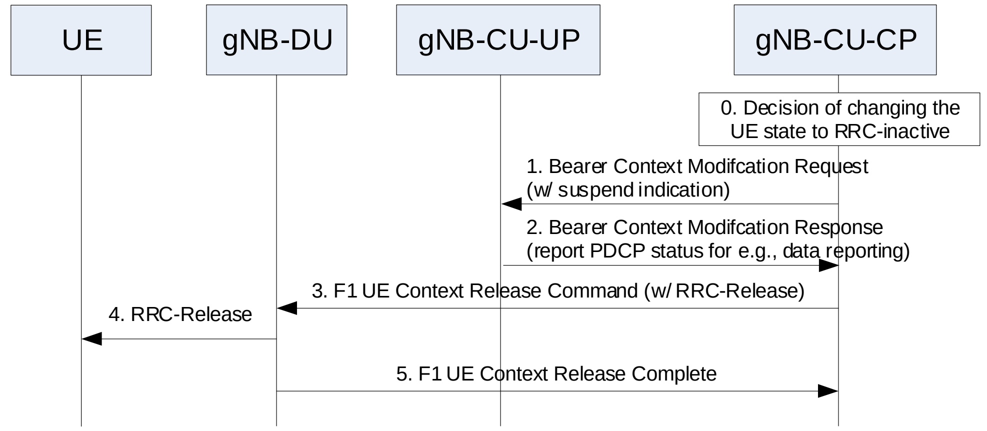


Figure 8.9.6.1-1: RRC Connected to RRC Inactive state transition.

0. The gNB-CU-CP determines that the UE should enter RRC-inactive.
1. The gNB-CU-CP sends E1 Bearer Context Modification Request with a RRC Suspend indication to the gNB-CU-UP, which indicates that the UE is entering RCC-inactive state. The gNB-CU-CP keeps the F1 UL TEIDs.
2. The gNB-CU-UP sends the E1 Bearer Context Modification Response including the PDCP UL and DL status that may be needed for e.g., data volume reporting. The gNB-CU-UP keeps the Bearer Context, the UE-associated logical E1-connection, the NG-U related resource (e.g., NG-U DL TEIDs) and the F1 UL TEIDs.
3. The gNB-CU-CP sends the F1 UE Context Release Command to the gNB-DU serving the UE, together with the RRC-Release message to be sent to the UE.

NOTE: step 1 and 3 can be performed at the same time.

4. The gNB-DU sends the RRC-Release message to the UE.
5. The gNB-DU sends the F1 UE Context Release Complete to the gNB-CU-CP.

8.9.6.2 RRC Inactive to other states

The procedure for changing the UE state from RRC-inactive to RRC-connected is shown in Figure 8.9.6.2-1.

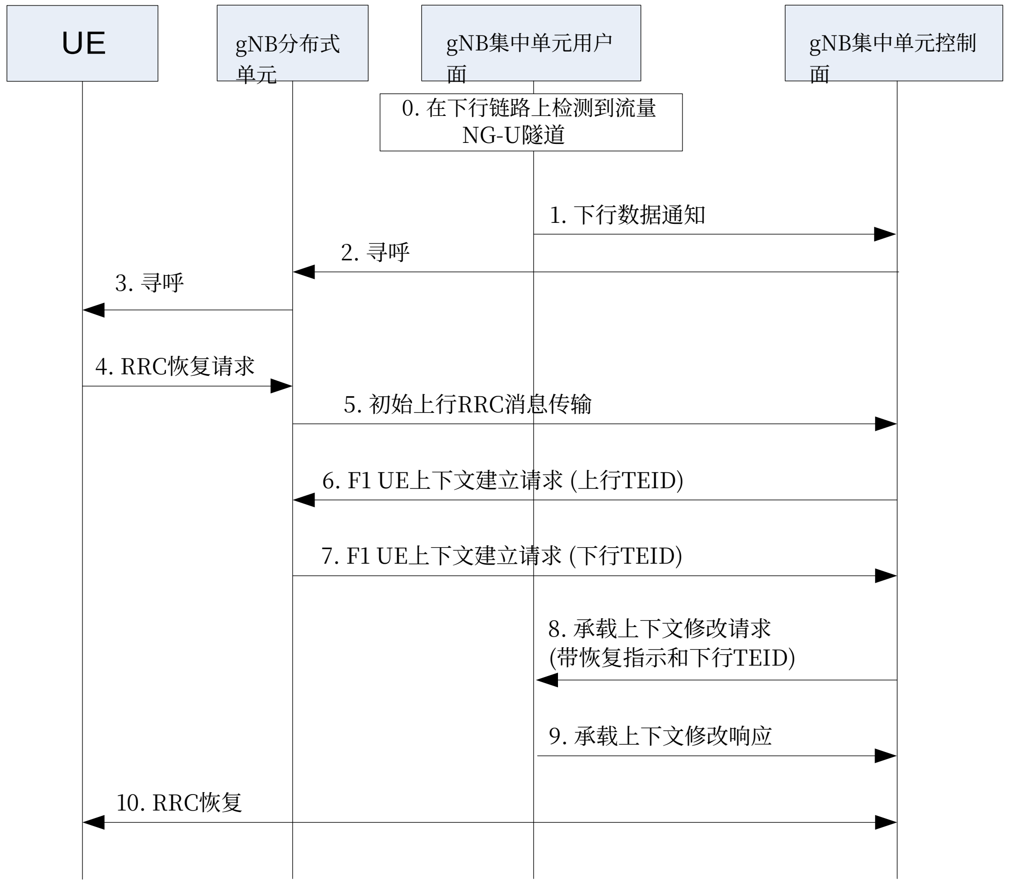


图8.9.6.2-1: RRC非活跃态到RRC连接态状态转换。

0. gNB-CU-UP在NG-U接口上接收下行数据。1. gNB-CU-UP向gNB-CU-CP发送E1下行数据通知消息。2. gNB-CU-CP发起F1寻呼过程。3. gNB-DU向UE发送RAN寻呼消息。注意：步骤0-3仅在下行数据情况下需要。4. UE在收到RAN寻呼或上行数据到达时发送RRC恢复请求。5. gNB-DU向gNB-CU-CP发送F1初始上行RRC消息传输消息。6. gNB-CU-CP发送F1 UE上下文建立请求消息，包括存储的F1上行TEID，以在gNB-DU中创建UE上下文。7. gNB-DU以UE上下文建立响应消息响应，其中包括为DRB分配的F1下行TEID。8. gNB-CU-CP发送带有RRC恢复指示的E1承载上下文修改请求，指示UE正在从RRC非激活状态恢复。gNB-CU-CP还包括从gNB-DU在步骤7中接收到的F1下行TEID。9. gNB-CU-UP以E1承载上下文修改响应。10. gNB-CU-CP和UE通过gNB-DU执行RRC恢复过程。注意：步骤8/9和10可以并行执行。

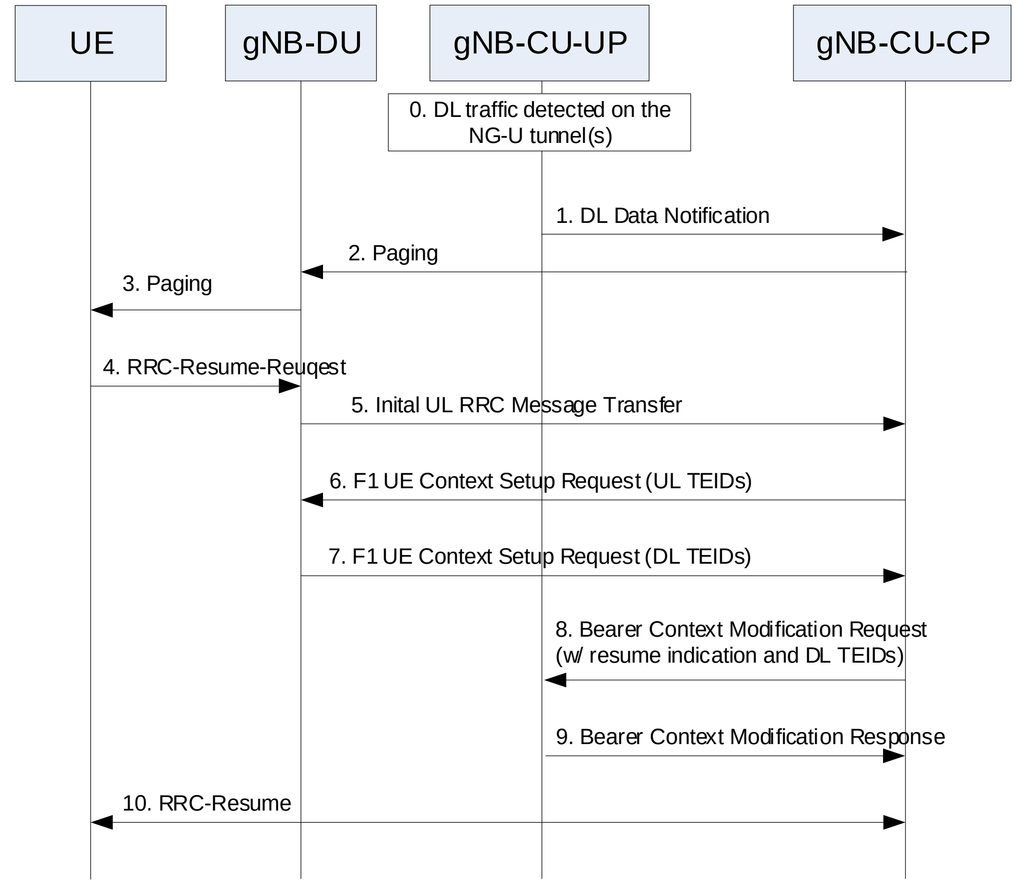


Figure 8.9.6.2-1: RRC Inactive to RRC Connected state transition.

0. The gNB-CU-UP receives DL data on NG-U interface.

1. The gNB-CU-UP sends E1 DL Data Notification message to the gNB-CU-CP.

2. The gNB-CU-CP initiates F1 Paging procedure.

3. The gNB-DU sends the RAN Paging message to the UE.

NOTE: step 0-3 are needed only in case of DL data.

4. The UE sends RRC-Resume-Request either upon RAN paging or UL data arrival.

5. The gNB-DU sends the F1 Initial UL RRC Message Transfer message to the gNB-CU-CP.

6. The gNB-CU-CP sends F1 UE Context Setup Request message including the stored F1 UL TEIDs to create the UE context in the gNB-DU.

7. The gNB-DU responds with the UE Context Setup Response message including the F1 DL TEIDs allocated for the DRBs.

8. The gNB-CU-CP sends E1 Bearer Context Modification Request with a RRC Resume indication, which indicates that the UE is resuming from RRC-inactive state. The gNB-CU-CP also includes the F1 DL TEIDs received from the gNB-DU in step 7.

9. The gNB-CU-UP responds with the E1 Bearer Context Modification Response.

10. The gNB-CU-CP and the UE preform the RRC-Resume procedure via the gNB-DU.

NOTE steps 8/9 and 10 can be performed in parallel.

9 同步

9.1 gNB同步

gNB应支持用于相位、时间和/或频率同步的逻辑同步端口。

用于相位和时间同步的逻辑同步端口应提供:

1) 精度应满足同步TDD单播区域内所有gNB对最大相对相位差的要求；2) 为同步TDD单播区域内所有gNB提供可溯源至公共时间参考的无闰秒连续时间。

例如，可为FDD时域小区间干扰协调同步区域内的所有gNB提供用于相位和时间同步的逻辑同步端口。

此外，应为同步TDD单播区域内的所有gNB提供公共SFN初始化时间。

基于此信息，gNB可根据以下公式推导SFN:

$$SFN = \{time\}\mathrm{mod}\{period(SFN)\},$$

其中:

time 经公共SFN初始化时间调整的时间，以10毫秒为单位，以匹配无线帧长度及相应精度；

周期(SFN) SFN周期。

如果gNB通过TDM接口连接，则可利用该接口对gNB进行频率同步。gNB中时钟的特性设计应考虑到接口上的抖动和漂移性能要求符合ITU-T建议书G.823 [8], 、ITU-T建议书G.824 [9] 中业务接口输出漂移的网络限制, 或符合ITU-T建议书G.825 [10], 中任何分层接口的最大输出抖动和漂移的网络限制，以适用者为准。

如果gNB通过以太网接口连接且网络支持同步以太网，则gNB可利用该接口获得频率同步。在此情况下，gNB时钟的设计宜考虑接口上的抖动和漂移性能要求符合ITU-T建议书G.8261/Y.1361 [11], 第9.2.1节中规定的EEC接口输出抖动和漂移要求。关于同步以太网建议和架构方面的进一步考虑在ITU-T建议书G.8261/Y.1361 [11]的第12.2.1节和附录A中定义。

在同步TDD单播区域中，所有gNB应支持可配置的LTE TDD起始帧偏移，以实现在共存场景中的互操作性。

10 NG-RAN接口

10.1 NG接口

3GPP TS 38.410 [14] 规定了NG接口总体方面和原则。

10.2 Xn接口

3GPP TS 38.420 [15] 规定了Xn接口总体方面和原则。

10.3 F1接口

第三代合作伙伴计划 TS 38.470 [16] 规定了F1接口的总体方面和原则。

9 Synchronization

9.1 gNB Synchronization

The gNB shall support a logical synchronization port for phase-, time- and/or frequency synchronization.

Logical synchronization port for phase- and time-synchronization shall provide:

1) accuracy that allows to meet the gNB requirements on maximum relative phase difference for all gNBs in synchronized TDD-unicast area;

2) continuous time without leap seconds traceable to common time reference for all gNBs in synchronized TDD-unicast area.

A logical synchronization port for phase- and time-synchronization may also be provided for e.g., all gNBs in FDD time domain inter-cell interference coordination synchronization area.

Furthermore common SFN initialization time shall be provided for all gNBs in synchronized TDD-unicast area.

Based on this information, the gNB may derive the SFN according to the following formula:

$$SFN = \{time\}\mathrm{mod}\{period(SFN)\},$$

where:

time time adjusted by the common SFN initialization time, in units of 10 ms to match the length of radio frame and accuracy accordingly;

period(SFN) SFN period.

In case gNB is connected via TDM interface, it may be used to frequency synchronize the gNB. The characteristics of the clock in the gNB shall be designed taking into account that the jitter and wander performance requirements on the interface are in accordance with network limits for output wander at traffic interfaces of either ITU-T Recommendation G.823 [8], ITU-T Recommendation G.824 [9] or network limits for the maximum output jitter and wander at any hierarchical interface of ITU-T Recommendation G.825 [10], whichever is applicable.

In case gNB is connected via Ethernet interface and the network supports Synchronous Ethernet, the gNB may use this interface to get frequency synchronization. In this case the design of the gNB clock should be done considering the jitter and wander performance requirements on the interface are as specified for output jitter and wander at EEC interfaces of ITU-T Recommendation G.8261/Y.1361 [11], defined in clause 9.2.1. Further considerations on Synchronous Ethernet recommendations and architectural aspects are defined in clause 12.2.1 and Annex A of ITU-T Recommendation G.8261/Y.1361 [11].

A configurable LTE TDD-offset of start frame shall be supported by all gNBs in synchronized TDD-unicast areas in order to achieve interoperability in coexistence scenarios.

10 NG-RAN interfaces

10.1 NG interface

3GPP TS 38.410 [14] specifies NG interface general aspects and principles.

10.2 Xn interface

3GPP TS 38.420 [15] specifies Xn interface general aspects and principles.

10.3 F1 interface

3GPP TS 38.470 [16] specifies F1 interface general aspects and principles.

10.4 E1接口

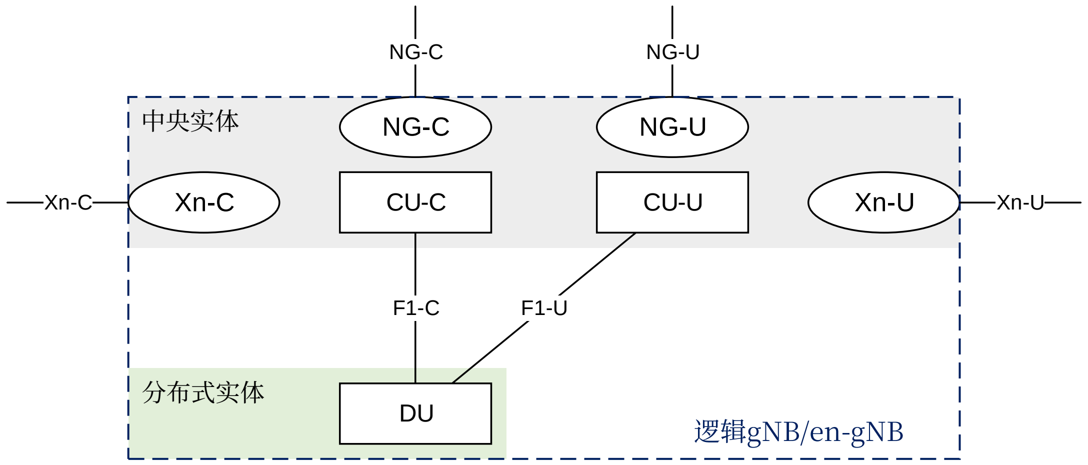
第三代合作伙伴计划 TS 38.460 [17] 规定了E1接口的总体方面和原则。

10.4 E1 interface

3GPP TS 38.460 [17] specifies E1 interface general aspects and principles.

附录A（资料性）： gNB/en-gNB的部署场景

图A-1显示了逻辑gNB/en-gNB内部的逻辑节点（集中单元控制面、集中单元用户面和分布式单元）。NG和Xn接口的协议终止点在图A-1中被描绘为椭圆。图A-1中所示的术语"中央实体"和"分布式实体"指的是物理网络节点。



图A-1: 逻辑gNB/en-gNB的部署示例

Annex A (informative): Deployment scenarios of gNB/en-gNB

Figure A-1 shows logical nodes (CU-C, CU-U and DU), internal to a logical gNB/en-gNB. Protocol terminations of the NG and Xn interfaces are depicted as ellipses in Figure A-1. The terms "Central Entity" and "Distributed Entity" shown in Figure A-1 refer to physical network nodes.

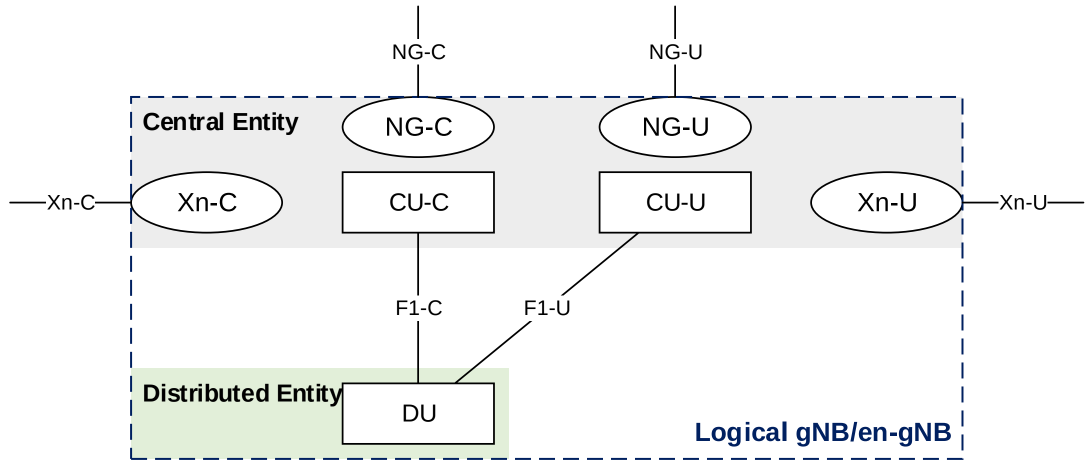


Figure A-1: Example deployment of an Logical gNB/en-gNB

附录B（资料性）：变更历史

变更历史							
Date	会议	TDoc	CR	修订			New 版本
2017年4月	RAN3第95次bis会i					初稿	0.0.1
2017年5月	RAN3第96次会议	R3-171962				反映已同意的文本提案在RAN3第96次会议. 新增第8条款，用于gNB-CU/DU的总体流程架构	0.1.0
2017年6月	RAN3新无线电会议临时会议	R3-172634				反映已同意的文本提案在RAN3#NR临时会议于2017年6月（青岛）	0.2.0
2017年8月	RAN3#97会议	R3-173449				在RAN3#97会议中反映已同意的文本提案	0.3.0
2017年10月	RAN3第97次双会i					在RAN3第97次双会议中反映已同意的文本提案	0.4.0
2017年10月	RAN3第97次双会议					删除10.3下的子条款（移至38.470）	版本0.4.1
2017年11月	RAN3第98次会议	R3-175054文档				反映RAN3#98会议同意的文本建议	版本0.5.0
2017年12月	RP-78文档	RP-172545文档				提交至RAN全体会议#78批准	版本1.0.0
2017年12月	RP-78文档					经RAN全体会议批准的TR	15.0.0
2018年3月	RP-79	RP-180468	0007 1		-	关于gNB-DU内切换的更正	15.1.0
2018年6月	RP-80	RP-181237	0001 5		B	引入 独立组网新空口 （38.401 基线变更请求 涵盖 RAN3工作组协议）	15.2.0
2018年6月	RP-80	RP-181240	0008 4		B	引入控制面与向上分离用于拆分选项2 (38.401 基线变更请求覆盖RAN3协议)	15.2.0
2018年6月	RP-80	RP-181238	0012 1		F	关于上行链路阻塞的TS38.401变更请求	15.2.0
2018年6月	RP-80	RP-181238	0014 1		F	针对38.401的DL用户数据到DU的变更请求	15.2.0
2018年6月	RP-80	RP-181238	0016 破折号		F	支持完整配置的38.401的技术提案	15.2.0

Annex B (informative): Change History

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2017-04	RAN3#95bis	R3-171307				First Draft	0.0.1
2017-05	RAN3#96	R3-171962				Reflected agreed TP in RAN3#96. Added new Clause 8 for overall procedure for gNB-CU/DU architecture	0.1.0
2017-06	RAN3#NR Ad Hoc	R3-172634				Reflected agreed TP in RAN3#NR Ad Hoc in 2017-06 (Qingdao)	0.2.0
2017-08	RAN3#97	R3-173449				Reflected agreed TP in RAN3#97	0.3.0
2017-10	RAN3#97bis	R3-174237				Reflected agreed TP in RAN3#97bis	0.4.0
2017-10	RAN3#97bis	R3-174258				Removed subclauses under 10.3 (moved to 38.470)	0.4.1
2017-11	RAN3#98	R3-175054				Reflected agreed TP in RAN3#98	0.5.0
2017-12	RP-78	RP-172545				Submit to RAN Plenary#78 for approval	1.0.0
2017-12	RP-78					TR Approved by RAN plenary	15.0.0
2018-03	RP-79	RP-180468	0007	1	-	Correction on intra-gNB-DU handover	15.1.0
2018-06	RP-80	RP-181237	0001	5	B	Introduction of SA NR (38.401 Baseline CR covering RAN3 agreements)	15.2.0
2018-06	RP-80	RP-181240	0008	4	B	Introduction of Separation of CP and UP for Split Option 2 (38.401 Baseline CR covering RAN3 agreements)	15.2.0
2018-06	RP-80	RP-181238	0012	1	F	CR to TS38.401 on UL blockage	15.2.0
2018-06	RP-80	RP-181238	0014	1	F	CR to 38.401 on DL User Data to DU	15.2.0
2018-06	RP-80	RP-181238	0016	-	F	TP for 38.401 in supporting full configuration	15.2.0

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